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Bayesian Phylolinguistic Inference under Cognate Dependence: A Study of Model Misspecification

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Abstract

In historical linguistics, Bayesian phylogenetic methods have quickly become the principal framework for reconstructing the evolution of languages from cognate data. One key assumption of such methods is conditional site independence: that cognate classes evolve independently from each other. However, given the relative complexities of language evolution, linguistic and cultural datasets often exhibit dependencies between sites. This causes model misspecification: a mismatch in assumptions made by the model and the true characteristics of the data, the consequences for which remain poorly understood.

This thesis investigates the impact of violating the conditional site independence assumption on Bayesian phylogenetic inference, applied in linguistic contexts. A simulation study was conducted to answer two questions: (1) the extent by which unknown dependencies amongst sites of the data had negative effects on tree reconstruction, and (2) whether including known correlated sites (and thus causing model misspecification) affected tree reconstruction. These questions were posed to provide an understanding of the implications caused by such dependency structures and a framework for decision making for retaining dependent data.

The results show that violating conditional site independence indeed reduces tree reconstruction quality relative to fully independent data. The depth of dependency, however, made little contribution to affecting the tree reconstruction quality. Instead, the greatest factor of reconstruction quality is the total *effective* information contained within the dataset. In low-information scenarios, unknown dependencies cause larger deterioration while additional dependent data generally improves reconstruction despite violating model assumptions. On the other hand, in high-information scenarios, the drop in reconstruction quality due to unknown dependencies is minor, while marginal benefits of additional dependent sites diminish and thus may not justify their inclusion.

These findings identify effective information as the underlying framework for understanding the effects of model misspecification in Bayesian phylolinguistics. They suggest that the implications of violating conditional site independence in such methods are a nuanced matter, where correlated sites present tradeoffs between introducing bias and additional effective information. Depending on the context at hand, these factors then determine whether their inclusion proves beneficial for phylogenetic reconstruction.

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Plagiarism statement

The work contained in this thesis is my own work unless otherwise stated.

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Chapter 1

Introduction

Language is, perhaps, one of the most fundamental aspects of human society. Beyond serving practically as a means of communication, languages embody culture, history and collective identity, preserving traditions, belief systems and social structures that have developed over generations. The modern world contains thousands of languages with a remarkable range of similarities and differences. By investigating how languages change and diversify over time, historical linguistics provides a window into the development of human societies themselves.

The task of reconstructing and explaining the evolution of all languages is no trivial undertaking. Szathmáry and Smith (1995) posit that language emergence in human societies is one of the major transitions in evolution [1], while Christiansen and Kirby (2003) describe languages as "by far the most complex behaviour we know of" [2]. Languages undergo evolution at varying timescales [2] while importing words from other languages (or *borrowing*) is commonplace [3]. On top of that, there are currently over 7000 languages in the world [4]. Furthermore, direct evidence about languages is not readily available and hypotheses about language evolution must be tested through an interdisciplinary analysis [2].

Despite these challenges, there have been many approaches to reconstruct language evolution. A common starting point is the observation that many languages exhibit systematic similarities, suggesting a shared common historical origin between them. Parsons (1767) first considered such similarities between numerals of languages in Europe and India, concluding that the languages must be related by a common ancestor [5]. These languages would later be proven to be from the same *language family* (the *Indo-European* family) [6] - where groups of languages which are said to derive from a common language are *genetically* related via a language family [7]. Today, there are now as many as 143 recognised language families [4], similarly identified through studying a plethora of similarities and systematic changes in languages over time [3] as well as employing more computational and quantitative methods in the age of big data [8]. These methods, which are commonly imported from other disciplines like phylogenetics, need to be carefully considered in the context of historical linguistics.

This chapter introduces the principal statistical methods used in historical linguistics, culminating in the advent of Bayesian phylogenetic analysis in language evolution, affectionately named *phylolinguistics*. Particular attention is given to the assumptions of the model, especially the assumption of site independence, and the extent to which these assumptions might be violated in linguistic data. Relevant previous work concerning site independence in both phylogenetics and linguistics is then reviewed.

1.1 History of Language Diversification Methods

We assume the starting point of analysing evolution within language families: where related languages share a common origin. A natural next step is: how do languages from the same family diverge from each other over time? We start by considering how language similarities are constructed.

1.1.1 Language Similarity and Data Generation

As mentioned, Parsons (1767) presented semantic similarities in basic numerals in what are now the languages of the Indo-European family, some of which are presented in Table 1.1, while other languages such as Malay and Turkish have very different words for numerals, concluding that the aforementioned languages should stem from a common ancestor [5].

Language	1	2	3	10
Latin	unus	duo	tres	decem
Spanish	uno	dos	tres	diez
German	eins	zween	drey	zehen
Dutch	een	twee	drii	tien
Danish	een	toe	tre	tie
English	one	two	three	ten
Bengali	ake	dho	teen	does
Persian	yek	do	see	da
Turkish	beer	echee	ewch	on
Hebrew	ehad	senaim	selosa	hasara
Malay	satoo	duo	teego	sepulo

Table 1.1 Numericals in languages as per Parsons’ sample [5], where languages which do not use the Latin alphabet have approximated transcriptions.

More sophisticated methods have since been used to justify systematic similarities and differences between languages for a given meaning [3]. While the exact methods used by linguists are outside the scope of this paper, we note that for a given meaning and set of languages, we can group languages together based on how similar the words (or *cognates*) used are. We define several terms to formalise our analysis:

Definition 1 (Cognate). Consider words (or more generally, lexical items) $w_1^m \in L_1$ and $w_2^m \in L_2$, from languages L_1 and L_2 respectively, that have meaning m . We can define a binary relation $\sim$ representing similarity of words of a given meaning m , such that if $w_1^m \sim w_2^m$, we say that w_1^m and w_2^m are cognates.

It is worth discussing that while in general, linguistic data is derived from lexical items, it is possible to consider other data that concern language diversification. For example, Buckley et al. (2025) considered different weaving technologies in addition to linguistics data to understand cultural and linguistic evolution amongst speakers of the Kra-dai language family [9].

Definition 2 (Cognacy class). Given meaning m and words $w_1^m \in L_1, \dots, w_n^m \in L_n$, the cognacy classes of a given meaning m are the sets $\{P_1^m, \dots, P_K^m\} =: P$ of $\{w_1^m, \dots, w_n^m\}$ that represent the equivalence classes of $\sim$ for a given meaning m . In other words, if P satisfies:

1. $\forall k = 1, \dots, K, \forall w_i^m, w_j^m \in P_k^m, w_i^m \sim w_j^m$ and

$$2. \forall k, l = 1, \dots, K, l \neq j, \forall w_i^m \in P_k^m, w_j^m \in P_l^m, w_i^m \approx w_j^m,$$

then $P_1^m, \dots, P_K^m$ are cognacy classes.

By assuming that all languages have exactly one word to describe each meaning m , the value of K in Definition 2 can vary between 1 (all languages have similar words for meaning m) and n (no languages have similar words for meaning m)¹. Each cognacy class can then be used to encode a vector of languages that belong to the cognacy class, which we will define as a site:

Definition 3 (Site). Given list of languages $\{L_1, \dots, L_n\}$ and cognacy class $P_k^m := \{w_{i_1}^m, \dots, w_{i_P}^m\}$, where $w_{i_1}^m \in L_{i_1}, \dots, w_{i_P}^m \in L_{i_P}$ and $1 \leq i_p \leq n, p = 1, \dots, P$. The corresponding site for cognacy class P_k^m is the vector $s \in [0, 1]^n$ such that $\forall j = 1, \dots, n$:

$$s_j = \begin{cases} 1 & j \in \{i_1, \dots, i_P\} \\ 0 & \text{otherwise} \end{cases}. \quad (1.1)$$

As such, for a given meaning m , there are K sites corresponding to the K cognacy classes present amongst the n languages. We can compile all sites across all meanings m to form a binary data matrix $\mathbf{D} \in [0, 1]^{n \times S}$, where S is the total number of sites, encoding all similarities and differences between languages $L_1, \dots, L_n$. Under such a construction, it is clear that the similarity of rows i and j of $\mathbf{D}$ corresponds to how similar languages L_i and L_j are, given a set of meanings. An example of this encoding process can be found in Table 1.2.

Language	'to laugh'		'wing'		Language	'to laugh'		'wing'		
	Word	Class	Word	Class		A	B	A	B	C
Fijian	dredre	tl-A	taba-na	w-A	Fijian	1	0	1	0	0
Marquesan	kata	tl-B	peheu	w-B	Marquesan	0	1	0	1	0
Hawaiian	'aka	tl-B	'heu	w-B	Hawaiian	0	1	0	1	0
Maori	kata	tl-B	parirau	w-C	Maori	0	1	0	0	1
Tahitian	'ata	tl-B	pererau	w-C	Tahitian	0	1	0	0	1

Table 1.2 Binary encoding of cognacy classes from five Oceanic languages. Adapted from [10].

A natural question that arises is, then, how one defines a suitable list of meanings, say the set $\mathcal{M}$, to base these similarities from. This is not merely a practical issue of data collection, but fundamentally shapes the signal that is recovered from a language dataset. A canonical example of such a set $\mathcal{M}$ was put forward by Swadesh (1952) in an attempt to estimate the age of the most recent common ancestor (see Section 1.1.3) [11]. Swadesh put forward 200 meanings which, in his view, were considered universal and broad enough for use in most languages. This set of meanings became known as the Swadesh list (which can be found in Table A.1). Since then, more analysis has gone into the choices behind $\mathcal{M}$. Zhang and Gong (2016) considered the reliability of the Swadesh list and the statistical invariability of the 100- and 200-Swadesh lists [12], while Dediu and Cysouw (2013) considered the stability of meanings in relation to evolution rates, noting the irreducible complexity of linguistic features [13]. Many more examples of $\mathcal{M}$ have been generated through larger databases like the Intercontinental Dictionary Series [14]. The choice of $\mathcal{M}$ is thus paramount to stability and reliability of any results obtained from linguistic data, and thus to making statements regarding relationships between languages.

¹This is not true in general: languages often have multiple words for a given meaning. However, this allows for building intuition on defining cognacy classes and sites.

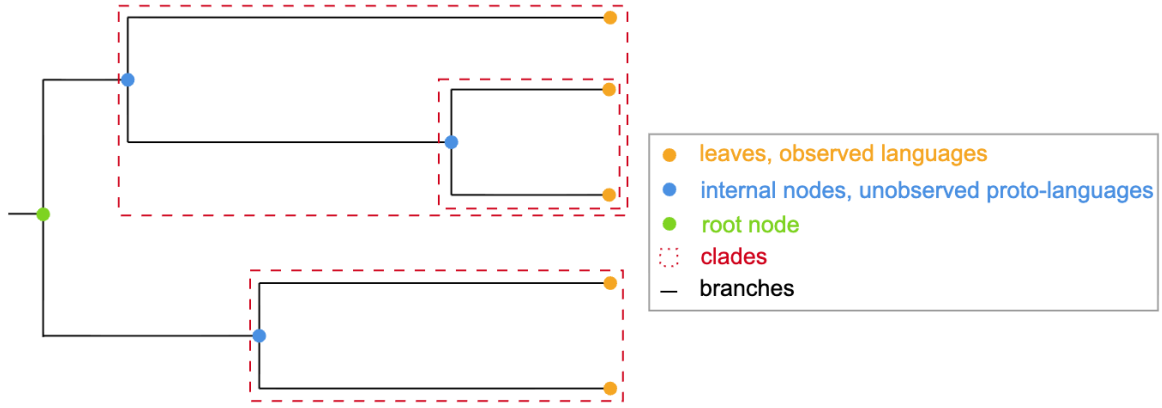


Figure 1.1 A language tree and its parts.

1.1.2 Language Trees

Such relationships between languages are represented, most commonly, in the form of a language tree [15], popularised by Schleicher (1861-62) [16].

Definition 4 (Language Tree). Let $T = (V, E, r)$ be a rooted directed tree, where V is a finite set of vertices, $E \subseteq V \times V$ is a set of directed edges and $r \in V$ is a designated root node. In the context of historical linguistics, T is a language tree if the underlying undirected tree is connected and acyclic (it is a directed acyclic graph), and every vertex $v \in V, v \neq r$ has exactly one parent $u \in V$.

An example of a language tree can be found in Figure 1.1. We continue defining the parts of a language tree:

Definition 5 (Observed and Unobserved Languages). Let $T = (V, E, r)$ be a language tree. The set of vertices $L := \{v \in V \mid \deg^+(v) = 0\}$ with outdegree zero (also known as the leaves or terminal nodes) as the observed languages of the tree. Additionally, the remaining vertices $W := V \setminus L$, or the internal nodes, represent the unobserved proto-languages that gave rise to events of divergence between languages.

Definition 6 (Clade). Let $T = (V, E, r)$ be a language tree. For some node $v \in V$, define its descendants as $\text{Desc}(v) := \{u \in V \mid v \rightarrow u\}$. A clade $C(v)$ with ancestor v is defined as $C(v) = \text{Desc}(v) \cup v$, a subgroup including ancestor v and all its descendants.

While other models exist, the tree model has persisted, with good reason, as the primary method in historical linguistics for representing language evolution. The model assumes that linguistic features are inherited with slight changes over time, and that related languages hail from a single common ancestor via divergence of cognacy classes [15]. The tree model can be further constrained such that the tree has a *bifurcating* structure (that is, that each internal node has exactly two out-branches) and has branches which have lengths proportional to time. This makes comparing events of divergence relevant: the tree model is able to hypothesise when two languages first diverged, and quantify how closely related languages are to each other. Although it is, like any model, unable to represent the full complexity of linguistic change, the model is well suited to cases of well-differentiated and geographically isolated languages [15], thus being applicable to studying language diversity within a language family.

1.1.3 Tree Reconstruction Methods

Having considered the construction of linguistic data $\mathbf{D}$ in Section 1.1.1 and the tree model T in Section 1.1.2, a natural question is whether the underlying evolutionary tree T can be inferred from the observed data $\mathbf{D}$. In particular, given a collection of related languages and meanings, quantities of interest include the topology of T (the genealogical relationships between languages) and the ages of internal and root nodes corresponding to ancestral languages.

These questions, among many others, have been traditionally answered by hand by linguists [17]. One of the first quantitative approaches to estimating root ages was glottochronology, introduced by Swadesh (1952), assuming a constant rate of evolution r within meanings in the Swadesh list [11]. However, this method has many limitations, particularly in the expressivity of the model in comparison to the complexity of language diversification [18, 19], leading to the need to explore more complex methods.

1.2 Bayesian Phylogenetic Methods

Advanced quantitative methods have gained in popularity in recent years, in part due to advances in statistical methods, increased computational power and larger datasets [8]. Through the seminal work of Gray and Atkinson (2003), Bayesian frameworks have become the primary method to infer topologies T from $\mathbf{D}$. In particular, Gray and Atkinson (2003) compared languages to genes, thus applying phylogenetic methods from evolutionary biology to linguistic data. Such a method was applied on cognate data from 87 Indo-European languages to justify the age of the Proto-Indo-European root [20].

The application of phylogenetic analysis in language evolution is a natural extension to the language tree diagrams (which we will denote phylogenetic/phylolinguistic trees or phylogenies from here on) introduced in Section 1.1.2. Historical linguistics has long understood language diversification as a branching process where descendant languages emerge through gradual divergence from common ancestral forms [15], making language evolution almost analogous to genetic evolution in the diversification process. Consequently, computational techniques originally developed in evolutionary biology, such as likelihood-based inference and Bayesian tree reconstruction can be adapted to linguistic data [21].

A central advantage of Bayesian phylogenetic methods is that they treat language history as a stochastic process on a space of possible trees $\mathcal{T}$ with varying rates of evolution upon different branches, allowing both the tree topology and parameters relating to the stochastic process to be inferred jointly from the data $\mathbf{D}$. Notably, these methods are robust to effects of horizontal transfer and borrowing [20]. These advantages, along with other advantages explored more rigorously in Chapter 2, allow the model to accommodate the difficulties associated with language genealogy reconstruction while resolving the issues of glottochronology. As such, these methods have been employed extensively in analysis of other language families [21–28] with strong methodological justification.

However, it would be unwise to blindly apply these methods directly to linguistics data. In particular, the evolutionary processes underlying language are considerably less well understood than those governing genetic evolution. Christiansen and Kirby (2003) also note that language evolution must be understood as a complex adaptive system rather than a simple tree-like branching process [2]. Understanding the assumptions of the model and the extent by which historical linguistics violates these assumptions allows for stronger interpretation and thus application of the results.

1.2.1 Assumptions and Site Independence

We save most of the technical analysis of the Bayesian phylogenetic method for Chapter 2 and focus our attention here on the assumption of interest:

Definition 7 (Conditional Site Independence Assumption). *In Bayesian phylogenetic analysis, data $\mathbf{D} := (\mathbf{D}_1 \cdots \mathbf{D}_S) \in [0, 1]^{n \times S}$ is generated with respect to true tree T^* and parameters θ^* of the evolutionary model independently: such that sites $\mathbf{D}_i \in [0, 1]^n, i = 1, \dots, S$ are conditionally independent.*

While there are other assumptions made in the context of Bayesian phylogenetic methods, we consider the validity of the assumption of conditional site independence in the context of historical linguistics.

While other assumptions can be reconciled by methods like comparing results to archaeological and historical data [28], the aforementioned assumption is the most difficult to accommodate for. Assuming conditional site independence treats meanings and cognacy classes as evolving independently across lineages. While this simplification is crucial to computational tractability (see Section 2.3.3) [10], it is clear that this is an idealisation of the data $\mathbf{D}$. In reality, lexical items are not independent units of evolution, but rather embedded in semantic networks and cultural systems that induce dependence in their rates of evolution. This suggests the phenomenon of *epistasis*: that changes in one site of the data might be dependent on the identity of another site or sites [29]. As a result, the application of the Bayesian phylogenetic model misspecifies the dataset $\mathbf{D}$ and may overstate the amount of independent evidence present. This can lead to inflated confidence in certain inferred relationships, particularly in cases where cognacy classes are driven by shared processes rather than independent evolutionary events.

However, as will be presented in Chapter 2, implementing a method accounting for correlation between sites is difficult and often times infeasible. Moreover, there are a plethora of benefits in implementing a model assuming conditional site independence, not least reducing computational intractability. Thus, it is worth investigating how damaging this misspecification is in recovering the true tree T^* along with the parameters θ^* of the model.

1.3 Previous Work

The assumption of conditional independence between sites has long been recognised as a modelling misspecification in phylogenetics and comparative biology. Substantial work has investigated data which violate site-independence assumptions and have discussed the theoretical and practical implications across a range of phylogenetic settings.

In Comparative Biology and Phylogenetic Inference. An influential discussion of site dependence appears in Maddison’s (1993) analysis of correlated traits in comparative biology [30], now coined Maddison’s red/blue tail problem. In particular, Maddison demonstrates that when a hierarchially dependent character - such as tail colour - is coded as a separate character to the higher level structure (in this case, the presence of a tail) and scored as “missing”, phylogenetic models treat missing entries as independent entries, allowing long-distance between distantly related clades. This results in a dependency that is biologically unjustified, since the common ancestor of both groups was tailless and thus had no tail colour at all.

Epistasis is prevalent even in comparative biology. These dependencies can manifest as interactions between genes [31], between protein sites [32, 33] or between sites in RNA molecules [34–36]. Furthermore, Nasrallah et al. (2011) showed that disregarding epistasis decreases the

accuracy of phylogenetic inference in simulations [37], either by introducing bias or reducing the effective number of sites in the alignment [29].

There have been many methods attempting to detect and account for site dependence in evolutionary sequence data. Shapiro et al. (2006) proposed a phylogenetic approach for detecting positive epistasis by comparing substitution patterns across a reconstructed phylogeny, identifying cases where paired sites exhibit coordinated evolutionary changes that are unlikely under an independent-sites model [34]. In a related direction, Kryazhimskiy et al. (2011) developed a framework for quantifying epistatic interactions by modelling how substitutions at one site influence substitution probabilities at another, thereby explicitly incorporating dependence into the evolutionary process [33].

Complementary to these model-based approaches, Poon et al. (2007) introduced an evolutionary network model combined with bootstrap procedures to assess the robustness of inferred interaction structures under phylogenetic uncertainty, allowing for the identification of stratified or structured dependencies among sites [38]. More recently, posterior predictive approaches have been used to evaluate model adequacy directly: for instance, Magee et al. (2021) assessed whether standard independent-sites models sufficiently capture observed patterns of epistasis by comparing empirical data to replicated datasets generated under the fitted model, thereby providing a formal diagnostic for violations of site independence [29].

In Historical Linguistics. Site dependence is not only prevalent in biological contexts but also in historical linguistics. In particular, work in syntactic phylogenetics has shown that linguistic features may exhibit structured patterns of non-independence, reflecting underlying grammatical constraints rather than freely evolving units. Longobardi et al. (2013, 2016) demonstrate that syntactic parameters used in phylolinguistic reconstruction are not always statistically independent, but instead may cluster due to shared structural or functional properties of grammar [39, 40]. This introduces a potential mismatch between the theoretical structure of linguistic objects and the simplifying assumption of site independence commonly employed in Bayesian phylogenetic models.

There are many clear examples where data exhibits site dependent behaviour, most commonly found when non-language cultural factors are considered in phylolinguistic analysis. Jordan and Shennan (2003) studied basketry techniques, which had inherent dependencies between traits and methods, along with language traits of Californian languages [41]. Buckley et al. (2025), in perhaps the clearest instance of site dependence present within data for phylolinguistic reconstruction, considered loom data, in addition to language data, to uncover language phylogenies. This loom data had known dependency structures which were accounted for by applying weights to prioritise independent sites. Clearly, the presence of certain traits of a loom are dependent on other traits being present, and thus a tiered approach to classify dependency structure of traits was introduced and tested. However, the authors demonstrated that loom data alone was still able to recover phylogenies broadly consistently with those obtained by language data considered [9]. This thus presents an important methodological question: to what extent, then, does violating conditional site independence affect inference through Bayesian phylogenetic analysis?

Outstanding gap in Literature. Despite extensive evidence that site dependence arises across biological sequence evolution, syntactic structure, and broader cultural datasets, these strands of literature remain largely disconnected from the statistical assumptions underlying Bayesian phylolinguistic inference. In particular, while biological phylogenetics has developed a rich toolkit for detecting, modelling, and diagnosing violations of site independence (including epistasis models, network-based approaches, and posterior predictive checks), and while linguistic and cultural studies have documented clear cases of structured dependence among features, these insights have not been systematically incorporated into the likelihood framework

used in modern Bayesian phylolinguistic reconstruction of trees. Instead, most applications in linguistics data seem to disregard the violation of the site independence assumption and apply the standard phylogenetic models to the data. This creates a methodological gap that fails to account for the effect on posterior estimates and potential bias that is generated from model misspecification in linguistic and cultural data.

1.4 Research Questions and Contributions

This thesis attempts to address this gap by analysing the implications of violating conditional site independence in lexical datasets and assessing its impact on inferred phylogenies despite model misspecification. In particular, we ask two questions:

1. How does an increased proportion of site-dependent data affect tree reconstruction? That is, given data of fixed size S on n related languages, how does (unknown) site dependence affect reconstruction of true tree T^* ?
2. How does adding dependent data to independent data affect reconstruction? That is, given data that maintains site independence of size S on n related languages, how does adding data that is known to be site-dependent affect reconstruction of the true tree T^* ?

By addressing these questions through a simulation study, this thesis aims to clarify the extent to which violations of conditional site independence can distort phylogenetic inference in linguistic datasets. In particular, it seeks to quantify whether the presence of dependent structure influences accuracy and stability of inferred tree topologies, and whether introducing data that violates the assumptions of the model is still effective in tree reconstruction. In doing so, it provides a bridge between advances in model misspecification found in biological sequence evolution to applications in linguistic and cultural phylogenetics.

1.5 Thesis Structure

In Chapter 2, we present the technical background of the Bayesian phylogenetic model, describing the theoretical implications of removing the conditional site independence assumption. Next, we provide the methodology used to conduct the simulation study that answers the aforementioned two questions in Chapter 3, validating the simulation framework in the process. Chapter 4 contains results from the simulation study which will be analysed and discussed in Chapter 5. We conclude the thesis by summarising findings, considering limitations and possible extensions in Chapter 6.

As is the nature of a simulation study, the experiments are written in and simulated with code. The code written for the results in this thesis can be found in this [GitHub repository](#).

Chapter 2

Background

In Chapter 1, we established that Bayesian phylogenetic methods are the principal method for recovering language evolution in historical linguistics. However, the method hinges on the critical assumption of site independence. Before investigating how violations of site independence affect inference, we focus on understanding the technical framework underpinning these methods.

This chapter provides the mathematical foundation for the simulation study found in later chapters. We start by building on the definitions found in Chapter 1 to define the phylogenetic trees and evolutionary models that describe linguistic evolution in Section 2.1, while also describing the Bayesian inference methods on the space of phylogenetic trees $\mathcal{T}$, including the computational implementation on the evolutionary analysis program BEAST. We discuss the implications of imposing site independence on data in Section 2.2, before turning to the dependent data generation methods and thus implications on relaxing the site independence assumption on data in Section 2.3 and discuss the possible impacts of model misspecification. Finally, we analyse both tree recovery metrics and algorithm diagnostics to quantify accuracy and ensure quality of simulation in Section 2.4.

2.1 Bayesian Phylogenetic Inference

We start by considering the space of language (or in this case, phylogenetic) trees $\mathcal{T}$, building on the earlier definitions found in Chapter 1. In particular, we focus our discussion on the space of bifurcating trees that have branches with lengths proportional to time, and assume that lexical data $\mathbf{D}$ can be generated via evolutionary models with parameters $\theta^* \in \Theta$ on a true tree $T^* \in \mathcal{T}$.

When considering phylogenetic trees in a Bayesian setting, a natural starting point is to consider Bayes' theorem in the given context:

Definition 8 (Bayes' theorem). *For a given data set $\mathbf{D} \in \{0, 1\}^{n \times S}$ and model parameters $\theta \in \Theta$, we consider a tree $T \in \mathcal{T}$. Bayes' theorem states that the probability of the tree given the data and evolutionary model (the **posterior** probability) is proportional to the probability of observing the data given the tree and the evolutionary model (the **likelihood**) multiplied by the joint probability of the tree and evolutionary model (the joint **priors**):*

$$P(T|\mathbf{D}, \theta) = \frac{P(\mathbf{D}|T, \theta)P(T, \theta)}{P(\mathbf{D})} \quad (2.1)$$

$$\propto P(\mathbf{D}|T, \theta)P(T, \theta) \quad (2.2)$$

We note that (2.2) follows naturally since we can treat $P(\mathbf{D})$ as a (unknown) constant. Additionally, consider θ as the set of all model parameters from the evolutionary model and the tree prior. Our goal is to model the distribution of trees informed by the data and the parameters, or the posterior distribution $P(T|\mathbf{D}, \theta)$, either by direct calculation or by generating samples that follow it. However, due to the intractability of $P(\mathbf{D})$, it is unfeasible to explore the former method. We are therefore required to generate samples (of trees in space $\mathcal{T}$) that take on the posterior distribution. In many Bayesian methods, this requires defining and calculating likelihood $P(\mathbf{D}|T, \theta)$ and priors $P(T, \theta)$.

2.1.1 Tree Priors: Yule model

We first consider defining the prior of the model, by decomposing it further as

$$P(T, \theta) = P(T|\theta)P(\theta) \quad (2.3)$$

where $P(T|\theta)$ represents the tree prior, while $P(\theta)$ considers the (joint) distribution of all parameters of the model. While there exists many methods of generating trees, there are only a select few which are applicable in the context of historical linguistics [10]. We limit our discussion to an elaboration of a the Yule model, common tree prior used in Bayesian phylogenetic model.

The Yule model [42] can be defined as a pure birth process (see Definition 9) with a single intensity $\lambda \geq 0$, where N_t refers to the number of branches at time t , $N_0 = 1$. We make a distinction here that the model is generated from the ancestral lineage and thus encodes time 0 as the oldest point of the analysis, while final time T where there are n tips for n languages represents the present. This is contrary to the common representation of time in phylogenetic analysis, where time 0 represents the present age, or agglomerative tree generation methods like the coalescent process. Conditioning on the tree having n languages and root age T , the prior distribution takes on form [43]:

$$P(T|\lambda, n) = n! \lambda^{n-1} \prod_{i=1}^{n-1} \exp(-\lambda T) \quad (2.4)$$

where λ is the only relevant parameter in θ that affects the tree prior.

Definition 9 (Birth Process [44]). *A birth process with intensity $\lambda \geq 0$ is a continuous-time stochastic process $\{N_t\}_{t \geq 0}$ with values in $\mathbb{N}_0$ that satisfies:*

1. N_t is a non-decreasing process: $N_0 \geq 0$ and $s < t \implies N_s \leq N_t$,
2. The infinitesimal transition probabilities model single arrival: for $t \geq 0, \delta > 0, n, m \in \mathcal{N}_0$,

$$P(N_{t+\delta} = n + m | N_t = n) = \begin{cases} 1 - \lambda\delta + o(\delta) & m = 0 \\ \lambda\delta + o(\delta) & m = 1 \\ o(\delta) & m > 1, \end{cases} \quad (2.5)$$

3. N_t has conditionally independent increments: $\forall s < t$, $N_t - N_s$ is independent of all arrivals prior to s conditional on the value of N_s .

where function f is $o(\delta)$ if $\lim_{\delta \rightarrow 0^+} \frac{f(\delta)}{\delta} = 0$.

This model, as with all tree priors, incorporates the linguistic assumption that languages diversify through repeated divergence events without horizontal transfer. As such, this adds a slight complexity to data generation: words or lexical items that are borrowed must be removed, which may require further linguistic arguments to determine [3]. We also note that due to the single arrival characteristic of Yule model, only bifurcating trees are generated. Moreover, since the model does not allow for deaths to occur, the model generates rooted binary trees whose tips are all extant languages. This means that all trees generated by the Yule model are *ultrametric*, where the distance from (or age of) the root from every language is identical. This presents an additional assumption of employing such a method: all languages are assumed to be extant at the present age. While this reduces the complexity of the model and reduces the number of parameters introduced by the tree prior (and thus simplifying (2.4)), this means that any dataset that includes extinct languages in its dataset is misspecified by such a model and may introduce bias, especially with respect to rates of evolution and node ages [10]. As such, other tree priors like birth-death processes would be of interest [43]. For the purposes of this simulation study, we will assume that all languages are extant.

2.1.2 Evolution Model: Continuous-Time Markov Models

Having defined the tree prior, evolutionary models attempt to explain the presence or absence of sites across the languages of the model. Since the data is binary, we need only consider two types of states: the presence (1) or absence (0) of a site. As the model is defined on continuous time, the binary continuous-time Markov chain (CTMC) model is a natural method to define the evolutionary process.

The process assumes the Markov property (see Definition 10), which can be used to model historical linguistics, where the current state of cognates usually encapsulates much of the information which produce future evolutions. This induces the infinitesimal rate (or *generator*) matrix $\mathbf{G} := (g_{ij})$ (see Definition 11), which results in a time-dependent transition probability matrix to be the entries of the matrix [45]:

$$\mathbf{P}(t) = \exp(\mathbf{G}t). \quad (2.6)$$

Definition 10 (Markov Property [44]). *A continuous-time process $\{X_t\}_{t \geq 0}$ defined on state space $\mathcal{S} = \{0, 1\}$ satisfies the Markov property if*

$$P(X_{t_n} = i_n | X_{t_{n-1}} = i_{n-1}, \dots, X_{t_1} = i_{n-1}) = P(X_{t_n} = i_n | X_{t_{n-1}} = i_{n-1}), \quad (2.7)$$

for all $i_1, \dots, i_n \in E$, $0 \leq t_1 < \dots < t_n < \infty$, where $n \in \mathbb{N}$.

Definition 11 (Generator [44]). *The generator $\mathbf{G} := (g_{ij})_{i,j \in E}$ of the family of transition probability matrices $\{\mathbf{P}_t | t \geq 0\}$ is given as*

$$\mathbf{G} := \lim_{\delta \rightarrow 0^+} \frac{1}{\delta} (\mathbf{P}_\delta - \mathbf{I}), \quad (2.8)$$

where g_{ij} represents the exponential rate of moving from state i to state j where $i \neq j$, and $q_{ii} = -\sum_{j \neq i} q_{ij}$ are the outgoing rates.

The computation of these probabilities are key to calculating the likelihood $P(\mathbf{D}|T, \boldsymbol{\theta})$. To facilitate efficient computation, we assume the existence of a unique time-reversible stationary distribution $\boldsymbol{\pi} = (\pi_0 \ \pi_1)$, where

$$\boldsymbol{\pi} \mathbf{G} = \mathbf{0}, \quad \pi_0(\mathbf{P}(t))_{01} = \pi_1(\mathbf{P}(t))_{10} \ \forall t. \quad (2.9)$$

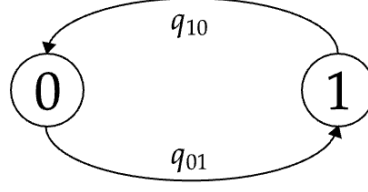


Figure 2.1 Transition Diagram of binary CTMC model.

While there are many other evolutionary models, we base our exploration in the binary CTMC model. The evolutionary model used in Gray and Atkinson’s seminal paper [20], the model assumes that the data $\mathbf{D}$ follows the transition diagram described in Figure 2.1. This implies exactly two parameters for the evolutionary model, given by $\{q_{01}, q_{10}\}$, where

$$\mathbf{G} = \begin{pmatrix} - & q_{01} \\ q_{10} & - \end{pmatrix} = \beta \begin{pmatrix} - & \pi_1 \\ \pi_0 & - \end{pmatrix}, \quad (2.10)$$

$$\boldsymbol{\pi} = (\pi_0, \pi_1) = \left(\frac{q_{10}}{q_{01} + q_{10}}, \frac{q_{01}}{q_{01} + q_{10}} \right), \quad (2.11)$$

where the $-$ are the conventional notation used implied diagonal entries as defined in Definition 11 and $\beta = q_{01} + q_{10}$ is a scalar constant¹.

We are thus able to calculate the likelihood $P(\mathbf{D}|T, \boldsymbol{\theta})$ for the model. For the sake of demonstration, we illustrate the likelihood only for a single site $\mathbf{D}_i$, saving full derivations for $\mathbf{D}$ in Sections 2.2 and 2.3. Given the binary phylogenetic tree $T = (V, E, r)$ with n observed languages, where sites evolve via a binary CTMC with generator $\mathbf{G}$ and stationary distribution $\boldsymbol{\pi}$, the likelihood for a site $\mathbf{D}_i$ is

$$P(\mathbf{D}_i|T, \mathbf{G}) = \sum_{\mathbf{x}_W \in \{0,1\}^{|W|}} P(\mathbf{D}_i, \mathbf{x}_W|T, \mathbf{G}, \boldsymbol{\pi}) \quad (2.12)$$

$$= \sum_{\mathbf{x}_W \in \{0,1\}^{|W|}} P(X_r = x_r) P(\mathbf{D}_i, \mathbf{x}_{W \setminus \{r\}}|T, \mathbf{G}, X_r = x_r) \quad (2.13)$$

$$= \sum_{\mathbf{x}_W \in \{0,1\}^{|W|}} P(X_r = x_r) \prod_{(u,v) \in E} P(X_v = x_v | X_u = x_u), \quad (2.14)$$

where X_u is the random variable of the state at node u , and we employed the Markov property of the CTMC model. As defined previously, the first term corresponds to the stationary distribution $\boldsymbol{\pi}$. This reduces the problem to calculating transition probabilities, where (2.6) is most relevant. This gives the expression

$$P(\mathbf{D}_i|T, \mathbf{G}) = \sum_{\mathbf{x}_W \in \{0,1\}^{|W|}} \pi_{x_r} \prod_{(u,v) \in E} \mathbf{P}(t_{u \rightarrow v})_{x_u x_v} \quad (2.15)$$

$$= \sum_{\mathbf{x}_W \in \{0,1\}^{|W|}} \pi_{x_r} \prod_{(u,v) \in E} \exp(\mathbf{G} t_{u \rightarrow v})_{x_u x_v} \quad (2.16)$$

This decomposition is inefficient for large values of n : the set of internal nodes W grows linearly with n , thus exponentially increasing the number of internal terms in the sum of (2.16).

¹Equation 2.10, that is, that the stationary distribution of a binary CTMC is proportional to the transition rates of the chain, is proven in Appendix B.1.

Felsenstein (1973) introduced an efficient “pruning” method [46], exploiting the structure of the tree to recursively evaluate likelihoods of subtrees [47]. This allows for efficient likelihood calculation that grows linearly rather than exponentially with n .

For a simulation study, the generation of data through such evolutionary processes require particular consideration. For a given tree T , at each site, the state at the root is sampled and propagated down the tree via the CTMC model along each branch that has transition probabilities $\mathbf{P}(t)$ via (2.6). This results in the generation of the site $\mathbf{D}_i$. If the assumption of conditional site independence is held, the process is repeated for every single site until the full binary data matrix $\mathbf{D} \in \{0, 1\}^{n \times S}$ is generated. This generative process is defined in Algorithm 1.

Algorithm 1: Data Generation from binary CTMC model.

Input: T , fixed topology tree with root r of n languages, where each edge $e := (u \rightarrow v)$ has branch length t_e ,
 $\mathbf{G}$, CTMC generator with parameters q_{01}, q_{10} ,
 π , stationary distribution of CTMC with generator $\mathbf{G}$,
 S , number of sites of data.

Output: $\mathbf{D} \in [0, 1]^{n \times S}$, where sites are columns $\mathbf{D}_1, \dots, \mathbf{D}_S$.

```

1 Function Propagate( $v, x_v$ )
2   for each child  $c$  of  $v$  do
3     Let  $e = (v \rightarrow c)$  with branch length  $t_e$ ;
4     Compute transition matrix  $P_e = \exp(\mathbf{G}t_e)$ ;
5     Sample:
6        $x_c \sim P_e(x_v, \cdot)$ 
7     Propagate( $c, x_c$ );
8   end
9 Function SiteGeneration( $T, \mathbf{G}, \pi$ )
10  Sample root state:
11     $x_r \sim \pi$ 
12  Propagate( $r, x_r$ );
13  return vector of language states  $(x_1, \dots, x_n)^T$ 
14 end
15 for  $i = 1, \dots, S$  do
16    $\mathbf{D}_i \leftarrow \text{SiteGeneration}(T, \mathbf{G}, \pi)$ 
17 end
18 return  $\mathbf{D} = (\mathbf{D}_1 \cdots \mathbf{D}_S)$ 

```

We note an additional constraint in the context of historical linguistics: sites are excluded if they are absent in all languages considered. This means that if certain sites are generated as a zero matrix $\mathbf{0}$, a process of *ascertainment* takes place, where these sites are removed [10]. This decreases the number of sites that are generated within the data. In order to account for this practically, an inflated value of S is set, before performing sampling without replacement to derive the dataset $\mathbf{D}^{n \times S}$.

The binary CTMC model is highly favoured since it is highly parsimonious and is computationally tractable. The evolution model introduces few parameters as discussed above, making likelihood calculations and thus Bayesian phylogenetic inference more efficient. However, these constraints insert more assumptions about the homogeneity and memorylessness properties of

transitions between only two states, which may be an oversimplification of the complexity of evolution in real languages. Studies have shown that more complex models such as the binary covarion model [48] perform better on linguistic data through allowing slow and fast rates of change [10].

For the context of the simulation study conducted in this thesis, we limit our discussion and implementation to the binary CTMC. This provides a powerful baseline model to study the effects of model misspecification, where more complex evolutionary models can be discussed in future work.

2.1.3 Rate Variation

As noted by Bergsland and Vogt (1962), the rates of evolution across lineages and subsets of data varies considerably [18]. For example, sites belonging to a meaning with few cognates have been shown to empirically evolve more slowly than a meaning with many cognates [49]. In order to accommodate for this, a site-specific rate multiplier r_i is included in (2.6). This includes S more parameters for model, where the method of generating these rates (and thus their priors) is most commonly a Gamma distribution with mean 1 and shape parameter α [10]. Such a method preserves the expected rates, while allowing for substantial variation across sites. In practice, this improves the model by measuring heterogeneity of site evolution, which may support fitting real language data.

For the sake of the simulation study contained within this thesis, rate variation is intentionally excluded - all sites are assumed to evolve with a fixed rate $r_i = 1$. This is a design choice, with an objective to isolate the effects of model misspecification through conditional site independence. Allowing the rate variation would introduce additional structure that could confound inference, making it difficult to attribute observed effects as due to the violation of conditional site independence. This reduces the model to a *homogeneous* CTMC model instead.

2.1.4 Posterior Sampling with Markov Chain Monte Carlo methods

Having discussed the prior and likelihood probabilities, we are now well equipped to discuss sampling from the posterior. As mentioned prior, the normalising constant of the posterior is often difficult to compute, which makes analytic expressions of the posterior distribution inaccessible. However, there are additional considerations in the context of historical linguistics, even if one were to apply a sampling based method. The size of the space of all possible binary phylogenetic trees $\mathcal{T}$ grows *super*-exponentially (that is, grows at a rate faster than exponential increase) with the number of languages n [50]. In fact, the problem of finding the optimal tree is *NP-hard* [51]. Moreover, the priors are defined on both discrete and continuous distributions, complicating the analysis of the model.

These challenges motivate the use of simulation-based approaches to approximate the posterior distribution. Bayesian phylogenetic inference commonly relies on Markov Chain Monte Carlo (MCMC) methods to generate samples from $p^* := P(T|\mathbf{D}, \boldsymbol{\theta})$ without the explicit computation of the normalising constant of (2.2). The method attempts to construct a Markov chain of trees (and parameters) whose stationary distribution π is equal to p^* , such that states of the chain are regarded as (dependent) samples from the posterior distribution. More formally, suppose we initialise the chain at some $\mathbf{z}^{(0)} = \{T^{(0)}, \boldsymbol{\theta}^{(0)}\}$, we generate chain $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(t)}, \dots\}$ such that $T^{(t)} \sim p^*$ after the chain is allowed to mix for an amount of time called the burnin.

This subsection aims to explain the methodology used in MCMC. In practice, these considerations within the algorithm have been neatly summarised in packages and applications such

as BEAST, which will be discussed in Section 2.1.5.

As the name of the algorithm suggests, the model generates a chain which fulfills the Markov property. This means that, given a current sample $\mathbf{z}^{(n)}$, in order to determine the next sample $\mathbf{z}^{(n+1)}$, we propose some $\mathbf{z}' \sim q(\cdot|\mathbf{z}^{(n)})$ by some *proposal distribution* q , before assessing whether the proposal is suitable.

This proposal distribution q is clearly multivariate in the context of historical linguistics, generating a new tree $T \in \mathcal{T}$ but also other parameters $\theta \in \Theta$. We assume that q can be decomposed into independent proposals, particularly of T , and that all other parameters θ can be proposed in a straightforward manner via conventional probability distributions. The proposal of a new tree T' given tree $T^{(n)}$ is clearly non-trivial due to its non-Euclidean nature and requires extra treatment. There are many methods, such as the Nearest Neighbour Interchange (NNI) [52], the Subtree Prune and Regraft (SPR) [53] and time-scaling operations, that generate a stochastic walk over the tree space $\mathcal{T}$.

Once the proposal $\mathbf{z}' = \{T', \theta'\}$ has been chosen via the proposal distribution q , it is evaluated based on an acceptance probability:

$$\alpha(\mathbf{z}^{(n)}, \mathbf{z}') = \min \left\{ 1, \frac{p^*(\mathbf{z}')q(\mathbf{z}^{(n)}|\mathbf{z}')}{p^*(\mathbf{z}^{(n)})q(\mathbf{z}'|\mathbf{z}^{(n)})} \right\} \quad (2.17)$$

$$= \min \left\{ 1, \frac{P(\mathbf{D}|\mathbf{z}')P(\mathbf{z}')q(\mathbf{z}^{(n)}|\mathbf{z}')}{P(\mathbf{D}|\mathbf{z}^{(n)})P(\mathbf{z}^{(n)})q(\mathbf{z}'|\mathbf{z}^{(n)})} \right\}, \quad (2.18)$$

that determines whether we accept the proposal as the next sample or not:

$$\mathbf{z}^{(n+1)} = \begin{cases} \mathbf{z}' & \text{with probability } \alpha(\mathbf{z}^{(n)}, \mathbf{z}') \\ \mathbf{z}^{(n)} & \text{with probability } 1 - \alpha(\mathbf{z}^{(n)}, \mathbf{z}'). \end{cases} \quad (2.19)$$

Equation 2.18 is paramount to the MCMC algorithm, state in full in Algorithm 2. The acceptance ratio is designed such that the resulting Markov chain $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(t)}, \dots\}$ has stationary distribution p^* [54]. Crucially, the method does not require the knowledge of marginal probability $P(\mathbf{D})$, which allows the acceptance probabilities to be easily calculated by decomposition of the posterior probabilities into the likelihood and prior probabilities.

As such, MCMC methods are elegant yet practical algorithms to resolve the issues faced in Bayesian phylogenetic inference. The method itself also allows for a natural quantification of uncertainty through the discretisation of the posterior distribution. Furthermore, the method is highly flexible and can incorporate models and priors more complex than has been discussed in Section 1.2 so far.

However, MCMC methods suffer from several limitations. Given the Markov structure of the chain, proposals cannot be conducted and evaluated in parallel. The Markov structure also causes samples to be highly dependent, requiring a large number of iterations. These two limitations constrain MCMC methods to require a large number of samples, thus requiring much computational runtime. Additionally, inference may be sensitive to poor mixing and multimodality of the posterior distribution, which is a common issue considering the space of phylogenetic trees $\mathcal{T}$.

In particular, the algorithm does not guard against model misspecification issues. For example, if the likelihood function is misspecified through the violation of the conditional site independence criterion, the model does not address the potential biases arising from violations of the underlying modelling assumptions.

Algorithm 2: Metropolis-Hastings (MCMC) algorithm, adapted from [54].

Input: N , number of samples to be generated,
 $\mathbf{z}^{(0)}$, starting point of Markov chain.

Output: $\mathbf{Z} = \{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(N)}\}$, resulting Markov chain with stationary distribution p^* .

```

1 for  $n = 1, \dots, N$  do
2   Sample proposal  $\mathbf{z}' \sim q(\mathbf{z}^{(n-1)})$ ;
3   Calculate acceptance probability  $\alpha(\mathbf{z}^{(n-1)}, \mathbf{z}')$  as in (2.18);
4   Sample  $u \sim \text{Uniform}[0, 1]$ 
5   if  $u < \alpha(\mathbf{z}^{(n-1)}, \mathbf{z}')$  then
6     | Accept Proposal:  $\mathbf{z}^{(n)} \leftarrow \mathbf{z}'$ ;
7   else
8     | Reject Proposal:  $\mathbf{z}^{(n)} \leftarrow \mathbf{z}^{(n-1)}$ ;
9   end
10 end
11 return  $\mathbf{Z} = \{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(N)}\}$ 

```

2.1.5 Implementation in BEAST

There are many packages available for practical implementations of MCMC methods on phylogenetic trees, one of which is the *Bayesian Evolutionary Analysis Sampling Trees*, or BEAST algorithm [55]. Given the observed cognate data, BEAST samples from the posterior distribution and constructs a Markov chain as described in this section. In particular, tree space $\mathcal{T}$ is explored through a collection of local and global proposal operators to facilitate sufficient exploration. The package incorporates high performance computing methods [56] outside the scope of this thesis that ensures an efficient practical implementation of the inference process.

In this thesis, BEAST is used as the inference engine for all inferences as detailed in Chapter 3. This enables a scalable pipeline for direct assessment of the effect of model misspecification through conditional site dependence.

2.2 Independent Paradigm

Having discussed the entire methodology to recover phylogenetic trees, we examine the conditional site independence assumption in more detail. In particular, the effects of imposing the assumption in data generation, likelihood and posterior calculations and effective information brought forward by the (generated) data $\mathbf{D}$ will be considered.

Under the *independent paradigm*, that is, where the conditional site independence assumption is imposed, data generation for the simulation study is conducted in a straightforward manner as discussed in Section 2.1.2. In particular, Algorithm 1 is implemented to generate independent sites $\mathbf{D}_i$ for $i = 1, \dots, S$, which ensures site independence throughout the data.

We return to the likelihood calculations in (2.16). Clearly, on considering the full dataset

$\mathbf{D}$, we can decompose the probability into a product of likelihood calculations at each site:

$$P(\mathbf{D}|T, \boldsymbol{\theta}) = \prod_{i=1}^S P(\mathbf{D}_i|T, \boldsymbol{\theta}) \quad (2.20)$$

$$= \prod_{i=1}^S P(\mathbf{D}_i|T, \mathbf{G}) \quad (2.21)$$

This reduces the likelihood calculations (and thus also the posterior calculations) related to the independent paradigm to the simpler problem of calculating likelihoods of individual sites, where we only require Felsenstein's pruning algorithm as described in Section 2.1.2. The algorithm propagates conditional likelihoods through the proposed tree T from the leaves to the root node using dynamic programming, effectively setting the time-complexity of likelihood calculation as $\mathcal{O}(nS)$, scaling linearly with respect to number of languages n and number of sites S .

The independent paradigm also suggests that every new site provides an equal amount of information. We formalise this idea by considering the Fisher information matrix derived from (2.25):

$$\mathcal{I}_{\text{data}} = -E [\nabla_{\boldsymbol{\theta}}^2 \ln P(\mathbf{D}|T, \boldsymbol{\theta})] \quad (2.22)$$

$$= -E \left[\sum_{i=1}^S \nabla_{\boldsymbol{\theta}}^2 \ln P(\mathbf{D}_i|T, \mathbf{G}) \right] \quad (2.23)$$

$$= - \sum_{i=1}^S E [\nabla_{\boldsymbol{\theta}}^2 \ln P(\mathbf{D}_i|T, \mathbf{G})] \quad (2.24)$$

$$= S\mathcal{I}_{\text{site}}. \quad (2.25)$$

using the fact that sites under the independent paradigm are also identically distributed. This suggests that the information is additive with respect to sites: the amount of (effective) information contained in the data scales linearly with the number of (independent) sites S contained in data $\mathbf{D}$. As such, an increase in S will result in increasingly peaked likelihoods that, under standard regularity conditions, will lead to an improved estimate of true tree T^* :

$$P(T = T^*|\mathbf{D}) \rightarrow 1 \text{ as } S \rightarrow \infty. \quad (2.26)$$

2.3 Dependent Paradigm

We turn our attention to the *dependent paradigm*, where sites are not assumed to be generated independently from each other. The main implication of relaxing this assumption is that the decomposition in (2.21) is now invalid - we now have to consider how sites may interact with each other to determine the correct likelihood structure.

While there are many approaches to construct such dependencies within simulated data $\mathbf{D}$, we focus our discussion on a methodology such that sites are generated in a hierarchical format. We construct L tiers of data, where data from Tier 1 refers to independently generated sites, while data from Tier $l = 2, \dots, L$ are sites that are dependent on certain sites from Tier $l - 1$. A diagram of the dependent data generation methodology is described in Figure 2.2.

This method thus requires us to answer two design questions:

1. How does each site in Tier l choose which site (which we will call the *parent* site) in Tier $l - 1$ to depend on? With reference to Figure 2.2, how are the start- and end-points arrows defining dependencies on previous tiers and sizes $N_2, \dots, N_L$ determined?

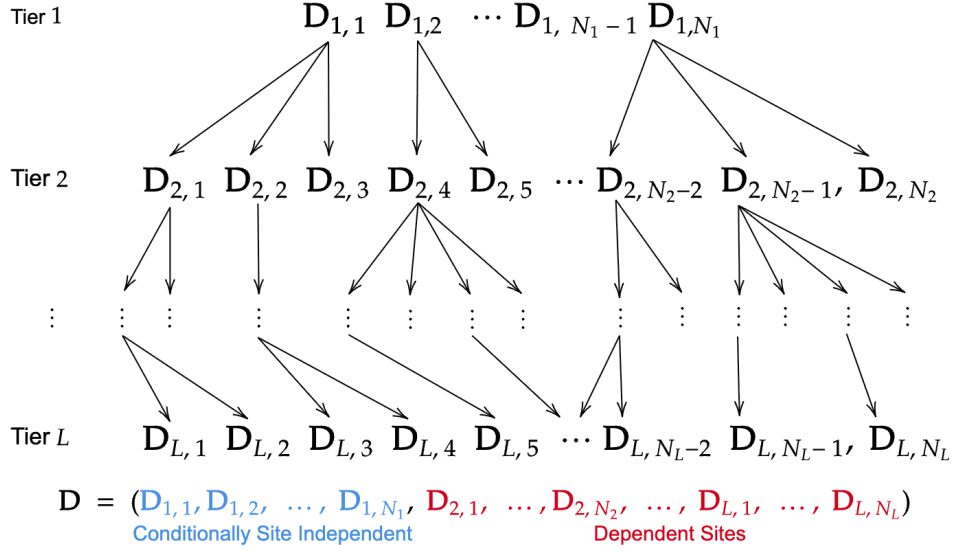


Figure 2.2 A hierarchical approach to dependent site generation with L tiers that generations data $\mathbf{D}$.

2. Upon choosing a parent site from Tier $l - 1$, how is each dependent site (which we call the corresponding *child* site) in Tier l generated given the parent site? With reference to Figure 2.2, how do the arrows transmit dependencies from the site in the previous tier?

The answers to these questions then define the manner in which data is generated, and thus how likelihoods will be calculated. For the purposes of this simulation study, we generate dependencies via the Zero-Inflated Poisson (ZIP), discussed in Section 2.3.1, and generate dependent site values by considering the product of the result of CTMC runs, discussed in Section 2.3.2.

This method in particular is highly relevant to related works on dependent data. This hierarchical dependency structure corresponds to the issue posed by Maddison’s red/blue tail problem [30], where $L = 2$. Additionally, in the context of Bayesian phylolinguistic analysis, Buckley et al. (2025) considered loom data that also exhibited the same structure [9], this time with $L = 4$. Consequently, investigating the extent of impact of applying Bayesian phylogenetic models, which assume conditional site independence, to hierarchially dependent data provides a frame of reference for assessing the consequences of model misspecification in settings where site dependencies are known or suspected to exist. Such analyses can thus address concerns of applying standard Bayesian phylogenetic models to data which exhibit such hierarchical dependency structures.

2.3.1 ZIP Construction

We start by specifying the *Zero-Inflated Poisson* (ZIP) random variable which generates the hierarchical dependencies, which will in turn describe the sizes of each tier of data generated.

Definition 12 (Zero-Inflated Poisson (ZIP)). *Let X be a discrete random variable. X is said to follow a Zero-Inflated Poisson with parameters $p \in [0, 1]$ and $\lambda \geq 0$, or $X \sim \text{ZIP}(p, \lambda)$ if its density is given by*

$$f_X(x) = \begin{cases} p + (1 - p) \exp(-\lambda) & x = 0 \\ (1 - p) \exp(-\lambda) \frac{\lambda^x}{x!} & x \neq 0 \end{cases} \quad (2.27)$$

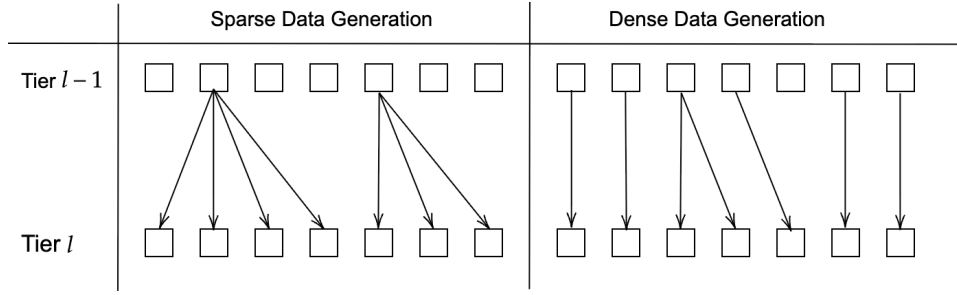


Figure 2.3 Sparse vs Dense Data Generation using the ZIP random variable, where boxes refer to sites. Note that the tier sizes used are for presentation purposes only: they need not be equal.

In other words, X can be represented as a product of independent random variables $X = YZ$, where $Y \sim \text{Bernoulli}(1 - p)$, $Z \sim \text{Poisson}(\lambda)$.

While dependency structure within linguistics data is widely acknowledged [40], the exact relationship between sites is not as clearly defined. The use of a ZIP random variable, instead of other conventional distributions like the standard Poisson distribution, is a design choice made with the context of historical linguistics in mind. The ZIP random variable is the generative model of choice as we require a flexible yet parsimonious model that allows both the absence of dependencies but also the substantial heterogeneity in number of dependent sites generated by a given site. By increasing p (and thus increasing the value of λ such that their expected values remain constant), the model generates dependencies much more *sparingly*: for a given site, there are either no dependencies at all, or there are many dependencies. An example can be found in Figure 2.3.

Proposition 1. Let X be a ZIP random variable with parameters p and λ . Then:

$$E(X) = (1 - p)\lambda =: \mu, \quad (2.28)$$

$$\text{Var}(X) = (1 - p)\lambda + p(1 - p)\lambda^2 = \mu(1 + p\lambda) =: \sigma^2 \quad (2.29)$$

Proof. This result follows directly from decomposing the expectation and variance of a product of independent variables:

$$E(X) = E(YZ) \quad (2.30)$$

$$= E(Y)E(Z) \quad (2.31)$$

$$= (1 - p)\lambda =: \mu, \quad (2.32)$$

$$\text{Var}(X) = \text{Var}(YZ) \quad (2.33)$$

$$= E(Y^2Z^2) - (E(YZ))^2 \quad (2.34)$$

$$= E(Y^2)E(Z^2) - (E(Y)E(Z))^2 \quad (2.35)$$

$$= E(Y)E(Z^2) - (E(Y)E(Z))^2 \quad (2.36)$$

$$= (1 - p)(\lambda + \lambda^2) - (1 - p)^2\lambda^2 \quad (2.37)$$

$$= (1 - p)\lambda + p(1 - p)\lambda^2 \quad (2.38)$$

$$= \mu(1 + p\lambda), \quad (2.39)$$

where we used $E(Y^2) = E(Y)$ of a Bernoulli distribution to derive (2.36). $\square$

This method of generating dependent data can also be seen as a branching process starting with L_1 individuals, with offspring distribution $\mathbf{d} = (P(X = k))_{k \geq 0}$, where $X \sim \text{ZIP}(p, \lambda)$.

As such, we can interpret dependent tiers (Tiers $2, \dots, L$) as later generations of the initial generation (Tier 1), thus reframing the issue of determining the properties of random variables $N_2, \dots, N_L$ in the context of finding the size of the population at a given generation. We can thus use results from branching processes to determine the size of the dataset at a given tier l :

Proposition 2. *Suppose independent data is generated at tier 1 with N_1 sites, where dependent data is generated via a branching process with offspring following a ZIP distribution. Then the size of tier l , given by N_l , is a random variable with mean and variance given as*

$$E(N_l) = N_1 \mu^{l-1}, \quad (2.40)$$

$$Var(N_l) = \begin{cases} N_1 \sigma^2 \mu^{l-2} \frac{\mu^{l-1}-1}{\mu-1} & \mu \neq 1 \\ N_1 \sigma^2 (l-1) & \mu = 1 \end{cases}, \quad (2.41)$$

where μ and σ^2 is as defined in Proposition 1.

Proof. We employ theory from branching processes [57], where our initial population is N_1 :

$$E(N_l) = \sum_{k=0}^{\infty} E(N_l | N_{l-1} = k) P(N_{l-1} = k) \quad (2.42)$$

$$= \sum_{k=0}^{\infty} E \left(\sum_{i=1}^{N_{l-1}} X_i^{(l-1)} | N_{l-1} = k \right) P(N_{l-1} = k) \quad (2.43)$$

$$= \sum_{k=0}^{\infty} E \left(\sum_{i=1}^k X_i^{(l-1)} \right) P(N_{l-1} = k) \quad (2.44)$$

$$= \sum_{k=0}^{\infty} k \mu P(N_{l-1} = k) \quad (2.45)$$

$$= \mu E(N_{l-1}), \quad (2.46)$$

where $X_i^{(l-1)} \sim \text{ZIP}(p, \lambda)$ representing the realisation of the offspring distribution for a single site i in generation $l-1$. This results in a recursive relationship that gives (2.40) as required. Similarly, for variance,

$$Var(N_l) = Var(E(N_l | N_{l-1}) + E(Var(N_l | N_{l-1})), \quad (2.47)$$

$$E(N_l | N_{l-1} = k) = \mu k, \quad (2.48)$$

$$Var(N_l | N_{l-1} = k) = Var \left(\sum_{i=1}^k X_i^{(l-1)} \right) = \sum_{i=1}^k Var(X_i^{(l-1)}) = \sigma^2 k, \quad (2.49)$$

$$\implies Var(N_l) = \mu^2 Var(N_{l-1}) + \sigma^2 \mu^{l-2} \quad (2.50)$$

$$= \begin{cases} n \sigma^2 & \mu = 1 \\ \sigma^2 \mu^{l-2} \frac{\mu^{l-1}-1}{\mu-1} & \mu \neq 1 \end{cases}. \quad (2.51)$$

□

With these estimates of N_l , we are able to calculate the expected size of data given number of independent sites N_1 , ZIP parameters (p, λ) and number of tiers L . We also note that these calculations are performed before ascertainment as introduced in Section 2.1.2. As a result, we expect values of N_l to be smaller due to the removal of 0 vector sites.

2.3.2 CTMC Product Construction

Next we turn to the second design question of Section 2.3 - determining the dependency approach. While there are many methods to create dependency, we do so by setting the dependent site as the product of two binary CTMCs, one of them being the parent site from the previous tier. More formally, for site $\mathbf{D}_{l,k}$ dependent on $\mathbf{D}_{l-1,j}$, we generate $\tilde{\mathbf{D}}_{l,k}$ via the same binary CTMC, and multiply the two vectors element-wise:

$$\mathbf{D}_{l,k} := \mathbf{D}_{l-1,j} \odot \tilde{\mathbf{D}}_{l,k}, \quad (2.52)$$

where we use $\odot$ as the element-wise product operator. This method was implemented for several reasons. Firstly, the method was chosen for its straightforward implementation in light of binary characters. Next, such a method which employs the same binary CTMC (with the same generator $\mathbf{G}$) reduces confounding factors, where any effects are caused by the dependency structure. Finally, sites will be present in a specific language if and only if both $(\mathbf{D}_{l-1,j})_i = (\mathbf{D}_{l,k})_i = 1$. This is consistent with Maddison's red/blue tail problem, where a given species would have to have a tail (set up as one site), as well as whether they have a blue or red tail [30].

We now prove the overall process induced from such a data generation method is a (non-binary) CTMC. The proofs as stated below will focus on the dependency from an independent site to site dependent on it. All other dependencies can be derived in a similar fashion.

Proposition 3. Consider the data generation method in (2.52) with generator $\mathbf{G} = \begin{pmatrix} - & q_{01} \\ q_{10} & - \end{pmatrix}$, then $\mathbf{B}_{l,k} := \{\mathbf{D}_{l-1,j}, \tilde{\mathbf{D}}_{l,k}\}$ and is also a CTMC of states $\mathcal{S} := \{00, 01, 10, 11\}$, with generator

$$\mathbf{G}_{\text{prod}} = \begin{pmatrix} - & q_{01} & q_{01} & 0 \\ q_{10} & - & 0 & q_{01} \\ q_{10} & 0 & - & q_{01} \\ 0 & q_{10} & q_{10} & - \end{pmatrix}. \quad (2.53)$$

Proof. We consider the transition probabilities over time $\delta > 0$ through (2.6) as a sum of matrices:

$$\mathbf{P}(\delta) = \exp(\mathbf{G}\delta) \quad (2.54)$$

$$= \mathbf{I} + \delta\mathbf{G} + o(\delta) \quad (2.55)$$

where we use $o(\delta)$ similarly to Definition 9 but in a matrix form. With the constructed process $\mathbf{B}_{l,k} := \{\mathbf{D}_{l-1,j}, \tilde{\mathbf{D}}_{l,k}\}$, we consider the transition probabilities over time δ from state $x = (x_1, x_2)$ to $x' = (x'_1, x'_2)$:

$$(\mathbf{P}_{\text{prod}}(\delta))_{x,x'} = P(\mathbf{B}_{l,k,t+\delta} = x' | \mathbf{B}_{l,k,t} = x) \quad (2.56)$$

$$= P(\mathbf{D}_{l-1,j,t+\delta} = x'_1, \tilde{\mathbf{D}}_{l,k,t+\delta} = x'_2 | \mathbf{D}_{l-1,j,t} = x_1, \tilde{\mathbf{D}}_{l,k,t} = x_2) \quad (2.57)$$

$$= P(\mathbf{D}_{l-1,j,t+\delta} = x'_1 | \mathbf{D}_{l-1,j,t} = x_1) P(\tilde{\mathbf{D}}_{l,k,t+\delta} = x'_2 | \tilde{\mathbf{D}}_{l,k,t} = x_2) \quad (2.58)$$

$$= (\mathbf{P}(\delta))_{x_1,x'_1} (\mathbf{P}(\delta))_{x_2,x'_2}. \quad (2.59)$$

This result shows that we can represent the resulting transition probabilities as a Kronecker product $\mathbf{P}_{\text{prod}}(\delta) = \mathbf{P}(\delta) \otimes \mathbf{P}(\delta)$ [58]. Using (2.55), we have that

$$\mathbf{P}_{\text{prod}}(\delta) = \mathbf{P}(\delta) \otimes \mathbf{P}(\delta) \quad (2.60)$$

$$= (\mathbf{I}_2 + \delta\mathbf{G} + o(\delta)) \otimes (\mathbf{I}_2 + \delta\mathbf{G} + o(\delta)) \quad (2.61)$$

$$= \mathbf{I}_2 \otimes \mathbf{I}_2 + \delta(\mathbf{G} \otimes \mathbf{I}_2) + \delta(\mathbf{I}_2 \otimes \mathbf{G}) + \delta^2(\mathbf{G} \otimes \mathbf{G}) + o(\delta) \quad (2.62)$$

$$= \mathbf{I}_4 + \delta(\mathbf{G} \otimes \mathbf{I}_2 + \mathbf{I}_2 \otimes \mathbf{G}) + o(\delta), \quad (2.63)$$

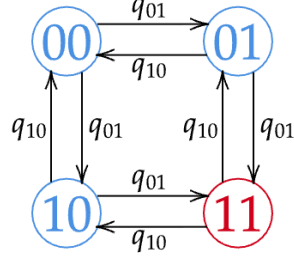


Figure 2.4 Transition Diagram of Product of CTMCs $\mathbf{B}_{l,k} := \{\mathbf{D}_{l-1,j}, \tilde{\mathbf{D}}_{l,k}\}$. Blue states represent where $D_{l,k,i} = 0$, red state represents where $D_{l,k,i} = 1$, where $D_{l,k,i}$ represents the i -th element of $\mathbf{D}_{l,k,i}$.

where $\mathbf{I}_n \in \mathbb{R}^{n \times n}$ is the identity matrix in the n -dimensional square matrix to avoid confusion of notation. It is then clear that via a similar expansion to retrieve (2.55), $\mathbf{G}_{\text{prod}} = (\mathbf{G} \otimes \mathbf{I}_2 + \mathbf{I}_2 \otimes \mathbf{G})$. We prove that this gives (2.53):

$$\mathbf{G} \otimes \mathbf{I}_2 = \begin{pmatrix} -q_{01}\mathbf{I}_2 & q_{01}\mathbf{I}_2 \\ q_{10}\mathbf{I}_2 & -q_{10}\mathbf{I}_2 \end{pmatrix} \quad (2.64)$$

$$= \begin{pmatrix} -q_{01} & 0 & q_{01} & 0 \\ 0 & -q_{01} & 0 & q_{01} \\ q_{10} & 0 & -q_{10} & 0 \\ 0 & q_{10} & 0 & -q_{10} \end{pmatrix}, \quad (2.65)$$

$$\mathbf{I}_2 \otimes \mathbf{G} = \begin{pmatrix} \mathbf{G} & \mathbf{0} \\ \mathbf{0} & \mathbf{G} \end{pmatrix} \quad (2.66)$$

$$= \begin{pmatrix} -q_{01} & q_{01} & 0 & 0 \\ q_{10} & -q_{10} & 0 & 0 \\ 0 & 0 & -q_{01} & q_{01} \\ 0 & 0 & q_{10} & -q_{10} \end{pmatrix}. \quad (2.67)$$

The addition of (2.65) and (2.67) gives the result as required. $\square$

A transition diagram of the resulting CTMC model can be found in Figure 2.4. We now derive the stationary distribution for such a model.

Proposition 4. *The unique stationary distribution of the resulting CTMC defined in Proposition 3 with generator given in (2.53) is*

$$\tilde{\pi} = \pi \otimes \pi \quad (2.68)$$

$$= (\pi_0^2, \pi_0\pi_1, \pi_0\pi_1, \pi_1^2), \quad (2.69)$$

where $\pi = (\pi_0, \pi_1)$ is the stationary distribution of the original CTMC defined in (2.9).

Proof. We use properties of a Kronecker product to propose $\tilde{\pi}$ as a stationary distribution of the CTMC derived in Proposition 3. We already know from (2.9), $\pi\mathbf{G} = \mathbf{0}$, and thus

$$(\pi \otimes \pi)(\mathbf{G} \otimes \mathbf{I}_2) = \mathbf{0} \otimes \pi = \mathbf{0}, \quad (2.70)$$

$$(\pi \otimes \pi)(\mathbf{I}_2 \otimes \mathbf{G}) = \pi \otimes \mathbf{0} = \mathbf{0}, \quad (2.71)$$

$$\implies \tilde{\pi}\mathbf{G}_{\text{prod}} = (\pi \otimes \pi)(\mathbf{G} \otimes \mathbf{I}_2 + \mathbf{I}_2 \otimes \mathbf{G}) \quad (2.72)$$

$$= (\pi \otimes \pi)(\mathbf{G} \otimes \mathbf{I}_2) + (\pi \otimes \pi)(\mathbf{I}_2 \otimes \mathbf{G}) = \mathbf{0}. \quad (2.73)$$

Thus $\tilde{\pi}$ is a stationary distribution of the CTMC with generator $\mathbf{G}_{\text{prod}}$. We now consider the uniqueness of $\tilde{\pi}$. Clearly all states are connected, making the Markov chain generated from the CTMC irreducible. Since the state space is finite, we have that the CTMC must admit a unique stationary distribution [54]. $\square$

We note that when considering the dependent site $\mathbf{D}_{l,k}$, we compress the CTMC $\mathbf{B}_{l,k}$ as

$$\mathbf{D}_{l,k} = \begin{cases} 0 & \mathbf{B}_{l,k} \in \{00, 01, 10\} \\ 1 & \mathbf{B}_{l,k} \in \{11\}. \end{cases} \quad (2.74)$$

While this stochastic process is not a CTMC ², we note that the stationary distribution of the model does carry over as

$$\tilde{\pi} = (1 - \pi_1^2, \pi_1^2). \quad (2.75)$$

We conclude this subsection with the full dependent data generation algorithm, presented in 3.

2.3.3 Likelihood Computation

having now discussed the intricacies of the dependent data generation model, we turn to analyse the difficulties of analysing the likelihood computations of such a model. Clearly, conditional site independence does not hold anymore, and we are not able to decompose the likelihood in a feasible manner. This comes with the additional cost of having to account for the latent CTMC variables that contribute to each dependent data generation.

Figure 2.2 provides a graphical interpretation, not just for understanding dependent data generation, but also of likelihood decomposition. This induces the dependency graph $\mathcal{G}$ (more precisely a directed acyclic graph), where edges (arrows) indicate how likelihoods of sites are dependent on another (parent) site. We present a high-level analysis that illustrates the complexity of likelihood computation under this model.

Given dependency graph $\mathcal{G}$, we can decompose the likelihood by separating independent sites, given by sets of sites $\mathcal{I}$ and $\mathcal{D}$ respectively:

$$P(\mathbf{D}|T, \theta) = \prod_{i \in \mathcal{I}} P(\mathbf{D}_i|T, \theta) \prod_{j \in \mathcal{D}} P(\mathbf{D}_j|\mathbf{D}_{\text{pa}(j)}, T, \theta), \quad (2.76)$$

$$P(\mathbf{D}_j|\mathbf{D}_{\text{pa}(j)}, T, \theta) = \sum_{\tilde{\mathbf{D}}_{l,k}} \mathbb{1}_{\{\mathbf{D}_j = \mathbf{D}_{\text{pa}(j)} \odot \tilde{\mathbf{D}}_j\}} P(\tilde{\mathbf{D}}_j|T, \theta), \quad (2.77)$$

where $\mathbb{1}$ is the indicator variable. The additional complexity of (2.77) comes from evaluating the latent sites $\tilde{\mathbf{D}}_{l,k}$ that give the required form of $\mathbf{D}_{l,k}$. This means that likelihood cannot be factorised site-wise, where exact likelihood evaluation requires considering the joint state of all sites within a dependency lineage. This increases the effective state space from a binary set of states to an exponentially increasing number of states, increasing the overall computational overhead of the calculation and thus the algorithm. Therefore, even with Felsenstein's pruning algorithm, the likelihood calculations make the dependency model infeasible for use.

However, assuming conditional independence on such dependent data will necessarily cause model misspecification. Additional dependent sites are unable to provide the same amount of additional information that independent sites provide. The key idea here is that information

²It does not satisfy the Markov property. This statement is justified in Appendix B.2.

Algorithm 3: Dependent Data Generation using ZIP branching process and product of binary CTMCs.

Input: T , fixed topology tree with root r of n languages, where each edge $e := (u \rightarrow v)$ has branch length t_e ,
 $\mathbf{G}$, rate matrix with parameters $\beta\pi_0, \beta\pi_1$,
 π , stationary distribution of binary CTMC,
 N_1 , number of independent sites (Tier 1),
 L , number of tiers,
 p , probability of extra 0s in the ZIP random variable,
 λ , the rate of the ZIP random variable.

Output: $\mathbf{D} \in [0, 1]^{n \times \hat{S}}$, where sites are columns $\mathbf{D}_1, \dots, \mathbf{D}_{\hat{S}}$.

```

1 Function DependentSiteGeneration( $T, \mathbf{G}, \pi, \mathbf{s}_{l-1}$ )
2   | Sample  $\tilde{\mathbf{s}}$  via SiteGeneration( $T, \mathbf{G}, \pi$ ); (see Algorithm 1)
3   | return  $\mathbf{s}_l := \mathbf{s}_{l-1} \odot \tilde{\mathbf{s}}'$ ;
4 end

5 for  $i = 1, \dots, N_1$  do
6   |  $\mathbf{D}_i \leftarrow \text{SiteGeneration}(T, \mathbf{G}, \pi)$ ; (see Algorithm 1)
7 end
8  $\mathbf{D}^{(1)} \leftarrow (\mathbf{D}_1 \dots \mathbf{D}_{N_1})$ ;
9 for  $l = 2, \dots, L$  do
10  |  $\mathbf{S} \leftarrow \mathbf{D}^{(l-1)}$ ;
11  |  $n_S = \text{ncols}(\mathbf{D}^{(l-1)})$ ;
12  | for  $\mathbf{s}_i^{(l-1)}$  in  $\mathbf{S}$  do
13    | Sample  $n_i \sim \text{ZIP}(p, \lambda)$ ;
14    | for  $j = 1, \dots, n_i$  do
15      |  $\mathbf{D}_{ij}^{(l)} \leftarrow \text{DependentSiteGeneration}(T, \mathbf{G}, \pi, \mathbf{s}_{l-1})$ ;
16    | end
17  | end
18  |  $\mathbf{D}^{(l)} \leftarrow (\mathbf{D}_{11}^{(l)} \dots \mathbf{D}_{1n_1}^{(l)} \mathbf{D}_{21}^{(l)} \dots \mathbf{D}_{n_S n_L}^{(l)})$ ;
19 end
20 return  $\mathbf{D} = (\mathbf{D}^{(1)} \dots \mathbf{D}^{(L)})$ ;

```

provided by new site $\mathbf{D}_{s+1}$ which has existing dependencies on $\mathbf{D} \in \{0, 1\}^{n \times S}$ will overlap with the information already obtained from $\mathbf{D}$, thus providing less additional effective information than an independent site would. As such, assuming independence of sites would artificially inflate the amount of effective information obtained by the model. We can see this through the fact that the Fisher information matrix under the dependent paradigm will decrease as dependent sites will, in general, have less degrees of freedom than in independent sites.

More formally, we present a simple example concerning the *mutual information* of single parent site $\mathbf{D}_1$ and one descendant $\mathbf{D}_2$ for a single parameter θ :

Proposition 5 (Effective Information of Dependent Sites). *Under the dependent data generation scheme of Section 2.3, define independent and identically distributed sites $\mathbf{D}_1, \tilde{\mathbf{D}}_2 \in \{0, 1\}^n$ generated from true tree T^* and parameters θ , and define dependent site $\mathbf{D}_2 := \mathbf{D}_1 \odot \tilde{\mathbf{D}}_2$. Therefore, the mutual information of $\mathbf{D}_1$ and $\mathbf{D}_2$ satisfies:*

$$I((\mathbf{D}_1, \mathbf{D}_2); T^*, \theta) \leq 2I(\mathbf{D}_1; T^*, \theta), \quad (2.78)$$

thus providing less information gain than two sites in the independent paradigm.

Proof. We use results from information theory. By the chain rule for conditional mutual information [59]:

$$I((\mathbf{D}_1, \mathbf{D}_2); T^*, \theta) = I(\mathbf{D}_1; T^*, \theta) + I(\mathbf{D}_2; T^*, \theta | \mathbf{D}_1), \quad (2.79)$$

thus it suffices to show that $I(\mathbf{D}_2; T^*, \theta | \mathbf{D}_1) \leq I(\mathbf{D}_1; T^*, \theta)$. From the data generation process, we note that we can represent the relationships between the aforementioned variables as a Markov chain conditional on $\mathbf{D}_1$:

$$(T^*, \theta) \rightarrow \tilde{\mathbf{D}}_2 \rightarrow \mathbf{D}_2, \quad (2.80)$$

since $P(\mathbf{D}_2 | \tilde{\mathbf{D}}_2, \mathbf{D}_1, T^*, \theta) = \mathbb{1}_{\{\mathbf{D}_2 = \mathbf{D}_1 \odot \tilde{\mathbf{D}}_2\}} = P(\mathbf{D}_2 | \tilde{\mathbf{D}}_2, \mathbf{D}_1)$ from (2.77). We can thus apply the Data Processing Inequality [60]:

$$I(\mathbf{D}_2; T^*, \theta | \mathbf{D}_1) \leq I(\tilde{\mathbf{D}}_2; T^*, \theta | \mathbf{D}_1). \quad (2.81)$$

We next consider the chain rule decompositions of $I(\tilde{\mathbf{D}}_2; \mathbf{D}_1, T^*, \theta)$:

$$I(\tilde{\mathbf{D}}_2; \mathbf{D}_1, T^*, \theta) = I(\tilde{\mathbf{D}}_2; \mathbf{D}_1) + I(\tilde{\mathbf{D}}_2; T^*, \theta | \mathbf{D}_1) \quad (2.82)$$

$$I(\tilde{\mathbf{D}}_2; \mathbf{D}_1, T^*, \theta) = I(\tilde{\mathbf{D}}_2; T^*, \theta) + I(\tilde{\mathbf{D}}_2; \mathbf{D}_1 | T^*, \theta) \quad (2.83)$$

$$\implies I(\tilde{\mathbf{D}}_2; \mathbf{D}_1) + I(\tilde{\mathbf{D}}_2; T^*, \theta | \mathbf{D}_1) = I(\tilde{\mathbf{D}}_2; T^*, \theta) + I(\tilde{\mathbf{D}}_2; \mathbf{D}_1 | T^*, \theta) \quad (2.84)$$

$$= I(\tilde{\mathbf{D}}_2; T^*, \theta) + 0 \quad (2.85)$$

$$\implies I(\tilde{\mathbf{D}}_2; T^*, \theta | \mathbf{D}_1) \leq I(\tilde{\mathbf{D}}_2; T^*, \theta) \quad (2.86)$$

since $\mathbf{D}_1$ and $\tilde{\mathbf{D}}_2$ are independently generated and information is non-negative. Finally, we use the fact that $\mathbf{D}_1, \tilde{\mathbf{D}}_2$ are identically distributed, we use the inequalities derived in (2.81) and (2.86) show that

$$I(\mathbf{D}_2; T^*, \theta | \mathbf{D}_1) \leq I(\tilde{\mathbf{D}}_2; T^*, \theta | \mathbf{D}_1) \quad (2.87)$$

$$\leq I(\tilde{\mathbf{D}}_2; \theta) = I(\mathbf{D}_1; \theta) \quad (2.88)$$

as required. $\square$

We note that this inequality are strict where $\mathbf{D}_1$ has at least one component equal to 0: $\mathbf{D}_1 \neq \mathbf{1}$, which in general holds almost surely. This means that information derived from dependent sites will always be smaller than that of independent sites:

$$I((\mathbf{D}_1, \mathbf{D}'_1); T^*, \theta) \geq I((\mathbf{D}_1, \mathbf{D}_2); T^*, \theta). \quad (2.89)$$

As such, by assuming independence between sites, the amount of effective information gained about true tree T^* is inflated. Consequently, information accumulates sublinearly relative to the independent case, which is the basis for the negative effects of model misspecification. This proposition provides the intuition that extends component-wise to the multivariate case, along with a more complicated dependency structure as seen in (2.76).

2.3.4 Model Misspecification

As seen in Section 2.2, the likelihood calculations decompose into simple evaluations using Felsenstein’s algorithm. On the other hand, as seen in Section 2.3.3, the time-complexity of the dependency structure causes any model that requires likelihood evaluation to be infeasible. As such, while applying the model that assumes conditional site independence causes dependent data to be misspecified, the model is still able to generate reconstructions of trees within a reasonable amount of computational runtime. This is the primary advantage of enforcing this assumption.

However, there are several other reasons why a model assuming conditional site-independence may be preferred. Firstly, when fitting to real linguistic data, the dependency structure is usually unknown. This makes any meaningful application of model relaxing such an assumption require the user to manually specify the dependency structure $\mathcal{G}$, which often times requires many design choices and other assumptions about the data $\mathbf{D}$. On top of that, such a model requires a data generation algorithm as specified in this section thus far, which adds another design choice but also puts the model at risk of overparameterisation. In certain cases, where the size of data is small, this may cause identifiability issues and may be fundamentally impossible to apply since there are more parameters than sites.

What remains to be seen, thus, is the effect of model misspecification under such models. While the assumption of conditional site independence yields substantial computational advantages, a model that is unable to recover the true tree T^* with an acceptable level of accuracy has limited practical value. Additionally, as seen from (2.77), if sites do exhibit dependency structure, marginal information obtained by including these sites is considerably lower than independent sites, thus decreasing the *effective size* of the dataset obtained. Blindly applying an model with this independence assumption would then inflate confidence or introduce bias in estimates as the model assumes a larger amount of information can be gained from the data. In other words, understanding the extent of the effect on reconstruction accuracy and bias in using the simplified model assuming conditional site independence allows one to understand the trade-offs involved in linguistic tree reconstruction.

2.4 Evaluation Metrics

We conclude this section with a discussion of evaluation metrics of both tree reconstruction and MCMC quality. The first section on tree recovery metrics (Section 2.4.1 considers methods for quantifying distances in the tree space $\mathcal{T}$, often dealing with a collection of trees $\tilde{T} := \{T_1, \dots, T_n\} \subseteq \mathcal{T}$ compared to some true tree $T^* \in \mathcal{T}$. The second section on MCMC diagnostics (Section 2.4.2) discusses the quantitative and qualitative methods to ensure the Markov chain is representative of the target posterior distribution p^* as detailed in Section 2.1.4.

2.4.1 Tree Recovery Metrics

To evaluate and compare the effects of violating conditional site-independence in linguistic data on phylogenetic tree reconstruction, it is essential to quantify discrepancies between a set of inferred trees $\tilde{T} := \{T_1, \dots, T_n\} \subseteq \mathcal{T}$ to the true tree T^* . While the space of trees $\mathcal{T}$ is a non-Euclidean space, there have been many specialised methods derived to quantify distances between trees.

The principal method for measuring distances between trees is the *Robinson-Foulds* (RF) metric [61]. The method as defined in Definition 13 considers the clades formed in each tree and calculates the number of clades that only exist in one of the trees. The definition in (2.90) is equivalent to $RF(T^*, T) = |\mathcal{C}(T^*)| + |\mathcal{C}(T)| - 2|\mathcal{C}(T^*) \cap \mathcal{C}(T)|$ [62]. Clearly, a distance of zero between two trees with the same number of leaves implies that the trees have the same set of clades and are topologically equivalent, whereas larger RF distances will imply increasingly different tree structures.

Definition 13 (Robinson-Foulds Distance). *The Robinson-Foulds (RF) Distance between two rooted trees T and T^* , with unique labelling in leaves, is defined as*

$$RF(T^*, T) = |\mathcal{C}(T^*) \setminus \mathcal{C}(T)| \cup |\mathcal{C}(T) \setminus \mathcal{C}(T^*)|, \quad (2.90)$$

where $\mathcal{C}(T) := \{C(v) | v \in V\}$ is the set of all clades in rooted tree $T = (V, E, r)$.

The RF distance is highly interpretable and simple to compute, the method is widely used to compare phylogenetic trees. However, the metric treats all clade disagreements equally, wherever they are on the tree. This method thus fails to consider biological or linguistic significance of clades [62]. For the purposes of the simulation study contained within this thesis, the RF distance is still used as it is standard across literature and is a computationally efficient algorithm, fitting the needs of scalable experiments.

Clearly, the RF distance has an upper bound, given as $2n - 6$, for comparing trees of n languages [63]. This allows us to compute the *normalised RF* (nRF) distance, an extension of the RF distance [62, 63], that normalises the RF-distance by its theoretical maximum, giving a distance score between 0 and 1 that allows comparison of pairs of trees with differing values of n . This is extremely useful in the context of the simulation study as detailed in Chapter 3 as trees of varying sizes are compared.

In addition to the RF distance, we consider a separate clade recovery metric, the *clade recovery rate*, that measures the proportion of internal clades of T^* that are recovered in inferred tree T . This is analogous to the recall metric in supervised machine learning contexts and can be defined as

$$CR = \frac{|\mathcal{C}_{\text{int}}(T^*) \cap \mathcal{C}_{\text{int}}(T)|}{|\mathcal{C}_{\text{int}}(T^*)|} \quad (2.91)$$

Unlike RF distances, the method notably considers the set of all internal clades $\mathcal{C}_{\text{int}}$ which thus explicitly evaluates the ability of the inference procedure to recover true groupings and thus explain distances in tree recovery more accurately. When considering a collection of trees $\tilde{T}$, one could then proportion the number of internal clades of T^* that are *highly* supported by $\tilde{T}$, that is, where the proportion of trees in $\tilde{T}$ that recover internal clade $c \in \mathcal{C}_{\text{int}}(T^*)$ exceeds a threshold like 0.9. We define the clade recovery metric for a collection of trees $\tilde{T}$ in Definition 14.

Definition 14 (Highly Supported Clade Recovery). *Given a collection of trees $\tilde{T}$ attempting to recover true tree T^* . Let $r(c) := \frac{|\{T \in \tilde{T} : c \in \mathcal{C}_{\text{int}}(T)\}|}{|\tilde{T}|}$ be the rate of recovery of clade $c \in \mathcal{C}_{\text{int}}$. The*

proportion of highly supported recovered clades is defined as

$$HSCR := \frac{|\{c \in \mathcal{C}_{int}(T^*) : r(c) > 0.9\}|}{|\mathcal{C}_{int}(T^*)|}. \quad (2.92)$$

Finally, it is worth considering how a collection of trees $\tilde{T}$ can be *summarised* into a single tree for comparison. While these methods can be applied to all trees before considering the summary statistics of the values, another approach would be summarising the trees into some representative tree $\hat{T}$ before applying these methods. These methods are most common in Bayesian phylogenetic methods like MCMC where a posterior sample of trees are generated, rather than a single estimate [10, 64, 65].

There are multiple techniques to define a representative tree, of which we will focus on two methods. Firstly, the majority-rule consensus (MRC) tree constructs a new tree T that includes all clades $\mathcal{C}(T) := \bigcup_{T \in \mathcal{T}} \mathcal{C}(T)$ that are supported in more than 50% of trees $\mathcal{T}$. This prioritises recovering as many of the highly supported clades, which may result in a non-binary tree instead.

A separate approach instead is the maximum clade credibility (MCC) tree, which chooses a tree $T \in \tilde{T}$ which maximises clade credibility:

$$CC(T) := \prod_{c \in \mathcal{C}(T)} P(c), \quad (2.93)$$

where $P(c)$ denotes the posterior probability of clade c . This results in choosing a (binary) tree from the posterior sample that has the highest “agreeability” with the rest of the posterior. These contrasts in priorities allow both methods to be reported frequently in Bayesian phylogenetic analyses to understand reconstruction accuracy [64].

2.4.2 MCMC Diagnostics

We conclude this section by considering the methods to ensure quality of the Markov chain. If the chain is unable to reach stationarity, the samples from the chain would not represent the target distribution, the posterior, to an acceptable degree. As such, methods to ensure proper mixing are required to validate the results of any Bayesian inference algorithm.

We introduce two methods to ensure quality of a given Markov chain

$$\left(\{\mathbf{z}^{(i)} = \{T^{(i)}, \boldsymbol{\theta}^{(i)}\}\} \right)_{i=1, \dots, N}. \quad (2.94)$$

Firstly, while slightly informal, it is often informative to consider the mixing behaviour of the MCMC chain visually, perhaps through a trace plot like in Figure 2.5. Poor mixing can arise due to a number of reasons such as ineffective hyperparameter tuning or high dimensionality of the parameter space. As such, visually verifying the mixing of a Markov chain allows one to identify trends, periods of poor exploration or failures to reach stationarity.

In general, a well-mixed Markov chain should fluctuate around a region of the parameter space without exhibiting long-term trends. However, this is subjective to certain cases, where the posterior distribution is multi-modal for example, a well-mixed Markov chain would exhibit jumps between regions of the parameter space.

Next, we consider the autocorrelation sequence $(\rho_\tau)_{\tau \geq 1}$ of samples, which give rise to Effective Sample Size (ESS) as a measure for quality of samples:

$$ESS = \frac{N}{1 + 2 \sum_{\tau=1}^{\infty} \rho_\tau}. \quad (2.95)$$

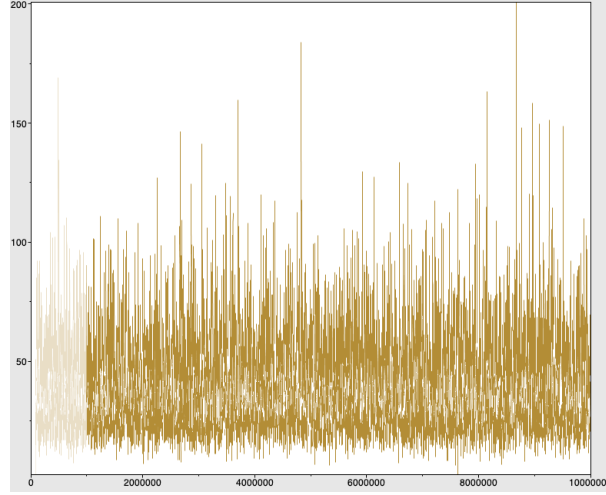


Figure 2.5 An example of a trace plot of a Markov chain which has mixed well.

The motivation for considering ESS is that the Markov chain generates dependent samples [66]. As such, when compared to an equally-sized independently generated data, the Markov chain holds less information. Through calculation of the ESS, we derive the size of independently generated samples from the posterior that would provide equal information as the Markov chain. As such, a chain that has highly correlated samples will have a much smaller ESS. For the context of this study, we shall set $\text{ESS} \geq 200$ as a pragmatic threshold for the quality of samples obtained, with reference to the application *Tracer* which sets the same benchmark for potentially problematic parameters [67].

While (2.95) is easily calculated for ordinary parameters θ , the autocorrelation sequence is much harder to define for the tree space. Instead, applications such as BEAST report the ESS for numerical quantities such as log-posterior and log-likelihood [55], which provide a Euclidean expression for the similarity between trees.

A common practice in Bayesian inference methods like MCMC is to perform *thinning* of the Markov chain which retains only every k -th sample. While reducing the overall dataset size, the resulting chain has much lower autocorrelation values and thus might improve ESS and practically reduce computational overhead [66]. Of course, this still requires a large number of samples once stationarity has been achieved, which may require the user to increase the total number of samples anyway to maintain ESS values.

Chapter 3

Methods

Having laid out the foundations of historical linguistics and Bayesian phylogenetic analysis, we turn our focus to assessing the impact of violating the conditional site independence assumption of the model. Two questions were raised in Section 1.4 to study this, namely:

1. How does an increased proportion of site-dependent data affect tree reconstruction? That is, given data of fixed size S on n related languages, how does (unknown) site dependence affect reconstruction of true tree T^* ?
2. How does adding dependent data to independent data affect reconstruction? That is, given data that maintains site independence of size S on n related languages, how does adding data that is known to be site-dependent affect reconstruction of the true tree T^* ?

As considered in Chapter 2, applying a model that does not assume independence to linguistic data causes likelihood calculations to be infeasible. As such, the study of model misspecification is relevant here. Question 1 aims to consider the impact of applying the model to data without knowing the true dependency structure of the data. This scenario is usually the case when fitting the model real linguistics data, as these dependencies are not only present but difficult to define without prior knowledge. Question 2, on the other hand, serves as a guide in situations where data is *known* to have a clear dependency structure, such as in the example provided by Buckley et al. [9], on whether including data that is misspecified by the model will affect tree reconstruction.

This section aims to explain the methods conducted in the simulation study to answer these questions. The design is split into two *experiments* that look to provide results that explain their respective questions. The data generation methods and simulation design for both experiments are described, before considering the implementation of Bayesian phylolinguistic analysis. Finally, several validation results are shown to provide a basis for the results that follow in Chapter 4. The full implementation of the simulation study can be found in this [GitHub repository](#).

3.1 Simulation Design

3.1.1 Experiment 1

Experiment 1 serves to answer Question 1 of the simulation study, that is, how manipulating proportion of site-dependent data q . As such, for a fixed size S of data across n related languages, a proportion of the data will be set as *independent* data, while the remaining proportion of data will be generated via the *dependent* generation method described by Section 2.3.

Additionally, within the dependent data generated, we vary the amount of tiers of the dependency scheme to consider the impact of the *depth* of dependency of data as well. Given a fixed number of tiers L (of which there are $L - 1$ tiers that are generated with dependence), the dependent component is uniformly distributed across the $L - 1$ dependent tiers. An algorithm detailing the dependent data generation in the context of Experiment 1 can be found in Algorithm 4.

Algorithm 4: Dependent Data Generation for Experiment 1.

Input: T^* , true topology tree with root r of n languages,
 $\mathbf{G}$, rate matrix with parameters q_{01}, q_{10} ,
 π , stationary distribution of binary CTMC,
 S , total number of sites,
 q , proportion of sites to be independent,
 L , number of tiers,
 p , probability of extra 0s in the ZIP random variable,
 λ , the rate of the ZIP random variable.

Output: $\mathbf{D} \in [0, 1]^{n \times S}$, where sites are columns $\mathbf{D}_1, \dots, \mathbf{D}_S$.

```

1  $N_{\text{dep}} \leftarrow \lfloor \frac{S(1-q)}{L-1} \rfloor$ ;
2  $N_1 \leftarrow S - N_{\text{dep}} * (L - 1)$ ;
3 for  $i = 1, \dots, N_1$  do
4    $\mathbf{D}_i \leftarrow \text{SiteGeneration}(T^*, \mathbf{G}, \pi)$ ; (see Algorithm 1)
5 end
6  $\mathbf{D}^{(1)} \leftarrow (\mathbf{D}_1 \dots \mathbf{D}_{N_1})$ ;
7 for  $l = 2, \dots, L$  do
8    $\mathbf{S} \leftarrow \mathbf{D}^{(l-1)}$ ;
9    $n_S = \text{ncols}(\mathbf{D}^{(l-1)})$ ;
10  for  $s_i^{(l-1)}$  in  $\mathbf{S}$  do
11    Sample  $n_i \sim \text{ZIP}(p, \lambda)$ ;
12    for  $j = 1, \dots, n_i$  do
13       $\mathbf{D}_{ij}^{(l)} \leftarrow \text{DependentSiteGeneration}(T^*, \mathbf{G}, \pi, s_{i-1})$ ; (see Algorithm 3)
14    end
15  end
16   $\mathbf{M}^{(l)} \leftarrow (\mathbf{D}_{11}^{(l)} \dots \mathbf{D}_{1n_1}^{(l)} \mathbf{D}_{21}^{(l)} \dots \mathbf{D}_{n_S n_L}^{(l)})$ ;
17  Sample  $N_{\text{dep}}$  sites from  $\mathbf{M}^{(l)}$  without replacement to form  $\mathbf{D}^{(l)}$ ;
18 end
19 return  $\mathbf{D} = (\mathbf{D}^{(1)} \dots \mathbf{D}^{(L)})$ ;

```

It is worth noting that because of ascertainment, the algorithm may return a smaller than requested amount of data. As such, a larger sample is generated before sampling without replacement is conducted, as found in Line 16 and 17 of Algorithm 4. As such, not all dependent data generated will be stored in the final data matrix $\mathbf{D}$. While this may distort the final dependency structure of the data (as opposed to the expected behaviour as found in Section 2.3.1), this approach ensures data generated contain the prescribed proportions of independent and dependent sites, which is the primary objective in this experiment.

For a given true tree T^* and fixed S , 13 sets of data are generated: where one set is completely independent (acting as the control), generated through Algorithm 1, while the remaining sets have some combination of proportion of independent data q and total number of tiers

used L :

$$q \in \mathcal{Q}, L \in \mathcal{L}, \quad (3.1)$$

generated through Algorithm 4. We repeat the experiment, varying the number of sites:

$$S \in \mathcal{S}, \quad (3.2)$$

for true tree T^* , before generating new trees:

$$T^* \in \mathcal{T}, \quad (3.3)$$

for a given n , and finally varying number of languages considered:

$$n \in \mathcal{N}. \quad (3.4)$$

This factorial design allows for a complete variation of relevant parameters, studying how model misspecification impacts tree reconstruction in a wide array of settings. The full list of parameters chosen for the implementation of Experiment 1 is found in Appendix C.

3.1.2 Experiment 2

Experiment 2 serves to answer Question 2 of the simulation study, that is, whether including dependent data affects tree reconstruction. Additionally, as there are a multitude of dependency structures even within the scheme described in Section 2.3, it is worth considering how differences in parameters (p, λ) affect tree reconstruction as well.

Under such an experimental setup, preserving the dependency structure of the model during data generation is essential. Additionally, in order to isolate the effects of different dependency structures, we require that all data generation methods generate similar amounts of data across all tiers, such that results are not systematically affected by the total number of sites generated. However, although ZIP random variables can have fixed expectations, through Propositions 1 and 2, the variance of sparse data generation schemes will be larger than dense data generation schemes. We treat this as an intrinsic property of the model chosen and incorporate it into the analysis.

A factorial design similar to Experiment 1 is adopted for a complete variation of relevant parameters. For a given true tree T^* and fixed *number of independent sites* N_1 , 13 sets of data are generated: one set with 1 tier of N_1 independently generated sites through Algorithm 1, while remaining sets are generated with dependent data sites using some dependency generation scheme $w \in \mathcal{W}$ and number of tiers used $L \in \mathcal{L}$, generated through Algorithm 3. In particular, the set of schemes $\mathcal{W}$ considered used ZIP random variables that have a fixed expected value of $\mu = 3$. The experiment is repeated for a given true tree T^* by varying the number of independent sites $N_1 \in \mathbb{U}$. The experiment is then repeated in the same fashion as Experiment 1 in (3.3, 3.4), and the full list of parameters chosen for this implementation can be found in Appendix C.

3.2 Implementation

In this section we consider the full pipeline for analysis, as detailed in Figure 3.1. Given the factorial design of the experiments, running simulations for every configuration of $\{\mathcal{S} \text{ or } \mathcal{U}, \mathcal{T}, \mathcal{N}\}$ can quickly become infeasible. However, this can be easily implemented with parallelisation on a High Performance Computing (HPC) cluster, where a single job only generates a single true tree T^* of fixed size n with fixed data sizes. This reduces the total time taken for all runs to

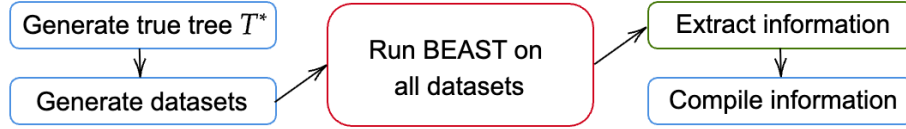


Figure 3.1 Pipeline for data generation (left), analysis (center) and data post-processing (right). The boxes are colour-coded by programme used: Python in blue, BEAST in red, and R in green.

complete, allowing the experiments to be run without compromising on exploring the parameter space, the number of simulations performed, or the statistical reliability of the resulting analyses.

The methods described in Section 3.1 are first implemented in Python to generate data. Having generated the relevant data required for each experiment, these datasets must be converted into relevant files that BEAST is able to read.

Upon completion of data generation, Bayesian phylogenetic analysis is performed through BEAST, where parallelisation can occur once again by running all datasets concurrently. All BEAST runs are performed to obtain 1,000,000 samples, of which the samples are thinned to obtain 10,000 remaining samples, where the first 10% of samples are (quite generously) discarded as burnin. The programme generates a number of files, most notably the `.trees` and `.log` files, which provide information of the samples obtained in the analysis. Useful information for analysing tree reconstruction can be collected from these files, namely:

- RF and nRF distances,
- Clade recovery rates and the proportion of well-recovered clades,
- ESS statistics, and
- Representative (MRC and MCC) trees, with the aforementioned metrics.

Due to the wealth of packages that exist in R for phylogenetic analysis such as `phangorn` [68, 69] and `phytools` [70], the analysis is completed in R and exported as a CSV file. Finally, all results from all jobs are compiled in Python to generate a complete data file with the required data for analysis.

3.3 Validation of Simulation Framework

3.3.1 Validation of Relationship between Data Size S and Tree Reconstruction

We now consider several validation plots relevant to the simulation study. We first empirically justify the claim made in Section 2.2 that larger amounts of (independent) data improves tree reconstruction of the Bayesian phylogenetic methods used. In particular, we attempt to show empirically that

$$P(T = T^* | \mathbf{D}) \rightarrow 1 \text{ as } S \rightarrow \infty \quad (3.5)$$

for a given true tree $T^* \in \mathcal{T}$ with fixed n and data $\mathbf{D} \in \{0, 1\}^{n \times S}$, where $T \in \tilde{\mathcal{T}}$ are samples derived from the Markov chain.

Through Figure 3.2, we note several observations about the reconstruction of trees with respect to independent data size. As expected, an increase in the amount of independent data

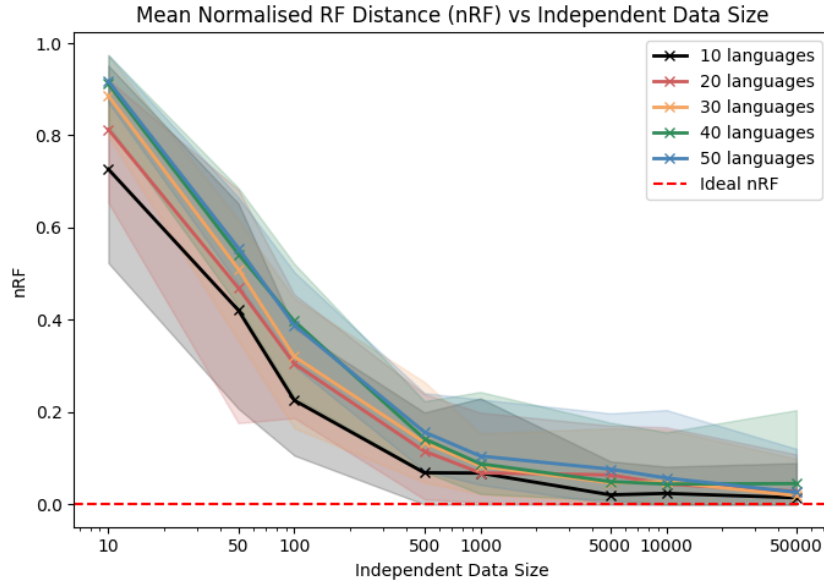


Figure 3.2 Mean nRF vs Independent Data Size, varying amount of languages.

does improve the model's ability to reconstruct the true tree T^* . It can also be seen that while the lines tend toward zero, it seems that 1000 sites and above are more than enough to reconstruct the trees. Additionally, we note that increasing amount of languages n , for a fixed number of sites S , decreases the model's ability to reconstruct trees accurately. This is intuitively true as well: as the tree space becomes more complex, more information is required to reliably perform reconstruction of the posterior distribution and thus obtain the true tree.

The same can be seen in Figure 3.3, where larger datasets result in higher clade recovery. The same analysis put forward for Figure 3.2 for the effect of number of languages is applicable here too: a larger set of languages generally requires more information (and thus more sites) to recover clades within the true tree.

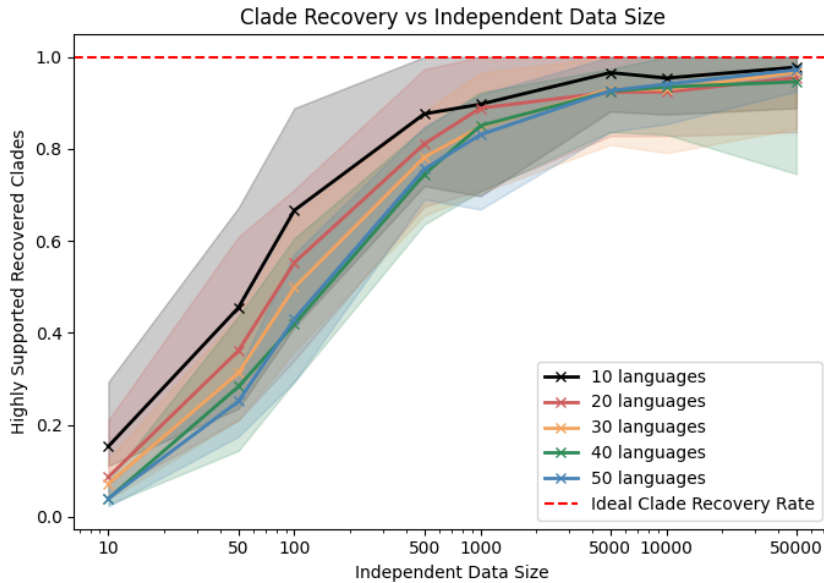


Figure 3.3 Clade Recovery Rate vs Independent Data Size, varying amount of languages.

We include a final figure for this first validation analysis, where we consider the ESS values

of the runs conducted for this exercise. From Figure 3.4 ESS appears to decrease for large datasets. One interpretation of this is that increasing the size of the data beyond a certain point leads to an overly-complex likelihood space, perhaps due to the programme being limited to smaller, local moves and thus increasing autocorrelation. Since there are instances in which ESS values drop below 200, there is a reasonable probability of the Markov chain failing to reach stationarity, and thus we are required to remove these simulations from the analysis process.

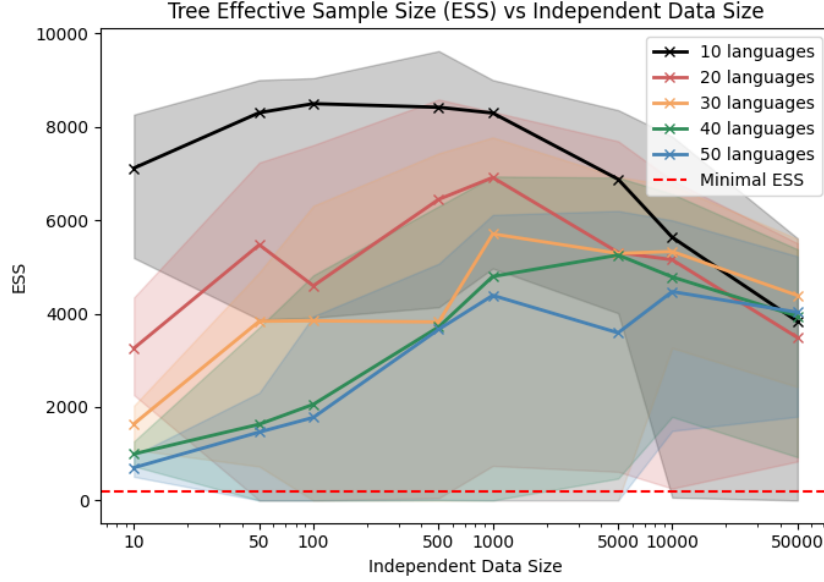


Figure 3.4 Effective Sample Size (ESS) vs Independent Data Size, varying amount of languages.

3.3.2 Validation of ZIP variables

We turn to validating the theoretical results derived in Section 2.3.1 of the mean and variance of ZIP variables. We do this by generating dependent data as per Algorithm 3, taking note of each tier's respective sizes before ascertainment. These are then plotted against the theoretical values described in Proposition 2. Additionally, we generate varying ZIP parameters (p, λ) that give an expected value of $\mu = 3$.

The results of this validation can be found in Figure 3.5. These results clearly show that the implementation of the ZIP random variable matches the theoretical expectation and variance of the branching process of N_l . In particular, each model's exponential growth in tier sizes, which is considered through the gradients of the graph Figure 3.5, matches the theoretical exponential growth of mean and variance predicted by Proposition 2.

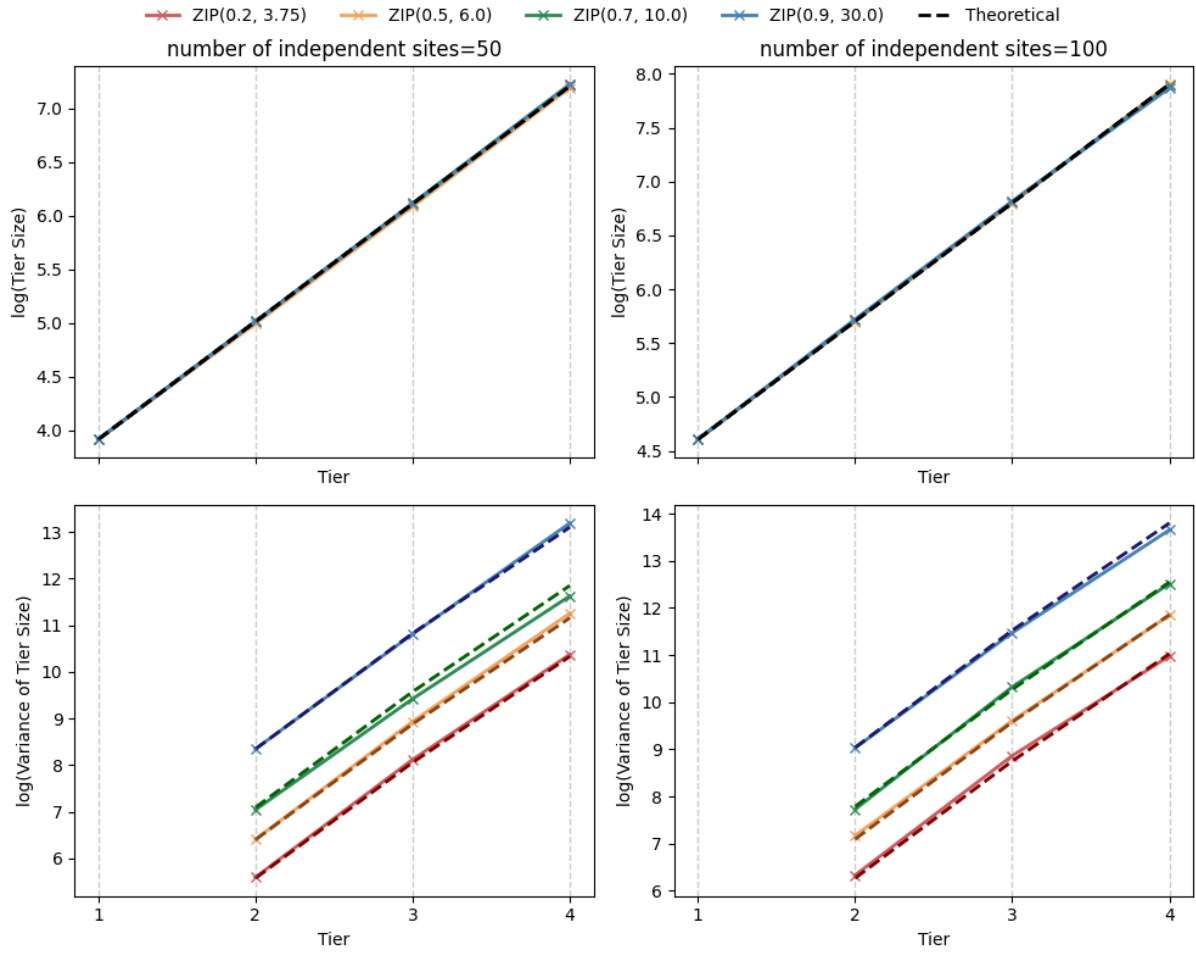


Figure 3.5 Mean and Variance of Tier Size against number of tiers, varying the number of independent sites N_1 and ZIP parameters (p, λ) .

Chapter 4

Results

Having described the simulation study methods and validated implementation results in Chapter 3, we discuss the results obtained from Experiments 1 and 2, in an attempt to answer Questions 1 and 2 as introduced in Section 1.4. This chapter will be split into two sections, presenting the results from each experiment. An in-depth analysis of each experiment’s results will be presented in Section 5.

In a similar vein to Section 3.3, the metrics of concern are the normalised RF distances and clade recovery of trees to their true tree T^* . Additionally, we consider the performance of the model not only on the full set of posterior trees $\tilde{T}$ but also on the two aforementioned representative trees, namely the MRC and MCC trees.

We note, however, as the factorial design covers analysis across a variety of parameters in the parameter space, we present relevant cases of value that portray the general trend observed throughout the study. We show the full results in Appendix A.2.

4.1 Experiment 1

We first study changes in tree reconstruction quality of the posterior sample in Experiment 1. Figure 4.1 shows a clear distinction between the independent control data and any of the dependent data used for tree reconstruction: even small departures from conditional site independence reduce reconstruction accuracy relative to the fully independent control.

On the other hand, the proportion of site-dependent data does not impact nRF values as much, varying minimally across the total number of tiers used. These differences in tree reconstruction quality, while minimal, are expected: having a higher proportion of data that satisfies conditional site independence reproduces posterior samples that have smaller nRF values. A plausible explanation would be that additional layers of dependence would contribute similar amounts of additional information loss, since all higher-order descendants are eventually linked to the same independent site.

Additionally, we note the increase in uncertainty shown in the 95% confidence regions of the nRF. Figure 4.2 tells a similar story: data that violates conditional site independence is clearly unable to recover as many internal clades as in the control, yet proportion of site dependency does not contribute as much to this decrease.

Moreover, we see that the effects of violating conditional site independence on tree reconstruction are more dramatic where the total number of sites S is low. In Figure 4.1, the biggest increase in nRF is where there S is low, increasing by 0.12-0.18, while the nRF barely changes where S is large. In Figure 4.2, where $S = 50$, there is a roughly 6% decrease in total clades

recovered on average, whereas this drops to below 2% where $S = 5000$.

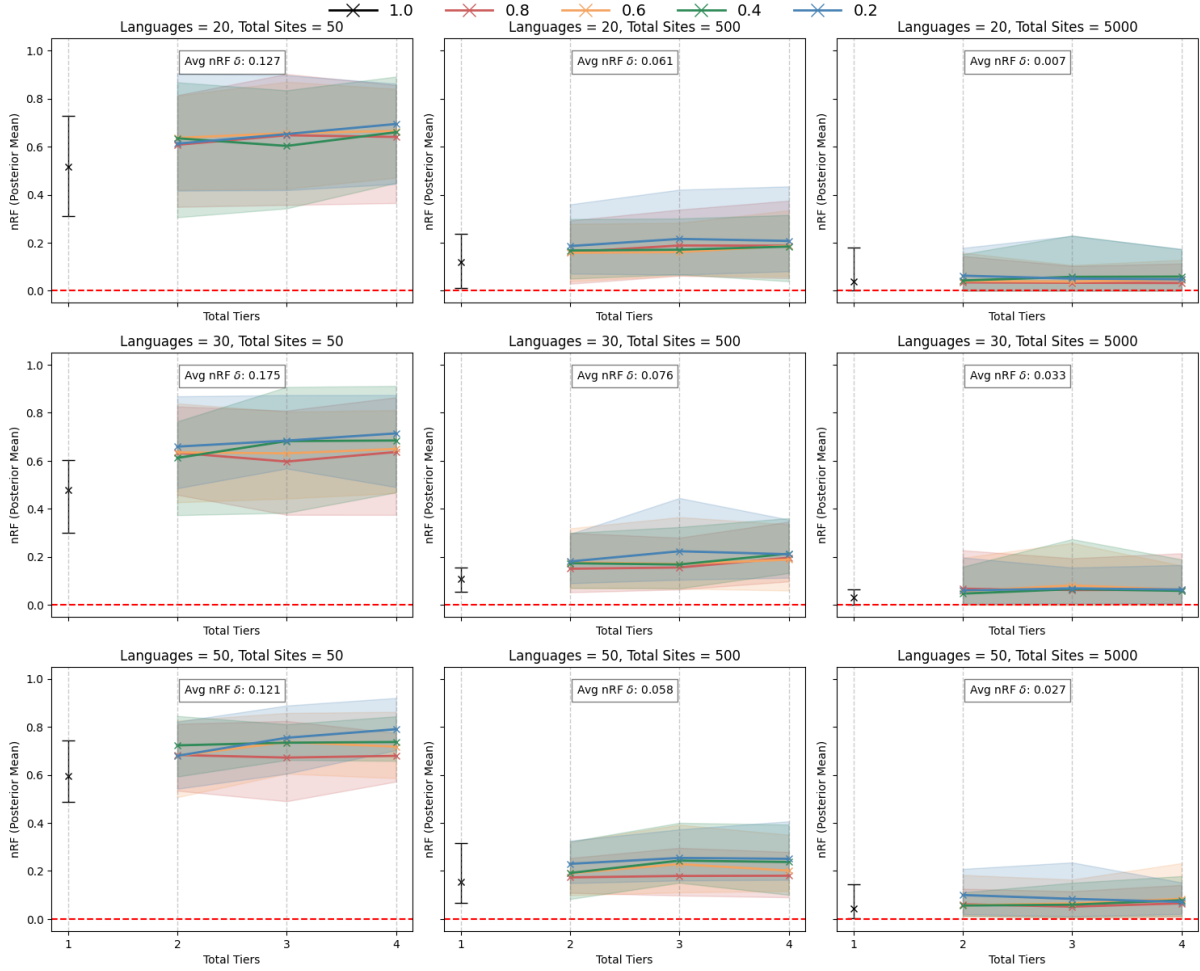


Figure 4.1 nRF of posterior samples against depth of dependencies within each subplot, varying proportion of independent data from 20% (blue) to 80% (red). This is an excerpt of the larger plot found in Figure A.1.

Despite the observed reduction in performance, the distributions remain highly overlapping in many settings, suggesting that the practical effect of moderate dependence may be limited. Perhaps the only times where the reconstruction is more extreme than in any of the independent cases is where there is low proportion of independent data (see the blue and green lines of Figures 4.1 and 4.2, in particular) and where total data is low.

These figures show that there are indeed marginal losses in tree reconstruction as a result of presuming the independence assumption on data that violates it. However, the limiting issue here is not the proportion of data that is dependent, but rather the amount of information obtained from the data. The largest improvements of tree reconstruction as found in Figures 4.1 and 4.2 are not by fulfilling site independence throughout the data, but instead by having a larger amount of data to start with, even if the data violates conditional site independence. These findings corroborate with the validation plots found in Figures 3.2 and 3.3.

We turn to the analyses of the MCC and MRC trees. Similar sentiments to the previous analysis can be found throughout Figures A.3, A.4, A.5 and A.6 (which are omitted from this section for brevity), where data with low proportions of independent sites negatively affect tree reconstruction, especially in scenarios where data size is small. In particular, it can be seen that the MRC tree performs better based on nRF, while the MCC tree performs better based on Clade

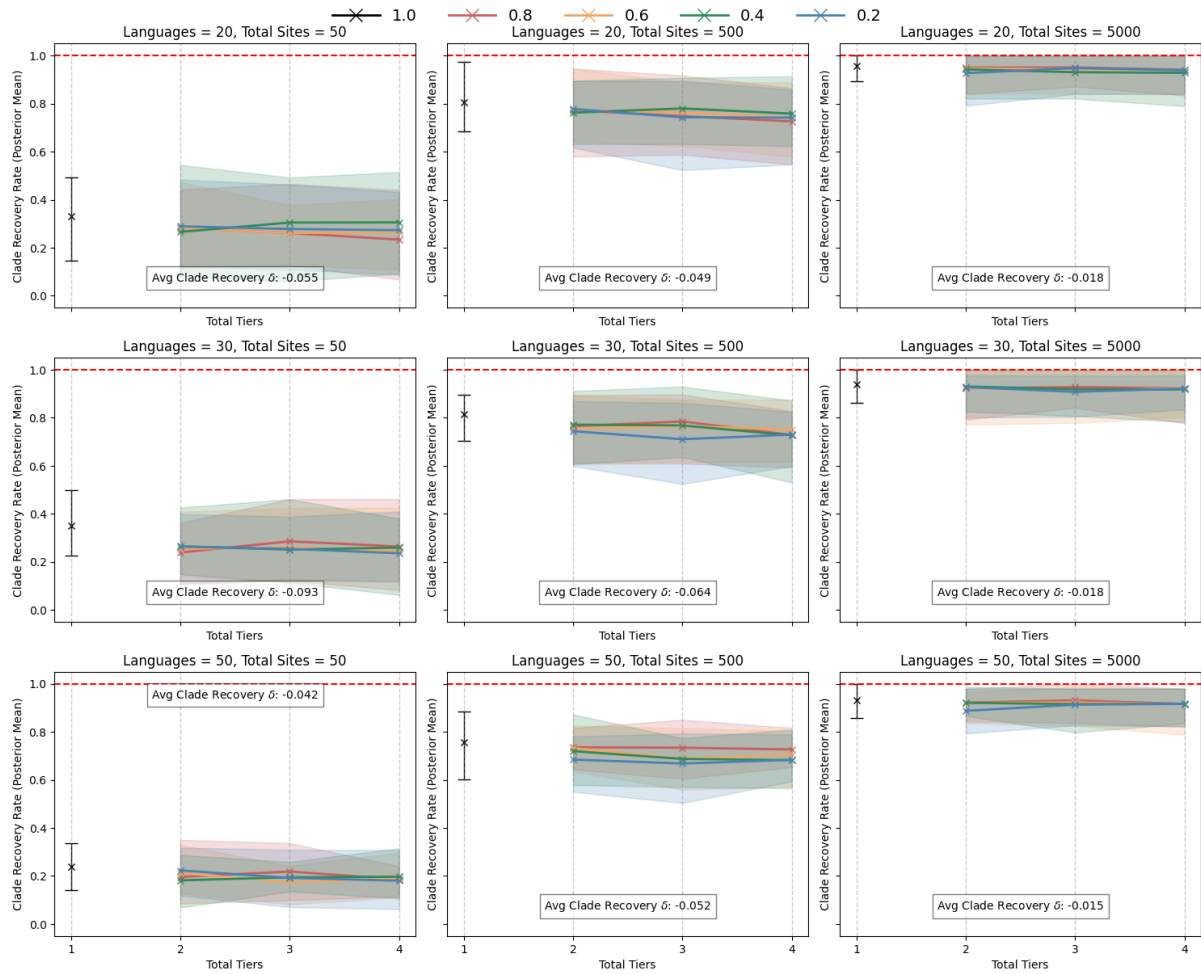


Figure 4.2 Clade Recovery Rate of posterior samples against depth of dependencies within each subplot, varying proportion of independent data from 20% (blue) to 80% (red). This is an excerpt of the larger plot found in Figure A.2.

Recovery. This is indicative of how each metric rewards tree structure: as nRF typically rewards conservative trees, the multifurcating MRC tree benefits much from absorbing uncertainty of clades into tree structure. Conversely, as clade recovery rate prioritises retaining true clades, MCC trees are more advantageous.

4.2 Experiment 2

We now examine whether adding data that has a known dependency structure to independent data is valuable to tree reconstruction. Figure 4.3 shows an excerpt of nRF values against total depth of dependency within the dataset. It can yet again be seen that the effects of such data sets are dependent on the amount of independent data as well as the complexity of the tree space considered. Clearly, in scenarios where the independent data size N_1 is small and the tree space is large (with say $n = 50$ languages in the bottom left plot), adding dependent data lowers nRF values. The opposite is true for small tree spaces and large independent data sizes: in the top right plot, adding any amount of additional data leads to lower quality tree construction. It can also be noted that as more dependency tiers are added, marginal gains on tree reconstruction are smaller, signalling that even though a large amount of data is being added to the dataset, little information about the tree is being recovered.

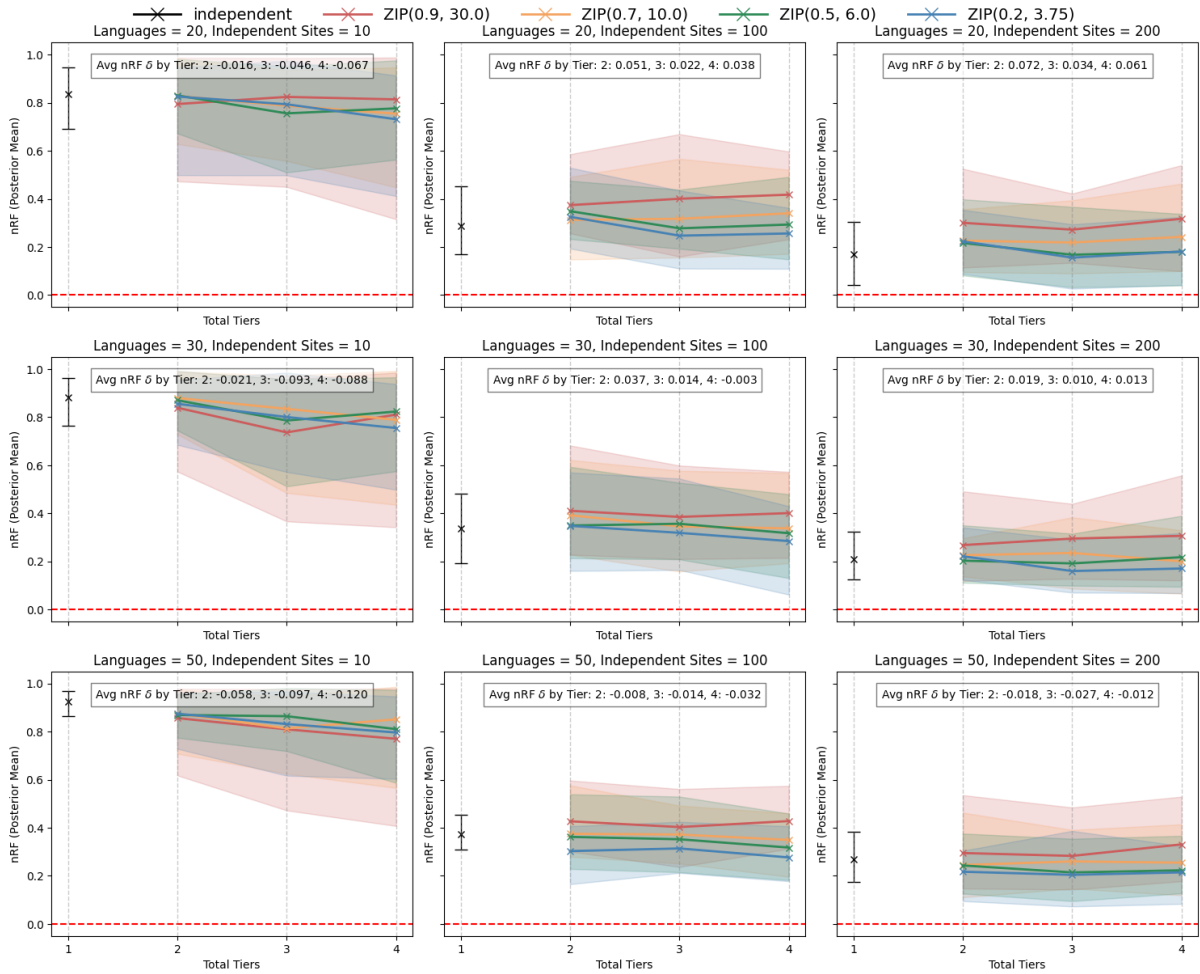


Figure 4.3 nRF of posterior sample against depth of dependency structure within each subplot, varying method of generation from most sparse (red) to most dense (blue). This is an excerpt of the larger plot found in Figure A.7.

The effect of different generation methods on tree reconstruction are clear to see. Firstly, we note that all generation methods already increase the uncertainty associated with estimating true tree T^* , suggesting that the violations of conditional site independence already introduces a degree of model misspecification that propagates into the posterior samples. Denser methods with low p and λ for the ZIP random variable leads to systematically lower nRF values as compared to sparser methods. These differences in nRF values are most pronounced where the independent data size is high. Additionally, the variance of posterior nRF means seem to follow closely with variance of data generation in Proposition 2: the amount of data generated in the sparse method varies the most and seems to have the largest 95% intervals, especially in low data scenarios. This is expected, as sparse structures create massive dependency clusters which reduce effective information more severely than in evenly distributed cluster sizes.

Perhaps the clearest evidence of a significant impact of adding dependent data on tree reconstruction is found in the clade recovery rates in Figure 4.4. We consistently see large improvements in clade recovery across all scenarios, even increasing as more dependency tiers are added, up to a maximum of 27.6% of true clades. Once again, these improvements are most pronounced where the original data size is low, and dense generations still exhibit better clade recovery rates than sparse generations.

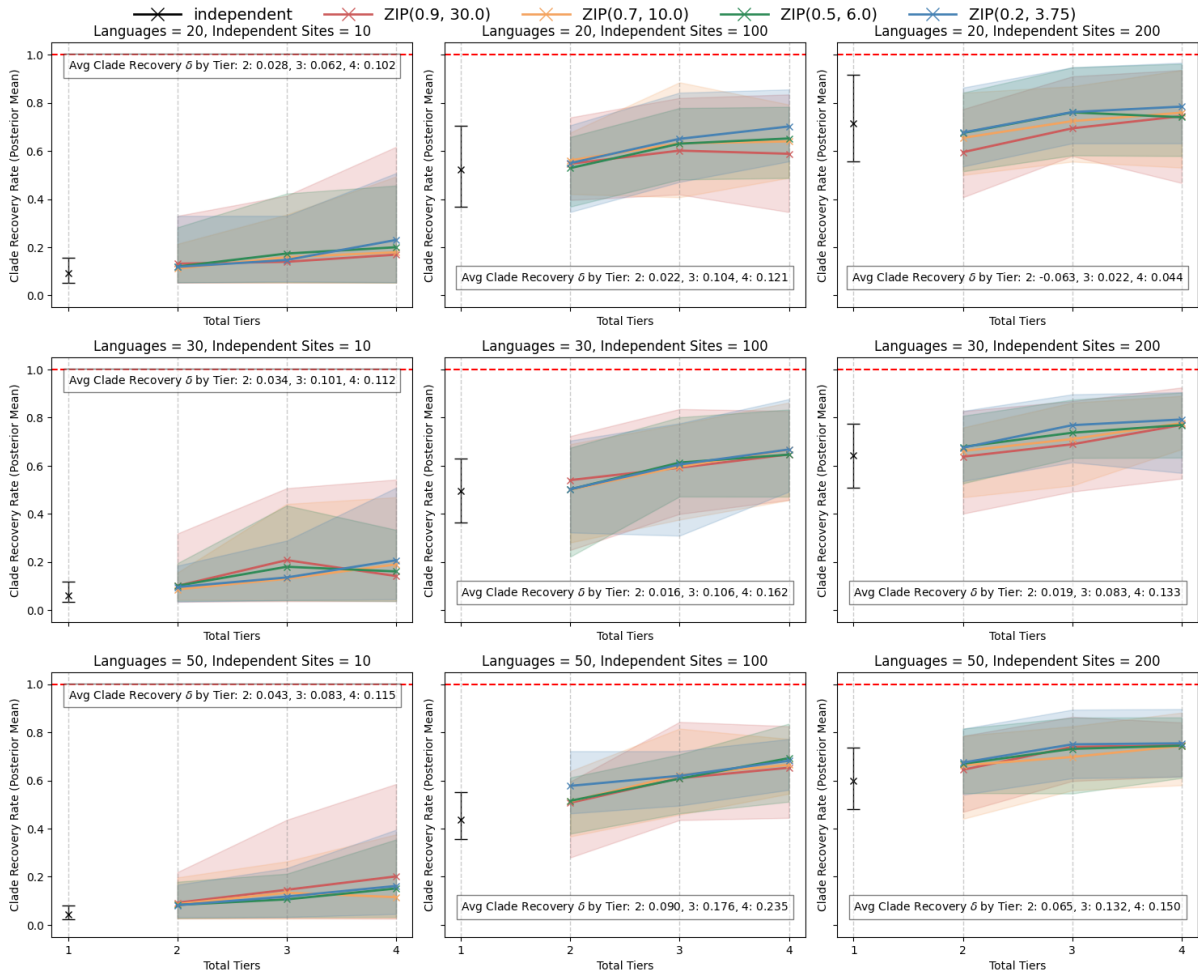


Figure 4.4 Clade Recovery Rate of posterior sample against depth of dependency structure within each subplot, varying method of generation from most sparse (red) to most dense (blue). This is an excerpt of the larger plot found in Figure A.8.

These findings tell a similar story to that of Experiment 1. While these improvements in

tree reconstruction are significant, one reason that these effects are due to the amount of data (and thus the amount of information) increasing as dependent sites are added. These effects are positive in scenarios where the amount of independent data is low to begin with, which slowly diminishes as more data is received. However, these effects are negative where there already exists enough independent data to make strong inferences about the true tree T^* . It appears that adding dependent data in this scenario provides much smaller benefits to tree reconstruction than losses in violating the conditional site independence assumption, such that improvements in the aforementioned metrics are either nullified or turned into deteriorations.

We finally turn our attention toward analysing the MRC and MCC trees in Figures [A.9](#), [A.10](#), [A.11](#) and [A.12](#). Similarly to Experiment 1, we see that the MCC trees perform better on the clade recovery statistic, while the MRC trees are still stronger on nRF values. The clearest difference in using these representative trees rather than considering the entire posterior sample is the reduction of improvements yet heightened deteriorations found in both metrics. The poorer performance observed likely arises from a loss of information incurred when only one (representative) tree is considered. While posterior samples are able to account for uncertainty in poorly resolved regions of the tree space, representative trees must make decisions on clades which lead to potentially incorrect topological decisions. These decisions thus outweigh the benefits gained from additional dependent data as noted in previous analyses, resulting in smaller improvements or larger deteriorations.

Chapter 5

Discussion

Having summarised the key results of the simulation study in Chapter 4, the results are interpreted in this chapter to assess the impact of violating conditional site independence on Bayesian phylogenetic inference and to address the research questions posed in Section 1.4.

5.1 Experiment 1 - Impact of Dependence on Reconstruction Accuracy

From the findings of Experiment 1, we answer the effects of applying Bayesian phylogenetic analysis on misspecified data - that is, data that does not satisfy conditional site independence. The principal result as detailed through Section 4.1 is that any violation of the independence assumption hurts tree reconstruction accuracy, but the amount of dependence matters less. The drop in performance from data fully satisfying the assumption to any amount of dependent data is ubiquitous across Figures 4.1, 4.2, A.3, A.4, A.5 and A.6. This represents the primary fault of model misspecification as detailed in Section 2.3.4: by assuming conditional site independence, the likelihood evaluations assume that the data provides more (effective) information about true tree T^* , whereas the dependency structure decreases the effective information gained from all dependent samples, as predicted by Proposition 5.

However, it is notable that varying the depth of dependency has little effect on tree reconstruction. The relative invariance across depth suggests that, within the parameter ranges considered, dependency depth has substantially less impact on reconstruction quality than the initial violation of independence. This behaviour is consistent with the result derived in Proposition 5: dependent sites contribute less effective information than independent sites. This suggests that tree reconstruction quality is largely robust to dependency depth due to the fact that effective information available to the model remains broadly similar across dependency depths. We can then note that the relevant quantity for reconstruction is not the total number of sites S but rather the total number of independent sites N_1 . This is seen through the differences in tree reconstruction quality between varying proportions of independent data, albeit having a smaller impact than the initial violation of conditional site independence.

We note that across all dependency configurations, the main improvement in tree reconstruction quality is due to increasing S . For example, reconstructing true tree T^* with n languages will always perform better with 5000 sites of data than 500, even if the data violates conditional site independence. This means dependent sites do carry phylogenetic signal that, while less informative than independent sites, still contributes meaningfully to likelihood calculations and outweighs the losses from model misspecification in high data regimes. At large S ,

the posterior is already tightly concentrated around the true tree, and thus misspecification is practically irrelevant. On the contrary, the issues posed by model misspecification are most obvious in scenarios where only a small amount of data for this language exists. This extends the conclusion found in Section 3.3.1: increasing data quantities has a larger effect than preserving independence within the parameter space considered.

Finally, we considered the differences in tree reconstruction for MRC and MCC trees. MRC trees tend to overperform with regards to nRF as the tree absorbs uncertainties into multifurcations rather than making clear choices. A possible explanation is that, under model misspecification, the posterior may be overconfident through treating correlated sites as independent ones, which MRC is able to account for by practicing conservative clustering. On the other hand, MCC trees outperform MRC trees with regards to clade recovery since the MCC trees prioritise posterior probability which allows the model to retain true clades more often, even under misspecification. As such, the choice of summary tree should therefore be determined based on priorities, choosing MRC for overall tree reconstruction accuracy, while choosing MCC when clades and groups of languages are of interest.

5.2 Experiment 2 - Efficacy of Additional Dependent Data

The findings of Experiment 2 highlight the efficacy of including known dependent data into Bayesian phylogenetic analysis. Additional information, albeit violating the conditional site independence assumption of the model, is most beneficial where independent size N_1 is small and the tree space is large. These effects are clearly seen across Figures 4.3, 4.4, A.9, A.10, A.11 and A.12, where both nRF and clade recovery improves in subplots nearer the bottom left of the figures. These gains can be substantial, providing up to a 27.6% increase in recovering true clades (when the independent data only reproduced about 25% in the first place), regardless of dependency scheme. As such, we can conclude that where the model is *starved* of information, a set of correlated sites still provides valuable information, albeit less so than truly independent sites. This information is able to outweigh the penalties from overestimating total information and thus be beneficial to analysis.

However, we note that as N_1 increases, the independent data alone produce a high level of tree reconstruction quality. This means that the marginal benefit by adding dependent data is much smaller than in the data sparse case. This means that the misspecification introduced by the dependent data introduces noise into the posterior by inflating total information gained from the dataset. We see that in the aforementioned figures, subplots nearer the top right either see much weaker improvements than in data sparse situations or even deteriorations in tree reconstruction. Crucially, this has implications on decisions made on including sites that violate conditional site independence. These results suggest that dependent data is most valuable where independent data alone cannot provide a strong enough signal for tree reconstruction.

These effects also vary with respect to dense and sparse dependency structures, identified through the associated ZIP random variable with parameters p and λ . We note that denser methods, which produce dependencies on sites more evenly, consistently produces better tree reconstruction quality than sparse methods. As noted in Section 4.2, this difference in tree reconstruction quality suggests that larger clusters of sites dependent on a single site will contribute less effective information about the tree than the same number of sites that are instead dependent on multiple sites. Put another way, increasing the total number of lineages present in the dependent sites leads to more effective information gained from dependent sites, increasing the benefits associated with appending such sites to the model. The contrary is true in sparser schemes, which also lead to higher variance amongst the posterior and increasing uncertainty in estimating T^* . This thus has implications not only on whether or not dependent

sites are retained for phylolinguistic analysis, but also on the dependency structure within the sites themselves.

Finally, we discuss the diminishing returns by including each additional tier of sites into the analysed data. As we increase the tiers, the construction method defined in Section 2.3.1 suggests that the total number of 0s in each site increases. As noted in (2.76), the repeated application of the dependency operator defined in Section 2.3 causes deeper sites to inherit large proportions of its state from ancestral sites leading to progressively less additional effective information gained. In particular, as l increases, while the exponential growth of N_l as derived in Proposition 1 is observed, the fraction of 1s at each side also decreases geometrically, thus decreasing the additional information by including each tier- l site. Coupled with the fact that the amount of information increases with each tier, deeper tiers would thus provide minimal effective information that in turn may hurt tree reconstruction. These effects are most clearly seen in the diminishing benefits from including the fourth tier of sites, providing negligible reconstruction benefit while continue to inflate the sample size assumed by the model.

5.3 Impact of Effective Information on Tree Reconstruction

While the findings conclude that data that violates conditional site independence has a small yet noticeable negative effect on tree reconstruction (as opposed to independently generated data), we note that these effects vary across different scenarios. The results of both experiments suggest an underlying factor that is key to tree reconstruction quality in Bayesian phylolinguistics: the effective information gained from the data at hand. This introduces a tradeoff between the bias caused by model misspecification, and the accuracy gained by introducing additional information into the model. This gives reason to why any negative effects of model misspecification on a fixed data set are magnified in low data scenarios, and why the positive effects of adding dependent data diminish as more independent sites are introduced.

This corroborates with the theoretical analysis derived in Chapter 2. With independent sites, we face no issues of model misspecification, and additional information is gained additively as found in (2.21, 2.25). However, as found in (2.76, 2.78), where dependent sites are concerned, the model introduces bias via incorrect likelihood calculations, and the amount of effective information gained is stunted by their dependency structures. These results are portrayed in the tradeoffs found in Experiments: where the benefits of effective information gained by correlated sites compete with the losses of bias induced by model misspecification. This relationship can also be seen in the type of dependency structure found in Experiment 2, where tiers of dependent sites generated by denser methods provide more effective information than tiers of sites generated by more sparse methods.

As such, Experiments 1 and 2 provide empirical support for the theoretical framework developed in Chapter 2. These experiments show that violations of conditional site independence cause model misspecification and reduce effective information. However, the present amount of effective information must be considered when making decisions on retaining correlated sites, which may still provide valuable information in low data scenarios.

5.4 Implications for Phylolinguistics

As such, we can conclude that while violations of the conditional site independence assumption in Bayesian phylogenetic analysis do produce measurable bias in tree reconstruction, the effects of these violations are conditioned on the amount of information present in the data itself.

Experiment 1’s findings suggest that in low information settings, any presence of dependence data within the data matrix $\mathbf{D}$ causes sizable deprecation of tree reconstruction quality, while these effects are much more trivial in high information settings (that is, where the total number of sites S is already high). Experiment 2’s findings extend these conclusions to provide evidence that in low (independent) information settings, the inclusion of data that is known to violate conditional site independence has much benefit in tree reconstruction, but is less useful or even harmful in high information settings.

These findings thus form a heuristic framework for analysing the efficacy of dependent data in Bayesian phylolinguistic analysis, which are heavily dependent on the amount of independent data already available in the dataset. This not only sets the expectation for the losses of model misspecification but also the amount of marginal benefit gained from including additional sites that violate conditional site independence. Additionally, when considering additional dependent sites, it is worth considering the likely dependency structure present within the data - dense structures would lead to larger improvements in tree reconstruction than sparse structures.

We consider the application of these results in the context of historical linguistics. Häuser et al. (2026) notes the current limitations of importing Bayesian phylogenetic analysis into historical linguistics - one of which being that high-quality cognate datasets are scarce and small and thus are unsatisfactory for phylogenetic analysis [71]. While this is an issue already being resolved through the generation of larger, standardised datasets for language families like the Indo-European language family [72], lack of data plagues less-documented language families in tree reconstruction. The results of the simulation study in this thesis can thus be applied in these respective scenarios, especially where there exist sites based on non-language cultural factors for language families to incorporate.

One such application of these results can be found in incorporating loom data into phylolinguistic analysis, as performed by Buckley et al. (2025) [9]. The paper compared the tree reconstruction ability of the Bayesian phylogenetic model on (1) the language data set, which comprised 653 sites, against that of (2) the loom data set, which comprised 216 sites which had a known dependency structure, similar to what has been described in Section 2.3, with only 36 independent sites. The data was collated across 29 language-loom pairs and the consensus trees of both data sets were compared. In the case of the language data set, the findings of Experiment 1 suggest that the potential violations of conditional site independence are not significant to tree reconstruction, although some divergences of clades can be explained via model misspecification. In the case of the loom data set, the inclusion of higher tiers (or levels) of data may be justified from the results obtained in this thesis: having only 36 independent sites suggests that correlated data is still able to improve clade recovery and tree reconstruction.

Chapter 6

Conclusion

6.1 Summary of Findings

This thesis investigated the effects of violating conditional site independence in Bayesian phylogenetic inference and evaluated whether such data can still improve tree reconstruction accuracy. These considerations were motivated from historical linguistics, noting the main challenges of applying such methods to linguistics data.

In particular, two questions were posed as found in Section 1.4. The first question considered how the proportion of data that contained dependency structures affected tree reconstruction accuracy. The results obtained from Chapters 4 and 5 show that any misspecification of the data consistently reduced tree reconstruction accuracy, as opposed to using independently generated sites fully in the data generation method. Increasing depth of dependency had minimal additional deterioration. In fact, the main source of reduction was mainly dependent on the amount of information available within the dataset.

The second question examined if dependent data should be retained, even where its dependency structure is known. The results obtained indicate that such data can still provide considerable benefits where independent data is scarce as the marginal benefit associated with phylogenetic signal still outweigh the negative effects of model misspecification. The contrary holds in data rich scenarios, where additional signal has diminished benefit that may not justify its inclusion into the data. Similar results hold for adding sites with deeper dependency structures, although these provide less benefit than lower tier sites.

When taken together, the results suggest that violations of conditional site independence, while introducing bias to the model, can also contribute additional information about the underlying true tree T^* . As such, the tradeoff between these competing effects, often times influenced by the total effective information of data, determines whether their inclusion is ultimately beneficial or detrimental to tree reconstruction and inference.

6.2 Contributions and Implications

The main contribution of this thesis is the identification of effective information as a heuristic for understanding the implications of violating conditional site independence in Bayesian phylolinguistics. The simulation results repeatedly demonstrate the influence of effective information obtained from the data, not only on reconstruction quality but in determining whether conditional sites are valuable.

This perspective reconciles the main findings of the study, harmonising the seemingly contradictory results between Experiments 1 and 2. In Experiment 1, dependency reduces effective information contained within a fixed dataset, which thus lowers reconstruction accuracy. In Experiment 2, introducing dependency in additional data still increases effective information (albeit smaller than additional independent data) and thus improves reconstruction accuracy. As such, the problem of model misspecification on the conditional site independence assumption is largely simplified into a heuristic of considering whether the data provides adequate effective information for tree reconstruction on Bayesian phylogenetic methods.

In the context of phylolinguistics, these results provide guidance for treatment of datasets that are suspected to have dependency structures. In highly-documented language families such as the Indo-European language family, there is no real need to include sites that violate conditional site independence as the independent signal already provides enough information. Conversely, in poorly documented language families, these sites are still valuable and should thus be retained to improve reconstruction quality, even if modelling assumptions are not met. These results were communicated to Buckley et al. (2025) who worked with loom data on Kra-dai language speakers [9], who similarly noted that these results could support the methods used in their analysis. Additionally, the team remarked that these results could also have relevance in linguistic phylogenies, especially when dealing with morphological features that do not satisfy conditional site independence.

6.3 Limitations and Future Work

We note several limitations of the simulation study. Firstly, the simulation study idealised several considerations related to phylolinguistic analysis. For example, rate variation (Section 2.1.3) was assumed to be constant, purely binary data matrices were considered instead of matrices that may include missing data, and sites of differing tiers assumed the same parameters on each evolutionary process. The study also focussed exclusively on a singular tree prior and evolutionary process. However, real linguistic datasets exhibit more complex behaviour which are not reflected in this study. Consequently, the quantitative results presented should not be interpreted as universal estimates of the effects of violating the conditional site independence assumption. Future work could extend the ideas portrayed in this study by implementing relevant complexities that better represent linguistic data, perhaps implementing such models to datasets of languages with known phylogenies. These implementations would help assess the practical relevance of the observed tradeoffs and how these findings generalise beyond simulated settings.

Next, the study focussed solely on the consequences of dependency using MCMC which assumes conditional site independence. Alternative modelling approaches that allow for dependency structures within sites were not considered. Future work could look into whether explicitly accounting for dependency indeed improves tree reconstruction quality, as opposed to the heuristics suggested in this thesis.

Finally, effective information was presented primarily as an interpretative framework, rather than a formally defined quantity. Developing a rigorous theoretical measure of effective information under different dependency structures, perhaps by quantifying empirical claims with additional theoretical results in information theory, would provide a stronger mathematical foundation for the findings presented. Such a framework may allow for quantitative decision-making on including dependent sites for inference.

Ultimately, this thesis shows that the consequences of violating conditional site independence on reconstruction quality are more nuanced than simply removing invalid data. While

these dependencies within the data introduce bias via incorrect likelihood calculations, they may still provide valuable phylogenetic information. As such, the effectiveness of including such data depends on the balance between these competing forces. As unknown dependencies between sites are often unavoidable in historical linguistics, understanding the tradeoffs is crucial for tree reconstruction of language evolution, which in turn provide a window into human history.

Appendix A

Additional Figures

A.1 Swadesh List

This table provides the list of meanings that Swadesh (1952) used in quantifying the age of the most recent common ancestor of languages [11], now known as the Swadesh list.

No.	Meaning	No.	Meaning	No.	Meaning	No.	Meaning	No.	Meaning
1	all	41	fall	81	in	121	root	161	that
2	and	42	far	82	kill	122	rope	162	there
3	animal	43	fat	83	know	123	rotten	163	they
4	ashes	44	father	84	lake	124	rub	164	thick
5	at	45	fear	85	laugh	125	salt	165	thin
6	back	46	feather	86	leaf	126	sand	166	think
7	bad	47	few	87	left	127	say	167	this
8	bark	48	fight	88	leg	128	scratch	168	thou
9	because	49	fire	89	lie	129	sea	169	three
10	belly	50	fish	90	live	130	see	170	throw
11	berry	51	five	91	liver	131	seed	171	tie
12	big	52	float	92	long	132	sew	172	tongue
13	bird	53	flow	93	louse	133	sharp	173	tooth
14	bite	54	flower	94	man	134	short	174	tree
15	black	55	fly	95	many	135	sing	175	turn
16	blood	56	fog	96	meat	136	sit	176	two
17	blow	57	foot	97	mother	137	skin	177	vomit
18	bone	58	four	98	mountain	138	sky	178	walk
19	breathe	59	freeze	99	mouth	139	sleep	179	warm
20	burn	60	give	100	name	140	small	180	wash
21	child	61	good	101	narrow	141	smell	181	water
22	cloud	62	grass	102	near	142	smoke	182	we
23	cold	63	green	103	neck	143	smooth	183	wet
24	come	64	guts	104	new	144	snake	184	what
25	count	65	hair	105	night	145	snow	185	when
26	cut	66	hand	106	nose	146	some	186	where
27	day	67	he	107	not	147	spit	187	white
28	die	68	head	108	old	148	split	188	who
29	dig	69	hear	109	one	149	squeeze	189	wide
30	dirty	70	heart	110	other	150	stab	190	wife
31	dog	71	heavy	111	person	151	stand	191	wind
32	drink	72	here	112	play	152	star	192	wing
33	dry	73	hit	113	pull	153	stick	193	wipe
34	dull	74	hold	114	push	154	stone	194	with
35	dust	75	how	115	rain	155	straight	195	woman
36	ear	76	hunt	116	red	156	suck	196	woods
37	earth	77	husband	117	right (correct)	157	sun	197	worm
38	eat	78	I	118	right (hand)	158	swell	198	ye
39	egg	79	ice	119	river	159	swim	199	year
40	eye	80	if	120	road	160	tail	200	yellow

Table A.1 Swadesh list of meanings, conveniently expressed in English [11].

A.2 Full Results of Simulation Study

Here we present the full results of the simulation study, encapsulating the summarised findings of Chapter 4. We note that most findings and discussions are similar to that of Chapters 4 and 5.

We note that throughout both experiments, there were runs that had limited tree ESS (at times only providing 1 effective sample). This likely suggests that the model was unable to propose efficiently and indicates poor mixing. As reliable inference assumes sufficient convergence of the Markov chains to the posterior distribution, we exclude these cases in the analysis provided in Chapters 4 and 5.

A.2.1 Experiment 1

Here we present the full results of Experiment 1. Figures A.1 and A.2 describe how the respective metrics are affected where a proportion of the data $\mathbf{D}$ have dependency structures. Each of the metrics are also tested on the consensus and MCC trees of each run as found in Figures A.3, A.4, A.5 and A.6.

A.2.2 Experiment 2

Similarly to Experiment 1, the full results of Experiment 2 are presented here. Figures A.7 and A.8 describe the effect on metrics when dependent data are added to independent data. The metrics are also tested on the consensus and MCC trees for each BEAST run as found in Figures A.9, A.10, A.11 and A.12.

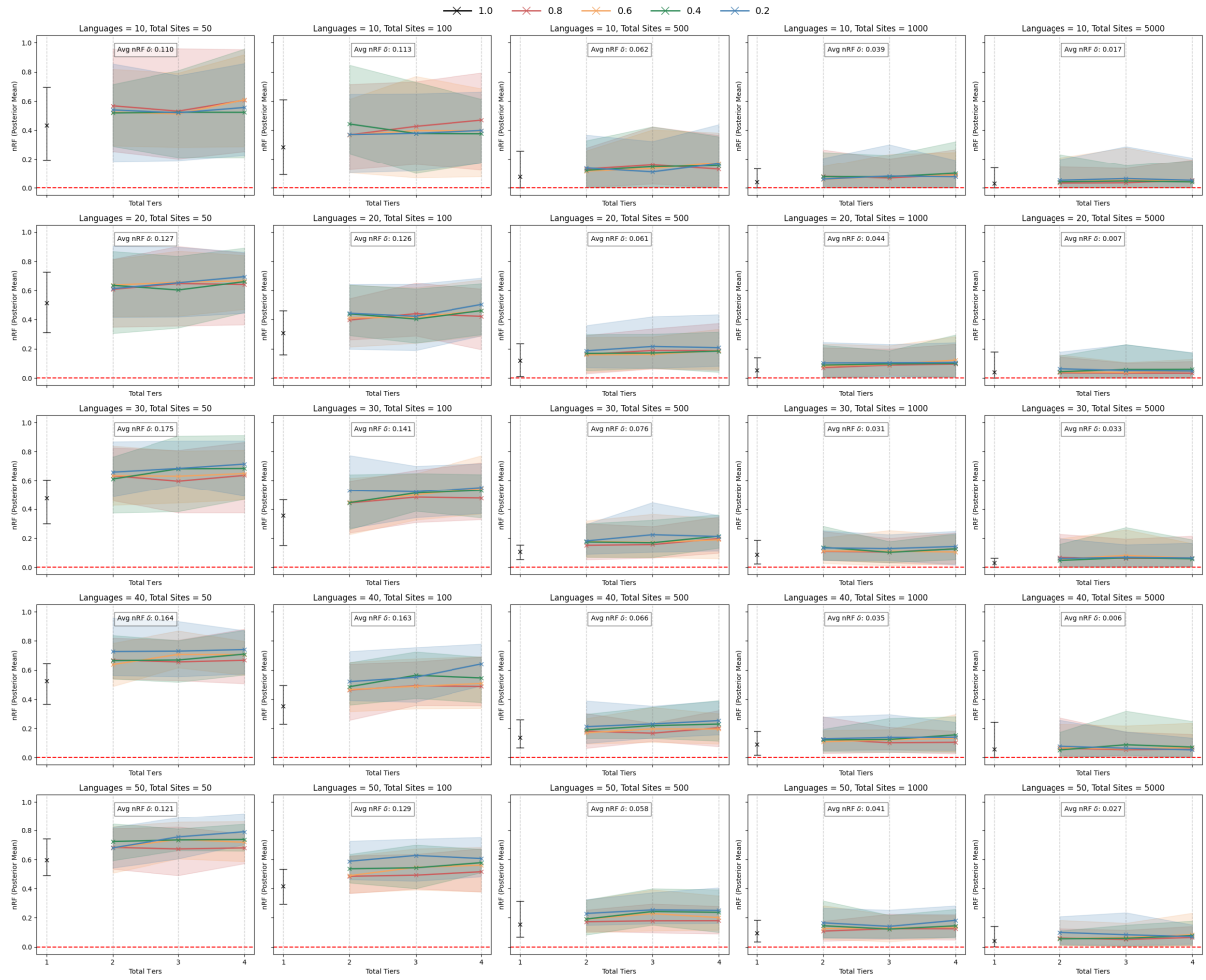


Figure A.1 nRF of posterior sample against depth of dependencies within each subplot, varying proportion of independent data from 20% (blue) to 80% (red).

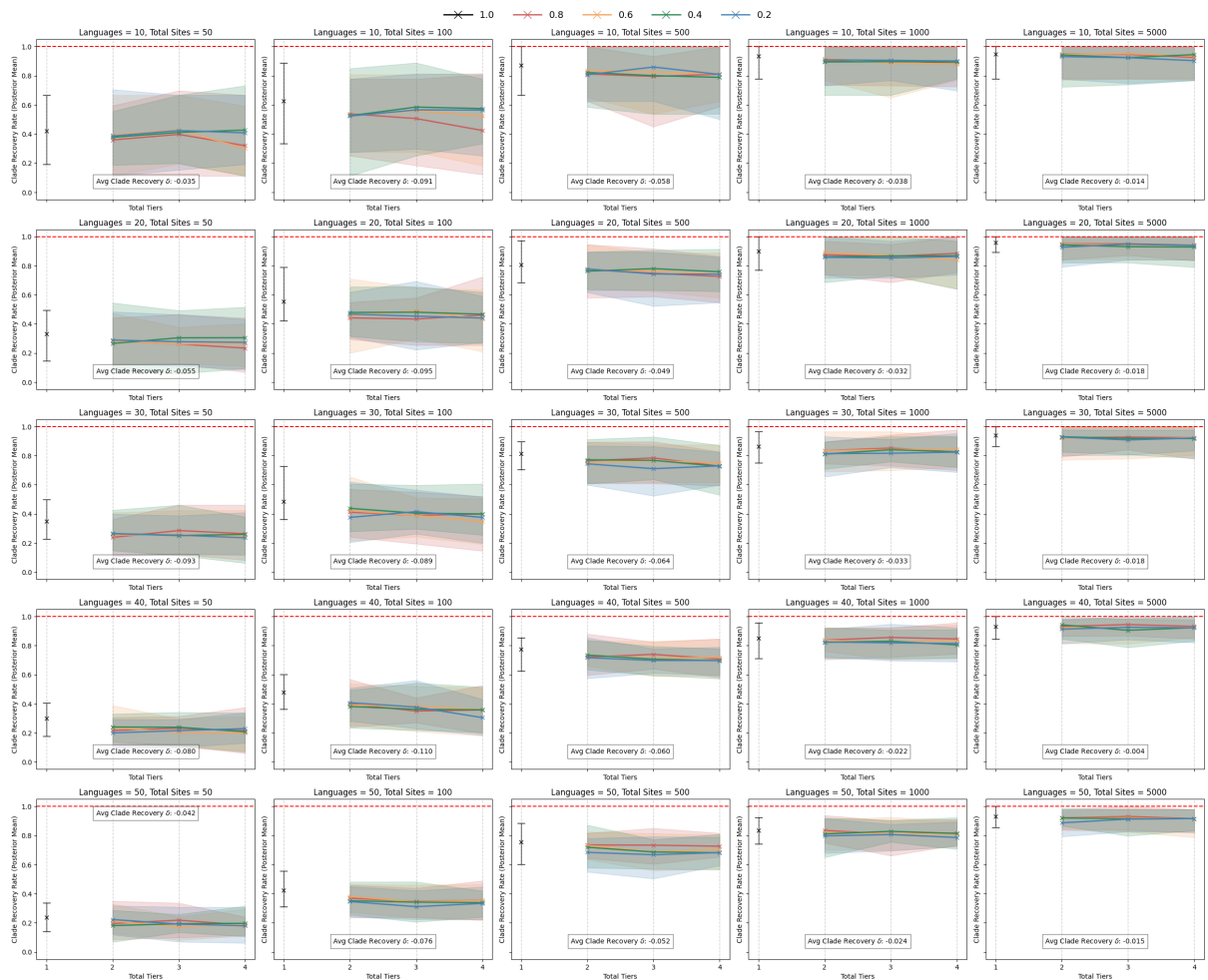


Figure A.2 High Support Clade Recovery Rate of posterior sample against depth of dependencies within each subplot, varying proportion of independent data from 20% (blue) to 80% (red).

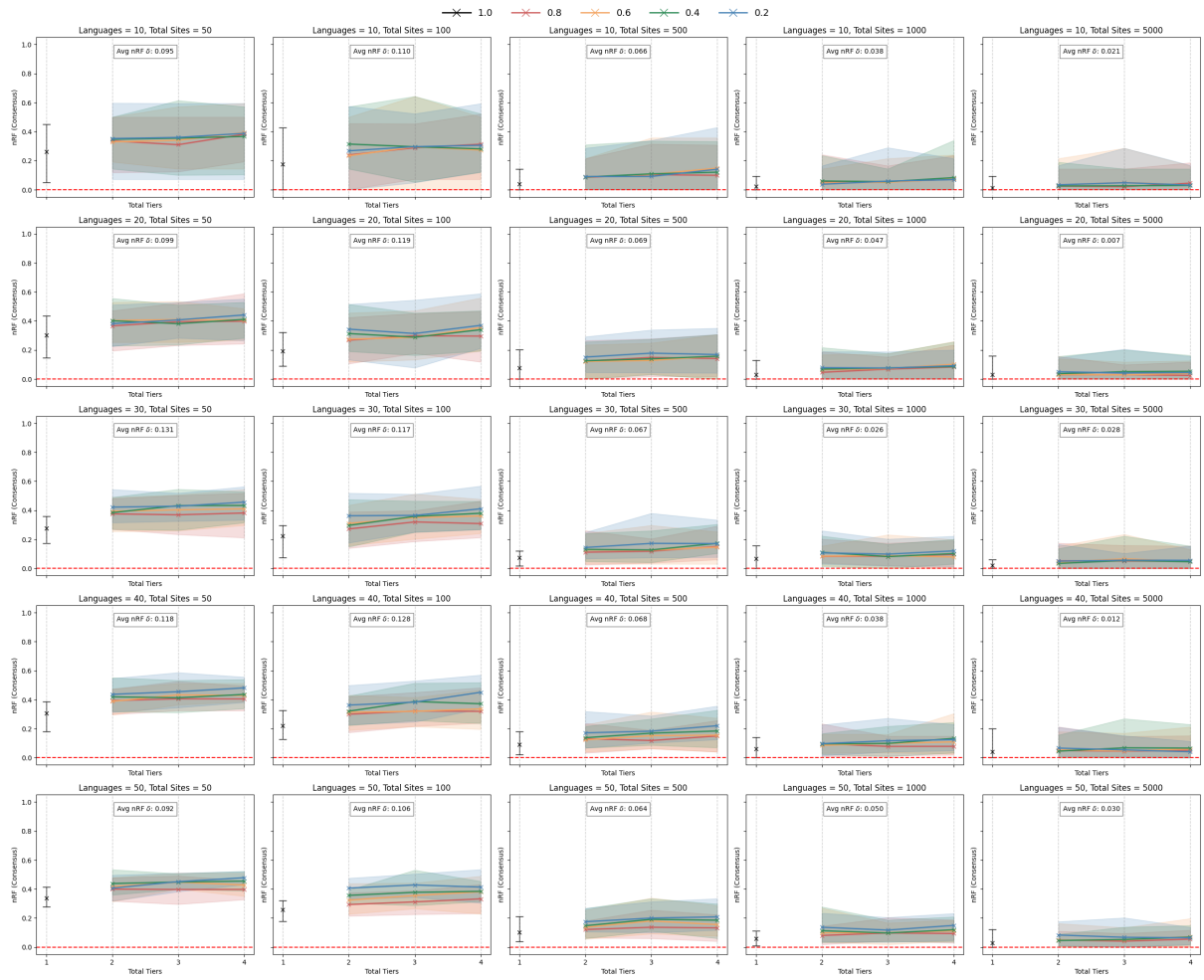


Figure A.3 nRF of MRC Tree against depth of dependencies within each subplot, varying proportion of independent data from 20% (blue) to 80% (red).

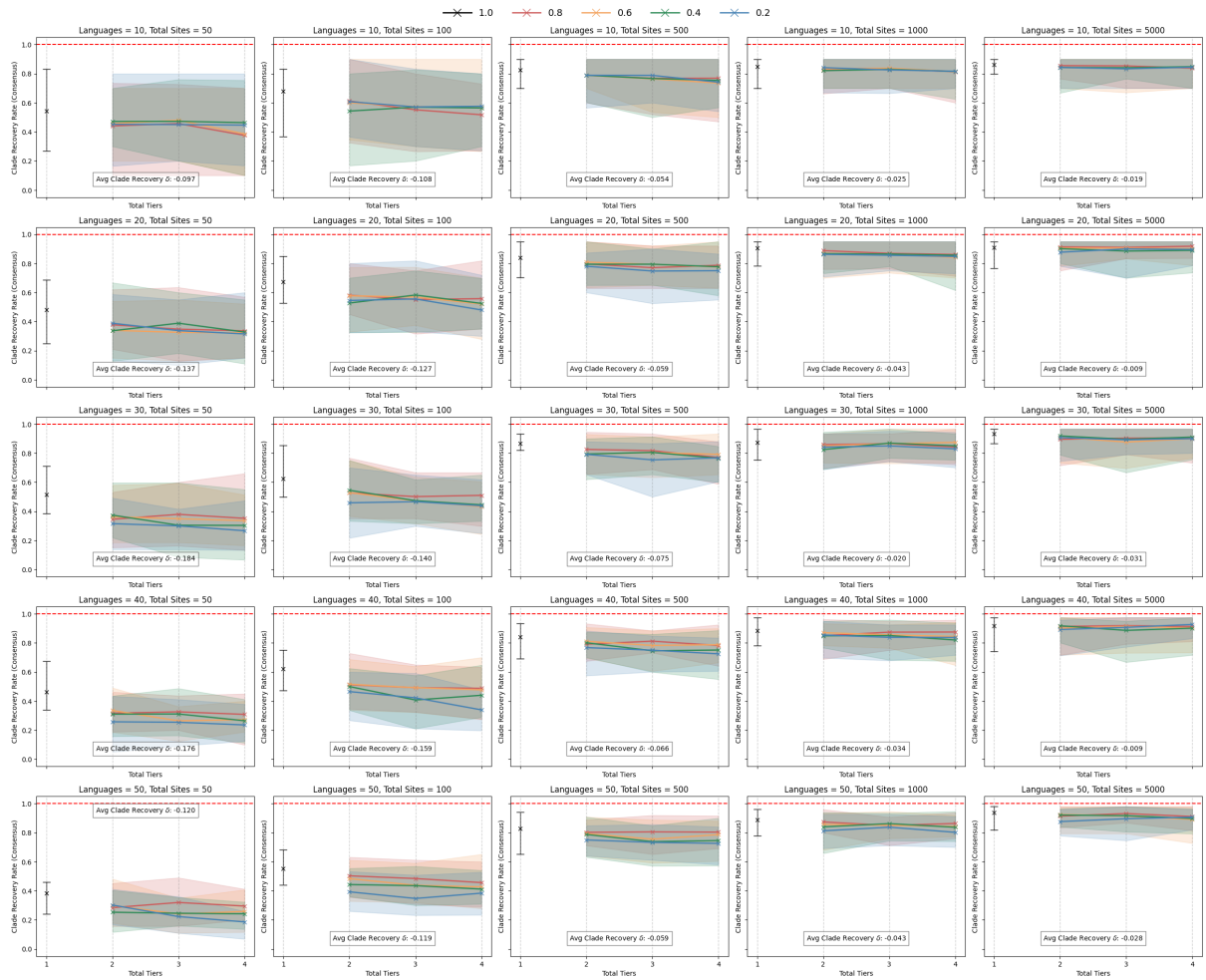


Figure A.4 Clade Recovery Rate of MRC Tree against depth of dependencies within each sub-plot, varying proportion of independent data from 20% (blue) to 80% (red).

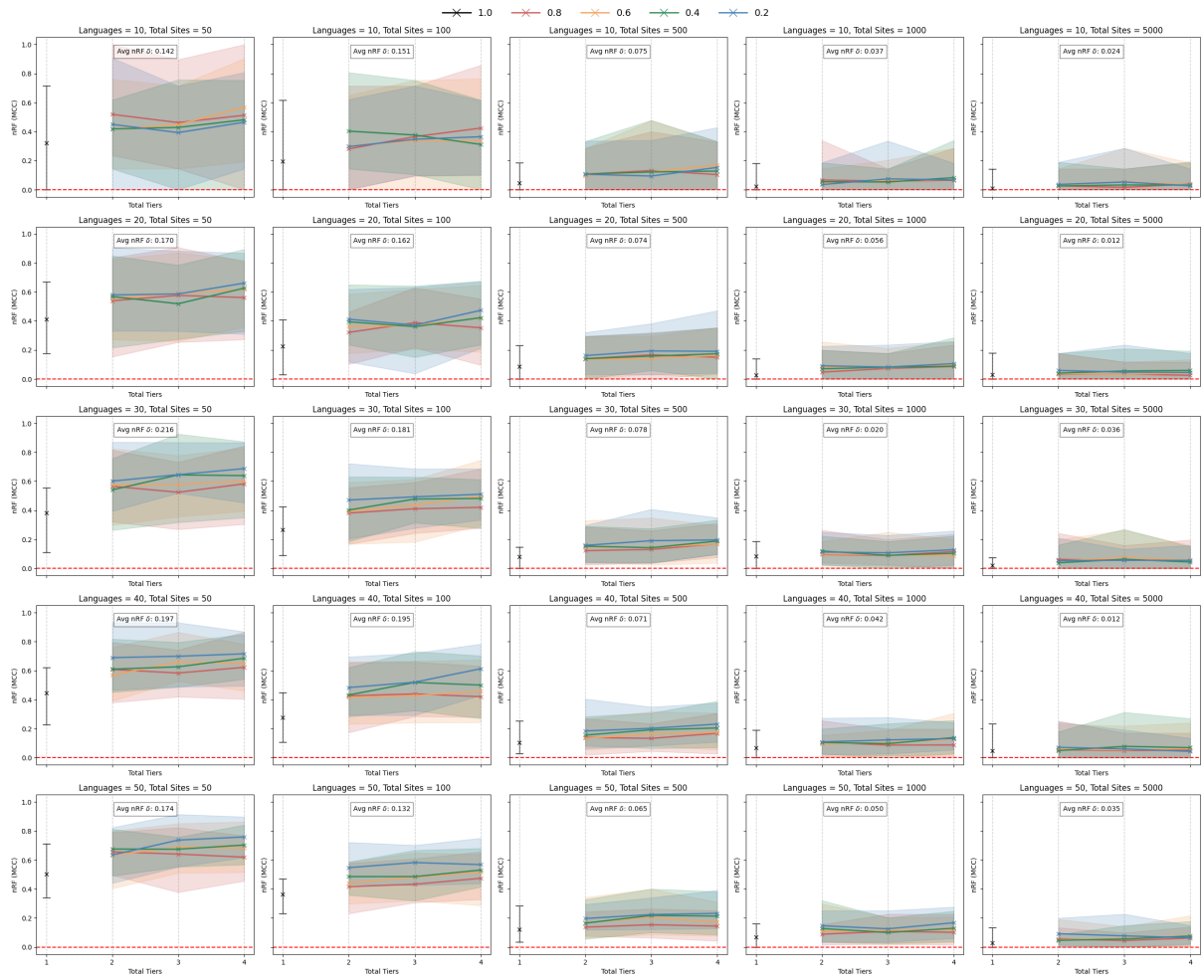


Figure A.5 nRF of MCC Tree against depth of dependencies within each subplot, varying proportion of independent data from 20% (blue) to 80% (red).

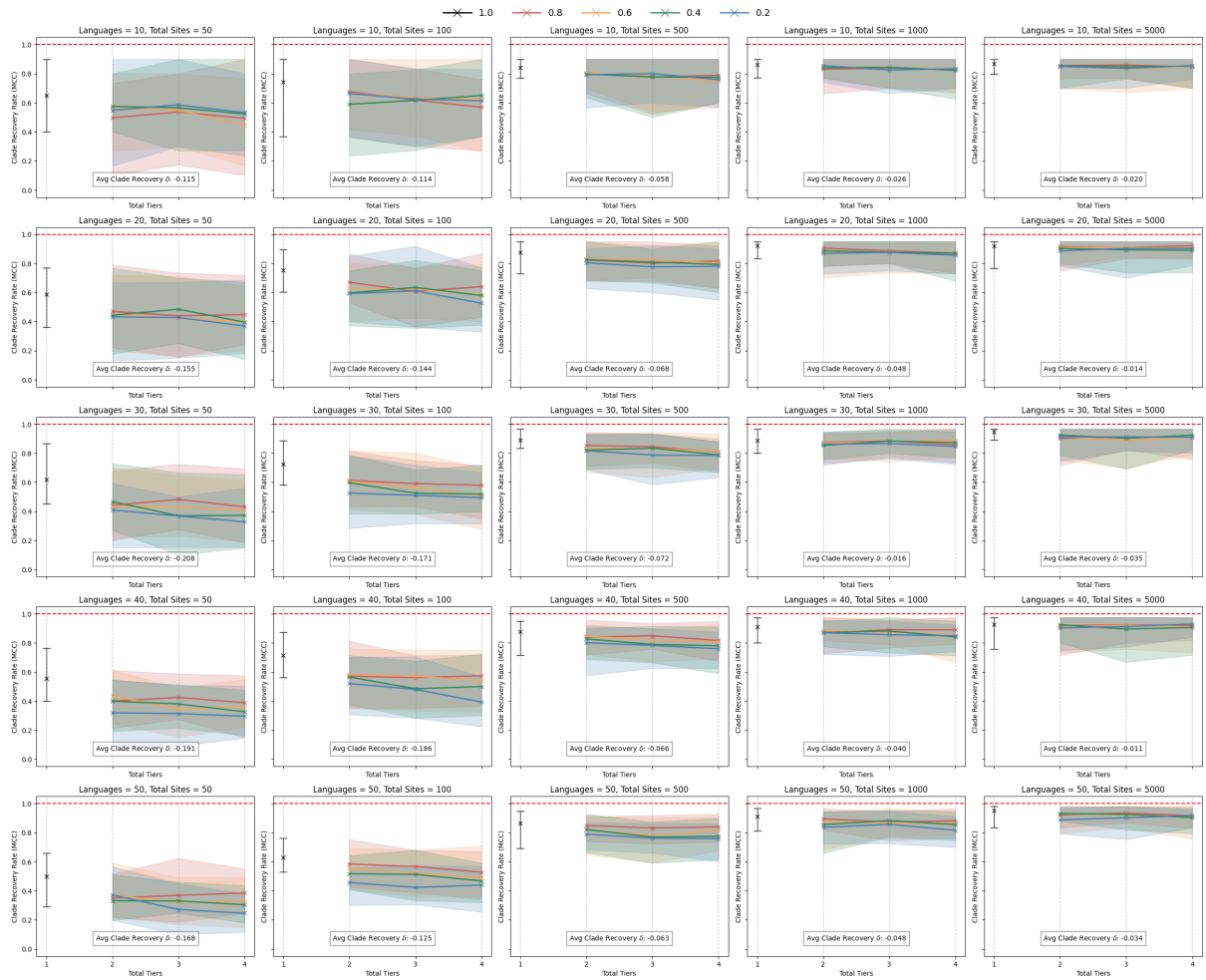


Figure A.6 Clade Recovery Rate of MCC Tree against depth of dependencies within each sub-plot, varying proportion of independent data from 20% (blue) to 80% (red).

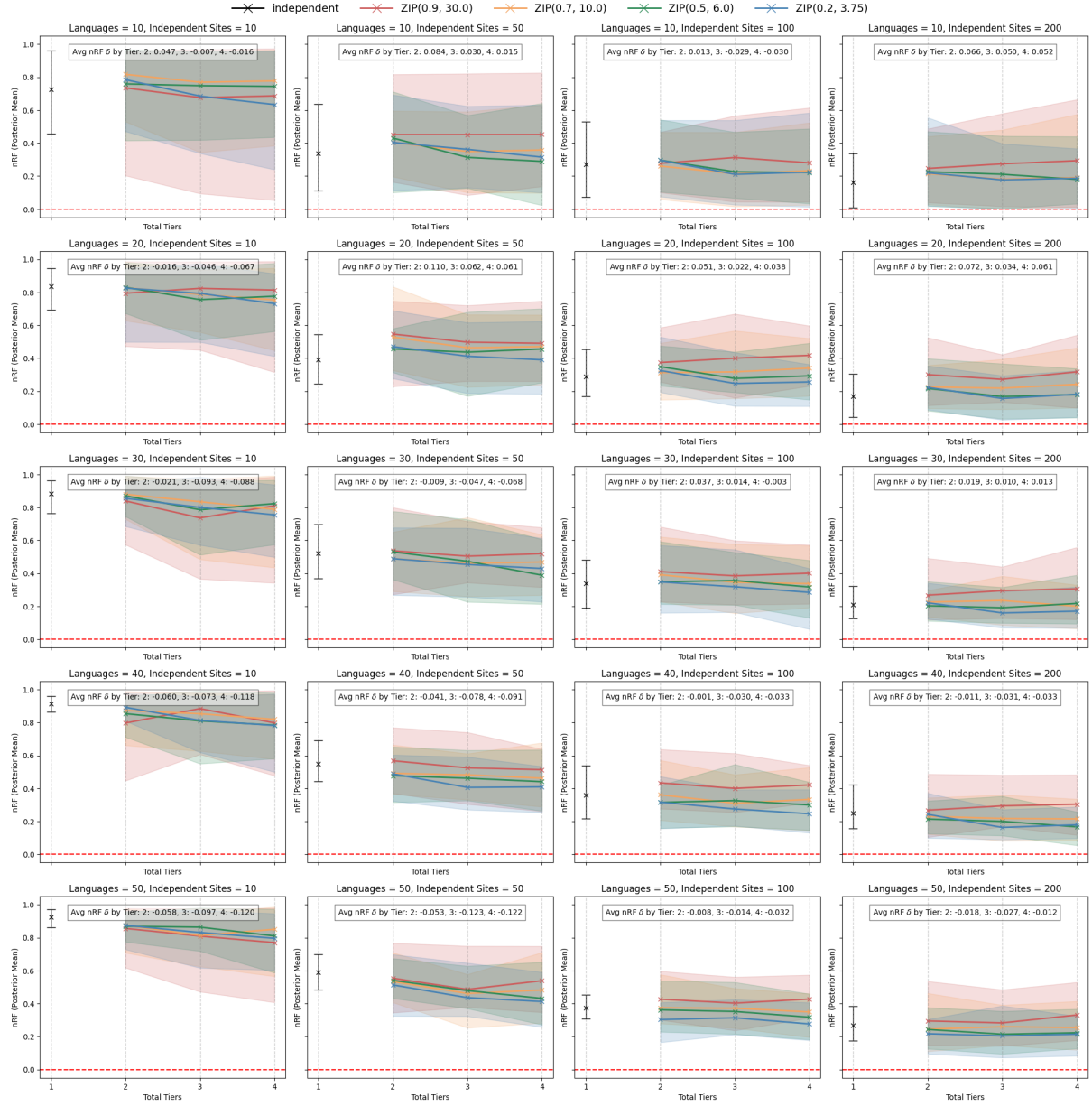


Figure A.7 nRF of posterior sample against depth of dependency structure within each sub-plot, varying method of generation from most sparse (red) to most dense (blue).

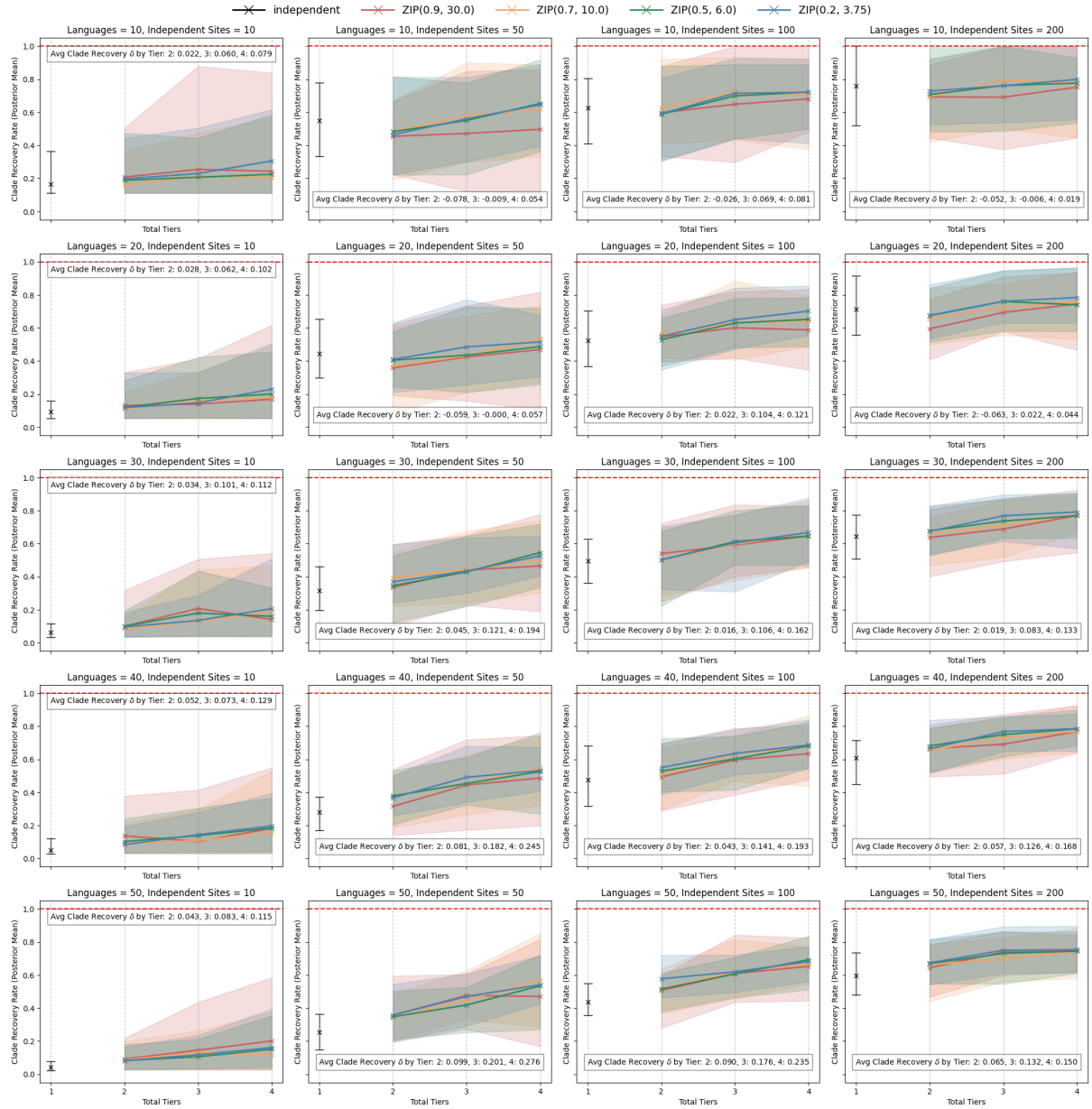


Figure A.8 High Support Clade Recovery rate of posterior sample against depth of dependency structure within each subplot, varying method of generation from most sparse (red) to most dense (blue).

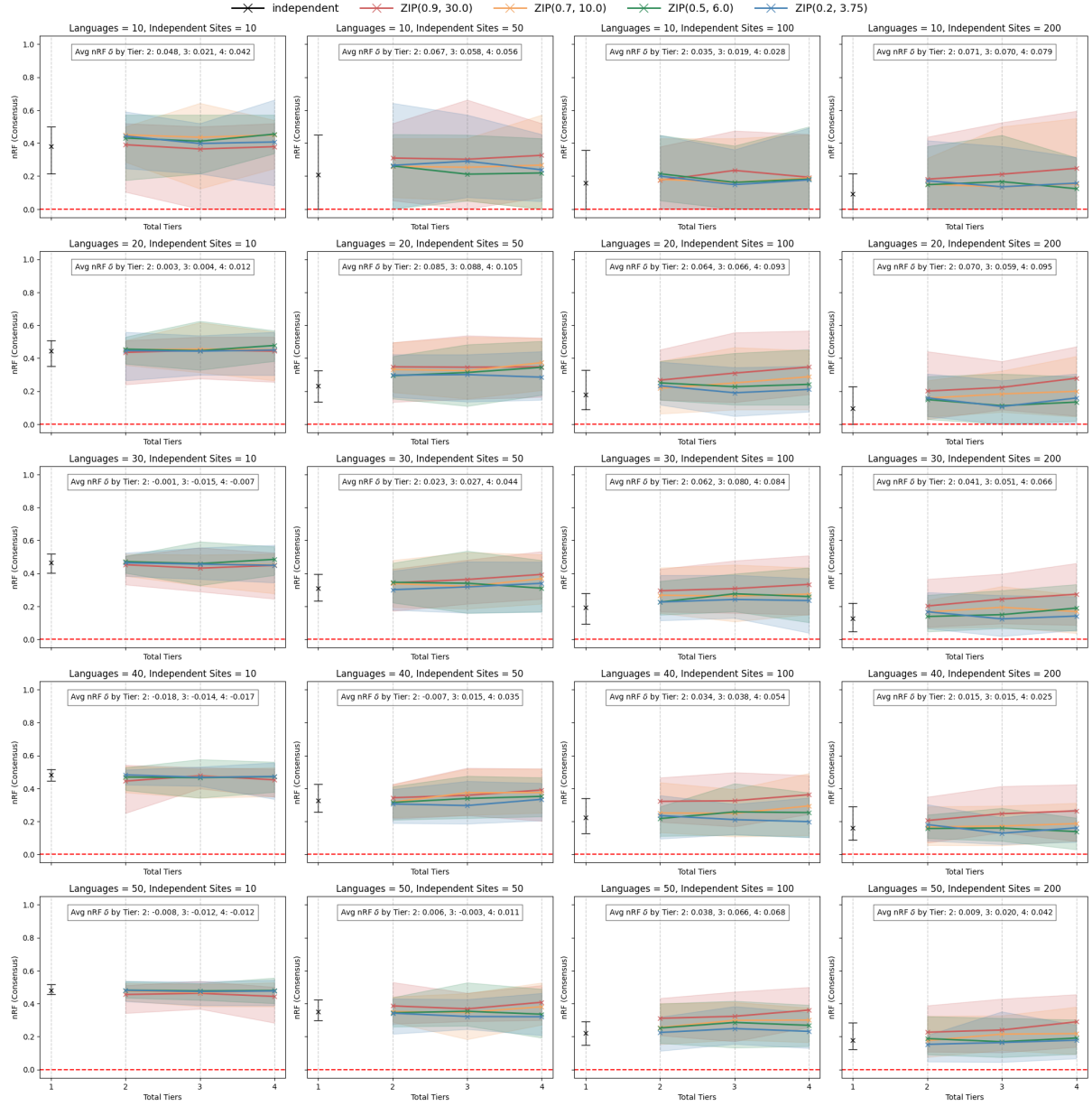


Figure A.9 nRF of MRC Tree against depth of dependency structure within each subplot, varying method of generation from most sparse (red) to most dense (blue).

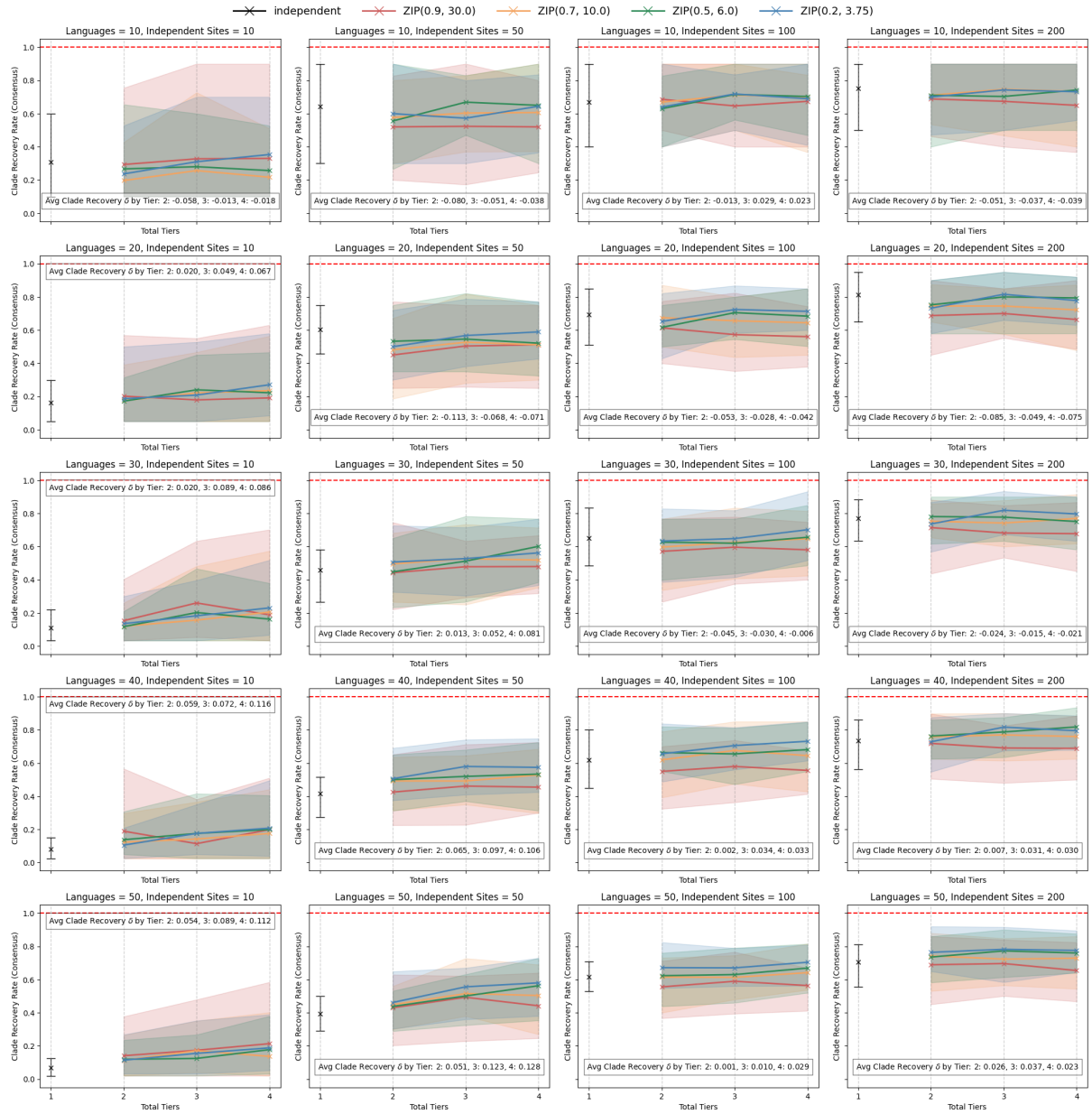


Figure A.10 Clade Recovery Rate of MRC Tree against depth of dependency structure within each subplot, varying method of generation from most sparse (red) to most dense (blue).

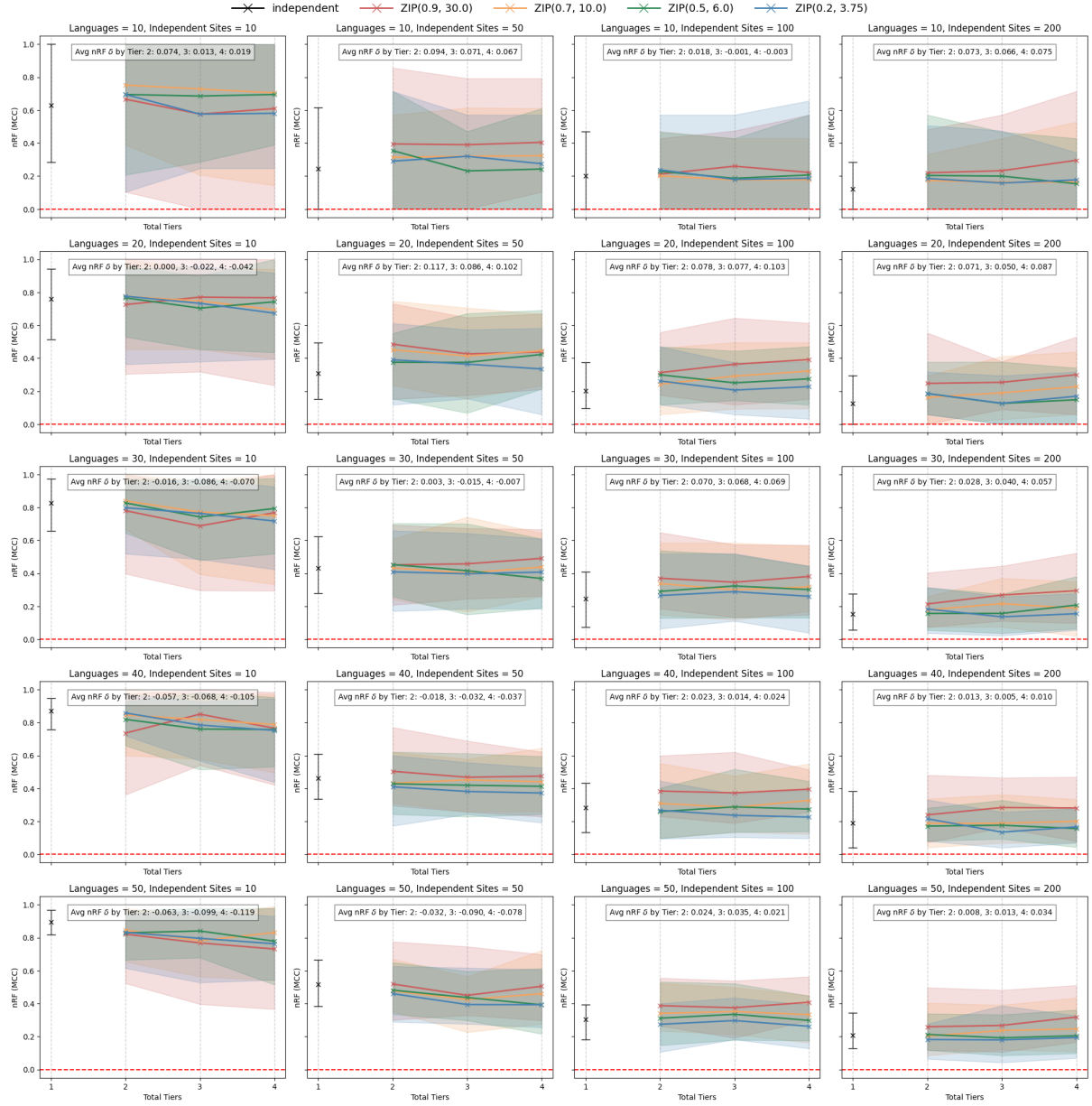


Figure A.11 nRF of MCC Tree against depth of dependency structure within each subplot, varying method of generation from most sparse (red) to most dense (blue).

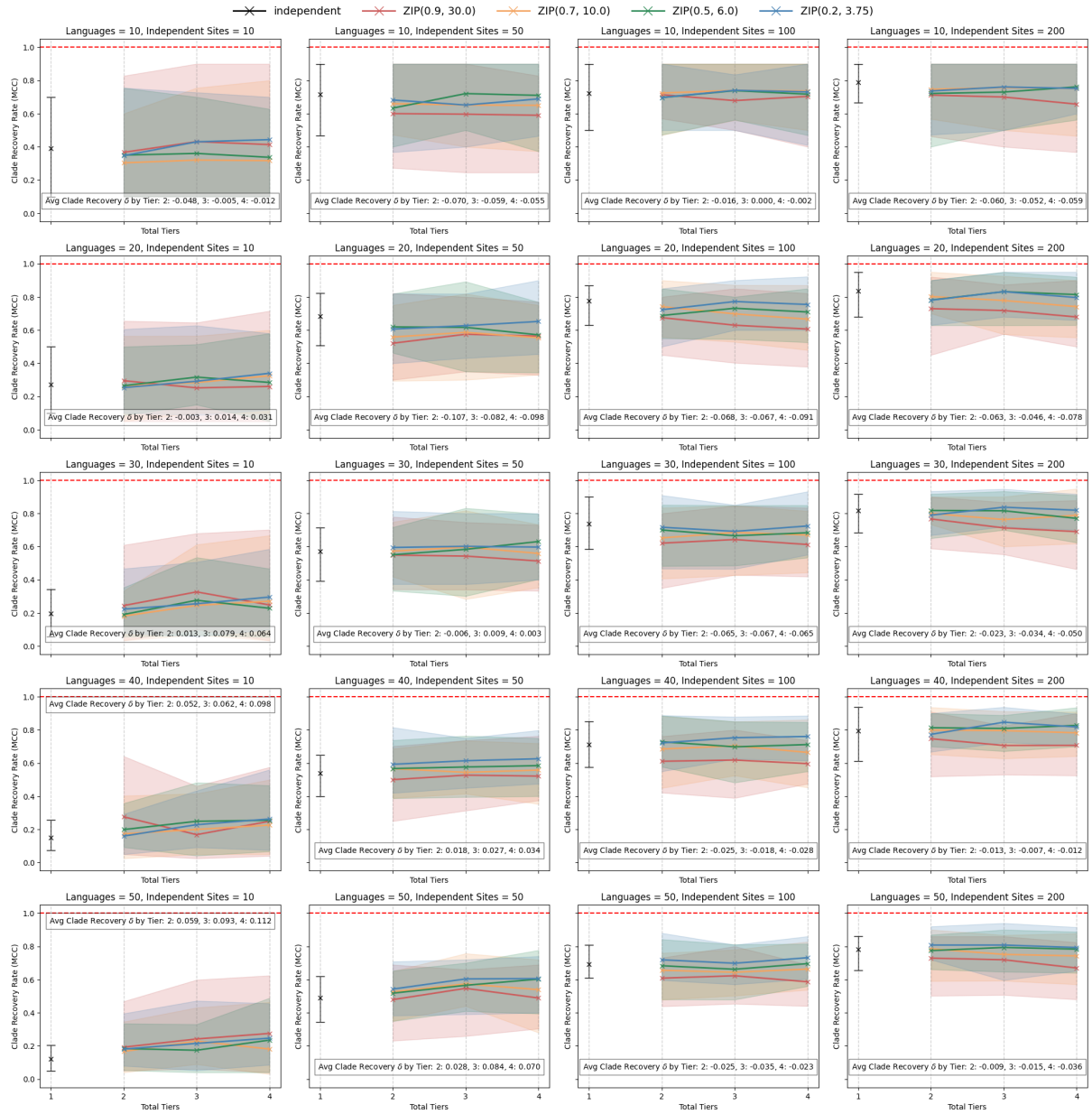


Figure A.12 Clade Recovery Rate of MCC Tree against method of dependency structure within each subplot, varying method of generation from most sparse (red) to most dense (blue).

Appendix B

Additional Mathematical Derivations

In this appendix, we consider mathematical derivations of results used throughout the thesis. These were derivations that were not relevant to the main thrust of considering the violation of conditional site-independence in models, and thus were omitted from the main text and included here.

B.1 Stationary Distribution of CTMC

Here we prove that the binary CTMC found in Figure 2.1 has stationary distribution

$$\pi = (\pi_0, \pi_1) = \left(\frac{q_{10}}{q_{01} + q_{10}}, \frac{q_{01}}{q_{01} + q_{10}} \right). \quad (\text{B.1})$$

We start by considering $\mathbf{G}$. We note that any stationary distribution π satisfies $\pi \mathbf{G} = \mathbf{0}$. Clearly, this is true of (B.1):

$$\pi \mathbf{G} = \left(\frac{q_{10}}{q_{01} + q_{10}}, \frac{q_{01}}{q_{01} + q_{10}} \right) \begin{pmatrix} -q_{01} & q_{01} \\ q_{10} & -q_{10} \end{pmatrix} \quad (\text{B.2})$$

$$= \left(\frac{-q_{10}q_{01} + q_{01}q_{10}}{q_{01} + q_{10}}, \frac{q_{10}q_{01} - q_{01}q_{10}}{q_{01}q_{10}} \right) = \mathbf{0}. \quad (\text{B.3})$$

Thus π is a stationary distribution of the CTMC with generator $\mathbf{G}$. We now consider the uniqueness of π . Clearly both states are connected, making the Markov chain generated from the CTMC irreducible. Since the state space is finite, we have that the CTMC must admit a unique stationary distribution [54]. This justification is also used for Proposition 3 to show that $\tilde{\pi}$ is also stationary.

B.2 Markov Property for Stochastic Process $\mathbf{D}_{l,k}$

We now turn to consider the fact that $\mathbf{D}_{l,k}$, the resulting binary encoding of the element-wise product of dependency $\mathbf{D}_{l-1,j}$ and simulated CTMC $\tilde{\mathbf{D}}_{l,k}$ as considered by (2.52) in Section 2.3.2, is not in fact a CTMC, because it does not satisfy the Markov property.

We define a partition on state space $\mathcal{S}$:

$$\mathcal{P} = \{\{00, 01, 10\}, \{11\}\} \quad (\text{B.4})$$

This is a clear case of reducing the size of the state space. We employ the idea of *Markov lumpability* [73]:

Definition 15 (Lumpability). Suppose $(X_t)_{t \geq 0}$ is a CTMC with finite state space $\mathcal{S}$ and generator matrix $\mathbf{G} = (g_{xy})_{x,y \in \mathcal{S}}$, and $\mathcal{P} := \{P_1, \dots, P_m\}$ is a partition of $\mathcal{S}$, then aggregated process $Y_t = i$ where $X_t \in P_i$, then we say that the CTMC is lumpable with respect to partition $\mathcal{P}$ if for every pair of blocks P_i, P_j , $\forall x, x' \in P_i$:

$$\sum_{y \in P_j} g_{xy} = \sum_{y \in P_j} g_{x'y}. \quad (\text{B.5})$$

This condition can be interpreted as ensuring that the jump probability from block P_i to block P_j is constant, regardless of the state within P_i .

Clearly, this is an important property the Markov chain must have in order for the reduced state space stochastic process to be considered a Markov chain as well. We state the formal result without proof (of which can be found in [73]):

Proposition 6 (Resulting process of lumped Markov chain is Markovian). Suppose $(X_t)_{t \geq 0}$ is a CTMC with finite state space $\mathcal{S}$ and generator matrix $\mathbf{G} = (g_{xy})_{x,y \in \mathcal{S}}$, and $\mathcal{P} := \{P_1, \dots, P_m\}$ is a partition of $\mathcal{S}$. Then the resulting aggregated process $(Y_t)_{t \geq 0}$ is also a CTMC if and only if the CTMC is lumpable.

As such, in order to prove that $\mathbf{D}_{l,k}$ does not satisfy the Markov property, we need only show that $\mathbf{B}_{l,k}$ is not lumpable with respect to the partition (B.4). This is true, as the generator $\mathbf{G}_{\text{prod}}$ as defined in (2.53) shows: we note that both $\{01, 10\}$ have an equivalent jump probability to $\{11\}$, but that is not the case with state $\{00\}$, which has a zero transition probability to $\{11\}$. As such, from Proposition 6, we can conclude that the lumped chain $\mathbf{D}_{l,k}$ with respect to the partition $\mathcal{P}$ defined in B.4 indeed does not follow the Markov property.

Appendix C

Simulation Parameters

We consider the parameters used for the simulation study put forward in Chapter 3. As mentioned prior, all simulations were conducted on a High Performance Computing (HPC) cluster on Imperial College London’s [Research Computing Service](#). Once again, all parameters and simulations can be found in the [GitHub repository](#) for the thesis.

C.1 Experiment 1

The parameters used in Experiment 1 can be found in Table C.1. We note that the primary method of analysing effect of proportion of independent sites on tree reconstruction is found in varying $q \in \mathcal{Q}$. The values $l \in \mathcal{L}$ determine the number of dependency tiers used in the hierarchical generation process, allowing for analysis between dependency depths. Finally, while not significant to the experiment, the ZIP parameters $(p, \lambda) = (0.1, 45)$ were a design choice made to ensure the amount of data generated would be sufficient after ascertainment.

Row	Parameter	Usage	Value(s)
1	$\mathcal{N}$	Total number of languages in tree	{10, 20, 30, 40, 50}
2	$\mathcal{S}$	Total number of sites	{50, 100, 500, 1000, 5000}
3	N_T	True trees per combination of above	30
4	$\mathcal{Q}$	Proportion of independently generated sites	{0.2, 0.4, 0.6, 0.8}
5	$\mathcal{L}$	Total dependency tiers considered	{2, 3, 4}
6	$\mathcal{W}$	ZIP parameters (p, λ) for dependent data*	(0.1, 45)
7	q_{01}	Gain rate of site	0.04
8	q_{10}	Loss rate of site	0.06
9	λ	Birth intensity of Yule Tree Prior	2.0

Table C.1 Parameter settings for Experiment 1.

Based on rows 4 and 5, and including the independent dataset, for each combination of $n \in \mathcal{N}$ (row 1), $s \in \mathcal{S}$ (row 2) and a given true tree T^* , there were 13 datasets generated, of which all were run using the pipeline described in Chapter 3. As such, a total of 9750 MCMC runs were conducted for Experiment 1.

C.2 Experiment 2

The parameters used in Experiment 2 can be found in Table C.2. Unlike Experiment 1, only the number of independently generated sites $N_1 \in \mathbb{U}$ are fixed, and dependency structures are varied across the family of ZIP random variables. One can empirically check that $E(W) = 3$ is constant for $w \in \mathcal{W}$, which allowed for the total amount of dependent data generated to be comparable across various settings. This allows analysis to be conducted primarily on the dependency structure of the data generated rather than having to reconcile difference in expected number of dependent sites.

Row	Parameter	Usage	Value(s)
1	$\mathcal{N}$	Total number of languages in tree	$\{10, 20, 30, 40, 50\}$
2	$\mathbb{U}$	Total number of independent sites N_1	$\{10, 50, 100, 200\}$
3	N_T	True trees per combination of above sets	30
4	$\mathcal{L}$	Total dependency tiers considered	$\{2, 3, 4\}$
5	$\mathcal{W}$	ZIP parameters (p, λ) for dependent data	$\{(0.2, 3.75), (0.5, 6), (0.7, 10), (0.9, 30)\}$
6	q_{01}	Gain rate of site	0.04
7	q_{10}	Loss rate of site	0.06
8	λ	Birth intensity of Yule Tree Prior	2.0

Table C.2 Parameter settings for Experiment 2.

Once again, based on rows 4 and 5, and including the independent dataset, for each combination of $n \in \mathcal{N}$ (row 1), $N_1 \in \mathbb{U}$ (row 2) and a given true tree T^* , there were 13 datasets generated, of which all were run using the pipeline described in Chapter 3. As such, a total of 7800 MCMC runs were conducted for Experiment 2.

C.3 MCMC Priors and Parameters

The priors and parameters used in the MCMC runs for both Experiments are found in Table C.3. All datasets were used to reconstruct posterior samples for their corresponding true tree T^* via Bayesian phylogenetic models through BEAST. The log-normal distribution was chosen for the Yule prior. For each dataset, a total of 10^7 iterations were run, only recording every 1000 iterations, with a burn-in percentage of 10%.

Parameter	Setting
Tree prior	Yule process
Prior on birth rate λ	$\log \mathcal{N}(\mu = 1.0, \sigma = 1.5)$, offset $\nu = 0$
MCMC chain length	10,000,000 iterations
Burn-in proportion	10%
Sampling frequency	Every 1000 iterations

Table C.3 Bayesian inference settings used for all analyses.

Bibliography

- [1] E. Szathmáry and J. Smith. The major evolutionary transitions. *Nature*, 374:227–32, 04 1995.
- [2] M. Christiansen and S. Kirby. Language evolution: Consensus and controversies. *Trends in cognitive sciences*, 7:300–307, 08 2003.
- [3] L. Campbell. *Historical Linguistics: An Introduction*. Edinburgh University Press, 2004.
- [4] G. F. S. Eberhard, David M. and A. J. Robinson. *Ethnologue: Languages of the world*. twenty-ninth edition., 2026.
- [5] J. Parsons. *Remains of Japhet*. Printed for the author; and sold by L. Davis and C. Reymers, in Holbourn; J. Whiston, at Boyle’s Head; B. White, at Horace’s Head, Fleet Street; and G. Faulkner, at Dublin, 1767.
- [6] F. Bopp and K. Windischmann. *Über das Conjugationssystem der Sanskritsprache in Vergleichung mit jenem der griechischen, lateinischen, persischen und germanischen Sprache: nebst Episoden des Ramajan und Mahabharat und einigen Abschnitten aus den Vedas*. Andreae, 1816.
- [7] B. Rowe and D. Levine. A concise introduction to linguistics: Fourth edition. *A Concise Introduction to Linguistics: Fourth Edition*, pages 1–413, 07 2015.
- [8] T. Bhattacharya, N. Retzlaff, D. E. Blasi, W. Croft, M. Cysouw, D. Hruschka, I. Maddieson, L. Müller, E. Smith, P. F. Stadler, G. Starostin, and H. Youn. Studying language evolution in the age of big data. *Journal of Language Evolution*, 3(2):94–129, 07 2018.
- [9] C. D. Buckley, E. Kopp, T. Pellard, R. J. Ryder, and G. Jacques. Contrasting modes of cultural evolution: Kra-dai languages and weaving technologies. *Evolutionary Human Sciences*, 7:e35, 2025.
- [10] K. Hoffmann, R. Bouckaert, S. J. Greenhill, and D. Kühnert. Bayesian phylogenetic analysis of linguistic data using BEAST. *Journal of Language Evolution*, 6(2):119–135, Sept. 2021. [_eprint: https://academic.oup.com/jole/article-pdf/6/2/119/41325380/lzab005.pdf](https://academic.oup.com/jole/article-pdf/6/2/119/41325380/lzab005.pdf).
- [11] M. Swadesh. Lexico-statistic dating of prehistoric ethnic contacts: With special reference to north american indians and eskimos. *Proceedings of the American Philosophical Society*, 96(4):452–463, 1952.
- [12] M. Zhang and T. Gong. How many is enough? statistical principles for lexicostatistics. *Frontiers in Psychology*, 7:1916, 2016.
- [13] D. Dediu and M. Cysouw. Some structural aspects of language are more stable than others: A comparison of seven methods. *PLOS ONE*, 8(1):e55009, 2013.

- [14] L. Borin, B. Comrie, and A. Saxena. The intercontinental dictionary series: A rich and principled database for language comparison. In L. Borin and A. Saxena, editors, *Approaches to Measuring Linguistic Differences*, volume 265 of *Trends in Linguistics*, pages 285–302. De Gruyter Mouton, Berlin, 2013.
- [15] T. Pellard, R. Ryder, and G. Jacques. The Family Tree model. In A. Ledgeway, E. Aldridge, A. Breitbarth, K. É. Kiss, J. Salmons, and A. Simonenko, editors, *The Wiley Blackwell companion to diachronic linguistics*. Wiley Blackwell, 2026.
- [16] A. Schleicher. *Compendium der vergleichenden grammatik der indogermanischen sprachen*. Number v. 1 in *Compendium der vergleichenden grammatik der indogermanischen sprachen*. H. Böhlau, 1861.
- [17] T. Pellard. *1. The Linguistic archeology of the Ryukyu Islands*, pages 13–38. De Gruyter Mouton, Berlin, München, Boston, 2015.
- [18] K. Bergsland and H. Vogt. On the validity of glottochronology. *Current Anthropology*, 3: 115 – 153, 1962.
- [19] M. A. Steel, M. D. Hendy, and D. Penny. Loss of information in genetic distances. *Nature*, 336(6195):118, Nov. 1988.
- [20] R. D. Gray and Q. D. Atkinson. Language-tree divergence times support the anatolian theory of indo-european origin. *Nature*, 426(6965):435–439, 2003.
- [21] S. J. Greenhill and R. D. Gray. Austronesian language phylogenies: Myths and misconceptions about bayesian computational methods. In A. Adelaar and A. Pawley, editors, *Austronesian Historical Linguistics and Culture History: A Festschrift for Robert Blust*, pages 375–397. Pacific Linguistics, Canberra, 2009.
- [22] A. Savelyev and M. Robbeets. Bayesian phylolinguistics infers the internal structure and the time-depth of the turkic language family. *Journal of Language Evolution*, 5(1):39–53, 02 2020.
- [23] M. Robbeets and R. Bouckaert. Bayesian phylolinguistics reveals the internal structure of the transeurasian family. *Journal of Language Evolution*, 3(2):145–162, 08 2018.
- [24] R. Grollemund et al. Bantu expansion shows that habitat alters the route and pace of human dispersals. *Proceedings of the National Academy of Sciences*, 112(43):13296–13301, 2015.
- [25] R. R. Bouckaert, C. Bowern, and Q. D. Atkinson. The origin and expansion of pama-nyungan languages across australia. *Nature Ecology & Evolution*, 2(4):741–749, 2018.
- [26] T. Honkola et al. Cultural and climatic changes shape the evolutionary history of the uralic languages. *Journal of Evolutionary Biology*, 26(6):1244–1253, 2013.
- [27] S. Lee. A sketch of language history in the korean peninsula. *PLOS ONE*, 10(5):e0128448, 2015.
- [28] F. Gerardi, T. Tresoldi, C. Aragon, S. Reichert, J. Gregorio de Souza, and F. Noelli. Lexical phylogenetics of the tupí-guaraní family: Language, archaeology, and the problem of chronology. *PLOS ONE*, 18:e0272226, 06 2023.
- [29] A. F. Magee, S. K. Hilton, and W. S. DeWitt. Robustness of phylogenetic inference to model misspecification caused by pairwise epistasis. *Molecular Biology and Evolution*, 38 (10):4603–4615, 10 2021.

- [30] W. P. Maddison. Missing data versus missing characters in phylogenetic analysis. *Systematic Biology*, 42(4):576–581, 1993.
- [31] O. Cohen, H. Ashkenazy, D. Burstein, and T. Pupko. Uncovering the coevolutionary network among prokaryotic genes. *Bioinformatics*, 28(18):i389–i394, 2012.
- [32] M. W. Dimmic, M. J. Hubisz, C. D. Bustamante, and R. Nielsen. Detecting coevolving amino acid sites using bayesian mutational mapping. *Bioinformatics*, 21(Suppl 1):i126–i135, 2005.
- [33] S. Kryazhimskiy, J. Dushoff, G. A. Bazykin, and J. B. Plotkin. Prevalence of epistasis in the evolution of influenza a surface proteins. *PLoS Genetics*, 7(2):e1001301, 2011.
- [34] B. Shapiro, A. Rambaut, O. G. Pybus, and E. C. Holmes. A phylogenetic method for detecting positive epistasis in gene sequences and its application to rna virus evolution. *Molecular Biology and Evolution*, 23(9):1724–1730, 2006.
- [35] X. Meyer, L. Dib, D. Silvestro, and N. Salamin. Simultaneous bayesian inference of phylogeny and molecular coevolution. *Proceedings of the National Academy of Sciences of the United States of America*, 116(11):5027–5036, 2019.
- [36] M. Golden, B. Murrell, D. P. Martin, O. G. Pybus, and J. Hein. Evolutionary analyses of base-pairing interactions in dna and rna secondary structures. *Molecular Biology and Evolution*, 37(2):576–592, 2020.
- [37] C. A. Nasrallah, D. H. Mathews, and J. P. Huelsenbeck. Quantifying the impact of dependent evolution among sites in phylogenetic inference. *Systematic Biology*, 60(1):60–73, 2011.
- [38] A. F. Y. Poon, F. I. Lewis, S. L. K. Pond, and S. D. W. Frost. An evolutionary-network model reveals stratified interactions in the v3 loop of the hiv-1 envelope. *PLoS Computational Biology*, 3(11):e231, 2007.
- [39] G. Longobardi, C. Guardiano, G. Silvestri, A. Boattini, and A. Ceolin. Toward a syntax-based phylogenetic analysis of modern indo-european languages. In L. Borin and A. Saxena, editors, *Approaches to Measuring Linguistic Differences*, pages 137–181. De Gruyter Mouton, 2013.
- [40] G. Longobardi, A. Buch, A. Ceolin, A. Ecay, C. Guardiano, M. A. Irimia, D. Michelioudakis, N. Radkevich, and G. Jäger. Correlated evolution or not? phylogenetic linguistics with syntactic, cognacy and phonetic data. *Proceedings of the 11th International Conference (EVOLANG11)*, 01 2016.
- [41] P. Jordan and S. Shennan. Cultural transmission, language, and basketry traditions amongst the california indians. *Journal of Anthropological Archaeology*, 22(1):42–74, 2003.
- [42] G. U. Yule. A mathematical theory of evolution, based on the conclusions of dr. j. c. willis, f.r.s. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 213(402–410): 21–87, 1925.
- [43] T. Gernhard. The conditioned reconstructed process. *Journal of Theoretical Biology*, 253(4):769–778, 2008.
- [44] R. Durrett. *Essentials of Stochastic Processes*. Springer Texts in Statistics. Springer, Cham, 3 edition, 2016.

- [45] W. J. Anderson. *Continuous-Time Markov Chains*. Springer Series in Statistics. Springer-Verlag New York, New York, NY, 1 edition, 1991.
- [46] J. Felsenstein. Maximum likelihood and minimum-steps methods for estimating evolutionary trees from data on discrete characters. *Systematic Zoology*, 22(3):240–249, 1973.
- [47] J. Felsenstein. *Inferring Phylogenies*. Sinauer Associates, Sunderland, MA, 2004.
- [48] C. Tuffley and M. Steel. Modeling the covarion hypothesis of nucleotide substitution. *Mathematical Biosciences*, 147(1):63–91, 1998.
- [49] M. Pagel and A. Meade. Estimating rates of lexical replacement on phylogenetic trees of languages. In P. Forster and C. Renfrew, editors, *Phylogenetic Methods and the Prehistory of Languages*, volume 1, pages 173–182. McDonald Institute for Archaeological Research, Cambridge, 2006.
- [50] E. Schröder. Vier combinatorische probleme. *Zeitschrift für Mathematik und Physik*, 15: 361–376, 1870.
- [51] L. Foulds and R. Graham. The steiner problem in phylogeny is np-complete. *Advances in Applied Mathematics*, 3(1):43–49, 1982.
- [52] M. Li, J. Tromp, and L. Zhang. Some notes on the nearest neighbour interchange distance. In *Computing and Combinatorics (COCOON ’96)*, volume 1090 of *Lecture Notes in Computer Science*, pages 343–351. Springer, 1996.
- [53] D. M. Hillis, C. Moritz, and B. K. Mable, editors. *Molecular Systematics*. Sinauer Associates, Sunderland, MA, 2 edition, 1996.
- [54] R. Douc, E. Moulines, P. Priouret, and P. Soulier. *Markov Chains*. Springer, Cham, 2019.
- [55] G. Baele, X. Ji, G. W. Hassler, J. T. McCrone, Y. Shao, Z. Zhang, A. J. Holbrook, P. Lemey, A. J. Drummond, A. Rambaut, and M. A. Suchard. Beast x for bayesian phylogenetic, phylogeographic and phylodynamic inference., 2025-08-01.
- [56] D. Ayres, M. Cummings, G. Baele, A. Darling, P. Lewis, D. Swofford, J. Huelsenbeck, P. Lemey, A. Rambaut, and M. Suchard. Beagle 3: Improved performance, scaling, and usability for a high-performance computing library for statistical phylogenetics. *Systematic biology*, 68:1052–1061, 11 2019.
- [57] R. P. Dobrow. *Branching Processes*, chapter 4, pages 158–180. John Wiley Sons, Ltd, 2016.
- [58] A. Graham. *Kronecker Products and Matrix Calculus with Applications*. Dover Books on Mathematics. Dover Publications, 2018.
- [59] R. L. Dobrushin. General formulation of shannon’s main theorem in information theory. *Russian Mathematical Surveys*, 14(6):1–63, 1959.
- [60] T. M. Cover and J. A. Thomas. *Elements of Information Theory*. John Wiley & Sons, Hoboken, NJ, 2 edition, 2012.
- [61] D. Robinson and L. Foulds. Comparison of phylogenetic trees. *Mathematical Biosciences*, 53(1):131–147, 1981.
- [62] A. P. Branco, C. Vaz, and A. P. Francisco. Computing rf tree distance over succinct representations. *Algorithms*, 17(1):15, 2024.

- [63] M. A. Steel and D. Penny. Distributions of tree comparison metrics—some new results. *Systematic Biology*, 42(2):126–141, 1993.
- [64] J. E. O’Reilly and P. C. J. Donoghue. The efficacy of consensus tree methods for summarizing phylogenetic relationships from a posterior sample of trees estimated from morphological data. *Systematic Biology*, 67(2):354–362, 03 2018.
- [65] N. Tahiri, B. Fichet, and V. Makarenkov. Building alternative consensus trees and supertrees using k-means and robinson and foulds distance. *Bioinformatics*, 38(13):3367–3376, 2022.
- [66] A. Gelman, J. Carlin, H. Stern, D. Dunson, A. Vehtari, and D. Rubin. *Bayesian Data Analysis, Third Edition*. Chapman & Hall/CRC Texts in Statistical Science. Taylor & Francis, 2013.
- [67] A. Rambaut, A. J. Drummond, D. Xie, G. Baele, and M. A. Suchard. Posterior summarisation in bayesian phylogenetics using tracer 1.7. *Systematic Biology*, 67(5):901–904, 2018.
- [68] K. Schliep. phangorn: phylogenetic analysis in r. *Bioinformatics*, 27(4):592–593, 2011.
- [69] K. Schliep, A. J. Potts, D. A. Morrison, and G. W. Grimm. Intertwining phylogenetic trees and networks. *Methods in Ecology and Evolution*, 8(10):1212–1220, 2017.
- [70] L. J. Revell. phytools 2.0: an updated R ecosystem for phylogenetic comparative methods (and other things). *PeerJ*, 12:e16505, 2024.
- [71] L. Häuser, G. Jäger, and A. Stamatakis. A systematic exploration of current limitations of cognate-based phylogenetic inference. *Open Research Europe*, 5:258, 2026.
- [72] C. Anderson, M. Scarborough, L. Jocz, M. J. Kümmel, T. Jügel, B. Irslinger, R. Pooth, H. Liljegren, R. F. Strand, G. Haig, U. Geupel, M. Macak, R. I. Kim, E. Anonby, T. Pronk, O. Belyaev, T. K. Dewey-Findell, M. Boutilier, C. Freiberg, R. Tegethoff, M. Serangeli, K. Stroński, A. Falileyev, N. Liosis, K. Schulte, G. K. Gupta, R. Izadifar, P. Markus, N. Williams, S. Loi, N. Sims-Williams, M. Findell, S. Adibifar, G. Abete, P. Atanasov, E. Baiwir, M.-R. Bastardas, A. Benkato, L. S. Bevevino, Buchi, G. Cadorini, C. Cathcart, L. Cheveau, C. Christodoulou, J. Delorme, S. N. Dworkin, D. Ekici, S. Farridnejad, M. Gheitasi, H. Hammarström, S. Hewitt, A. A. Khan, M. K. Khan, L. Khokhlova, D. Kim, C. Lewin, B. Lushaj, P. Mahmoudveysi, M. Mahommadirad, S. Mersch, B. Mustafa, F. Nemat, M. Nourzaei, P. Ó Muircheartaigh, V. Oogjen, M. Ourang, H. Pagan, T. S. Palmer, S. Pepper, M. Purandare, K. Rehman, G. Rhys, U. Røyneland, M. Z. Sagar, J. J. Sandstedt, L. Steensland, M. Taheri-Ardali, M. Talebi-Dastenaiei, S. Tittel, T. Tresoldi, M. de Vaan, A. Verkerk, A. Versloot, P. Videsott, N. Vuletić, M. Widmer, A. Zeini, H.-J. Bibiko, F. Runge, R. D. Gray, and P. Heggarty. The indo-european cognate relationships dataset. *Scientific Data*, 12(1):1541, 2025.
- [73] J. G. Kemeny and J. L. Snell. *Finite Markov Chains*. Springer-Verlag, New York, 2nd edition, 1976.